

# Finite Transfers and Open Energy Balances

Physical tests of loaded gears, switched returns, and prepared states

Hob Nilre & Bo C. Herlin

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## Abstract

A rotating planet brought to a fourth central shaft gives a specific apparatus for measuring how an attached receiver changes the other shaft works. The ideal single-receiver control changes carrier work by  $7/3$  joule in one second. We derive its local forces and identify the additional joint, support, stator and endpoint observations needed for a physical comparison. A dimensioned Oldham coupling resolves a different support model without assigning its forces to universal joints. An isolated-primary tapped-secondary circuit then makes switched return a definite connection problem, with persistent winding, cell, gate and thermal states. Exact controls distinguish clocked attraction from hybrid operation, terminal equivalence from hidden energy, and prepared energy from receiver work. Two series-cell preparations have the same entire terminal history but stores of 1 and 5 joules. Finite reconnection specifies binary observation and paired-work certificates; an LC/shunt example proves every charge-sign event on its full window. A six-channel shared output has a certified finite return interval. Supplied timing prevents full-state repetition in an insulated model, and loaded electrical-port observations have an exact  $1/100$  joule worst-case initial-energy error on a two-state family. Reset order, material states and physical supplies lead to separate measurable questions. Independent uncertainties accompany work and endpoint observations. The results are exact or explicitly conditional within declared models; no apparatus data are used.

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Latest on GitHub: <https://github.com/hobnilre/physics-gear-op>

## 1 One apparatus, then its physical tests

A planet rolls between a sun and ring while its centre travels with the carrier. Bring the planet's rotating stub through two correctly phased universal joints to a separately supported central shaft. There are now four accessible shafts: sun, carrier, ring and planet output. When a receiver loads that fourth shaft, which reactions and work readings change, and how can their complete physical balance be measured?

*Frames, Returns, and Port Power* (Nilre and Herlin, 2026) constructs the ideal four-shaft mechanism and two electrical realizations of its terminal relations. Here the same construction supplies the first experiment. A held-ring control predicts a  $7/3$  J change in carrier work when one output receives  $7/3$  J in one second. Actual joint forces, moving supports, bearing losses and preparation must be identified independently. A separate Oldham model makes that distinction tangible: its known slot forces cannot determine a Cardan cross's reactions.

The electrical investigation likewise begins with connections that can be drawn and measured. A switched full-to-tap return is a different circuit from a change of voltage zero. A series-to-parallel bank change can reveal energy that its earlier terminal history concealed. Following these operations requires individual cells, finite winding states, switch and probe loading, gate supplies and actual reset endpoints.

A shared magnetic receiver adds another connected problem. Six isolated input cores feed twelve branch windings on one output core. A prepared finite transition can return work to its sources while supplying a receiver; full-state repetition and the accuracy of inferred initial energy require different proofs. The declared model below establishes a finite witness, an insulated-cycle obstruction and a bounded observation-error obstruction.

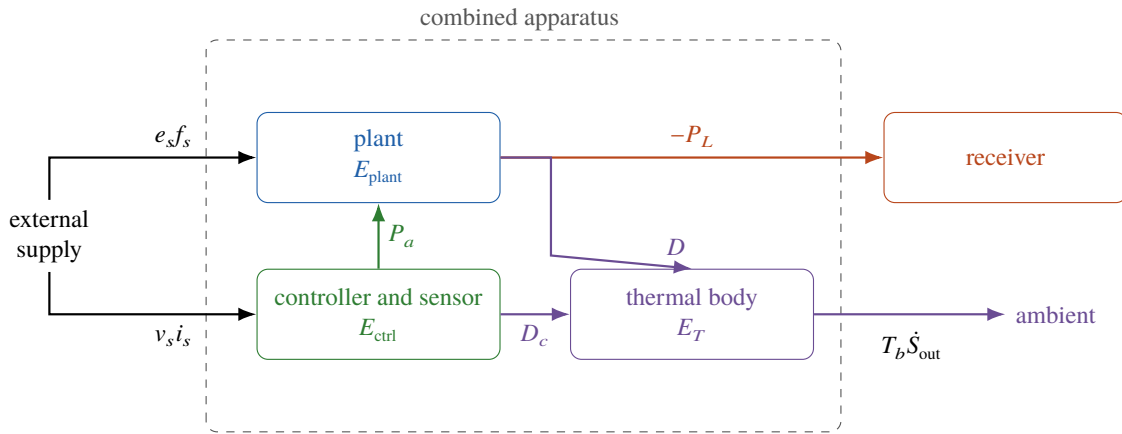
The article proceeds from the four-shaft test to its mechanical qualifications, then to windings, switched returns and signed cell states. Later sections develop material and supply laws and the independent uncertainties needed for a decision. Exact controls establish their own predictions; unknown constitutive laws require acquisition; a physical energy remainder requires simultaneous measurements of all included ports and endpoints. None of these tasks can supply the missing data for another.

## 2 Boundaries, signs, and exact integration

Power is positive into the named system. Mechanical products are  $\mathbf{F} \cdot \mathbf{v} + \mathbf{M} \cdot \boldsymbol{\omega}$  at the force application point and body; electrical products are complete voltage differences times oriented current. Thermal export is  $-T_b \dot{S}_{\text{out}}$  at boundary temperature  $T_b$ ; internal entropy production is not a second boundary heat port. On each explicit interval,

$$W_j(t_0, t_1) = \int_{t_0}^{t_1} e_j f_j dt, \quad r_E = E(t_1) - E(t_0) - \sum_j W_j. \quad (1)$$

For vector ports use the corresponding scalar product. Opposite sides of an internal interface are integrated separately and cancel only when their actual motions and efforts agree. An impulse represented by a declared jump law replaces a resolved finite edge; it is never added to that edge a second time. Both endpoints are constitutive evaluations, not residuals.



**Figure 1.** Schematic boundary hierarchy. Shared mechanical or electrical interfaces cancel only for the same realized path. Plant and controller conversion rates are  $D$  and  $D_c$ . Controller and thermal stores remain inside the combined apparatus; external supply, receiver, and ambient transfers remain measurable.

Figure 1 shows why enlarging a boundary changes the list of external transfers. All dimensional coefficients below are illustrative. In displayed worked paths,  $t$  denotes the numerical time in seconds and every coefficient has the SI units needed for the stated quantity. Unless another interval is specified, each worked path uses  $[0, 1]$  s. Equations using symbolic  $T$  retain its time units. Configuration I is the simple

train; II the linkage; III the compound geometries; IV the regular winding pair; V the prepared passive and switched circuits; VI the material and supply controls.

For later linear circuits a useful closed form integrates each port without solving for a missing work. Augment a constant, polynomial, or sinusoidal source by its exact differential equations, so that  $\dot{y} = Ay$ . For  $P_j = y^T Q_j y$ , with  $Q_j$  symmetric, define

$$\mathcal{I}(Q_j, A, y_0, T) = \text{vec}(Q_j)^T T \varphi_1(T(A \otimes I + I \otimes A)) \text{vec}(y_0 y_0^T), \quad \varphi_1(Z) = \sum_{n=0}^{\infty} \frac{Z^n}{(n+1)!}. \quad (2)$$

Here  $y(T) = e^{AT} y_0$  and  $Q_j = (e_j f_j^T + f_j e_j^T)/2$  when the two linear channel vectors are  $e_j, f_j$ . This entire matrix function is a closed form, including singular  $A$ ; no series is truncated.

**Lemma 1** (Independent port primitive). *Equation (2) equals the signed integral of that port.*

*Proof.* For  $Y = yy^T$ ,  $\dot{Y} = AY + YA^T$ . Column vectorization gives  $\text{vec } Y(t) = e^{t(A \otimes I + I \otimes A)} \text{vec } Y(0)$ . Integrating its convergent exponential series and contracting with  $Q_j$  gives the result without using any store.  $\square$

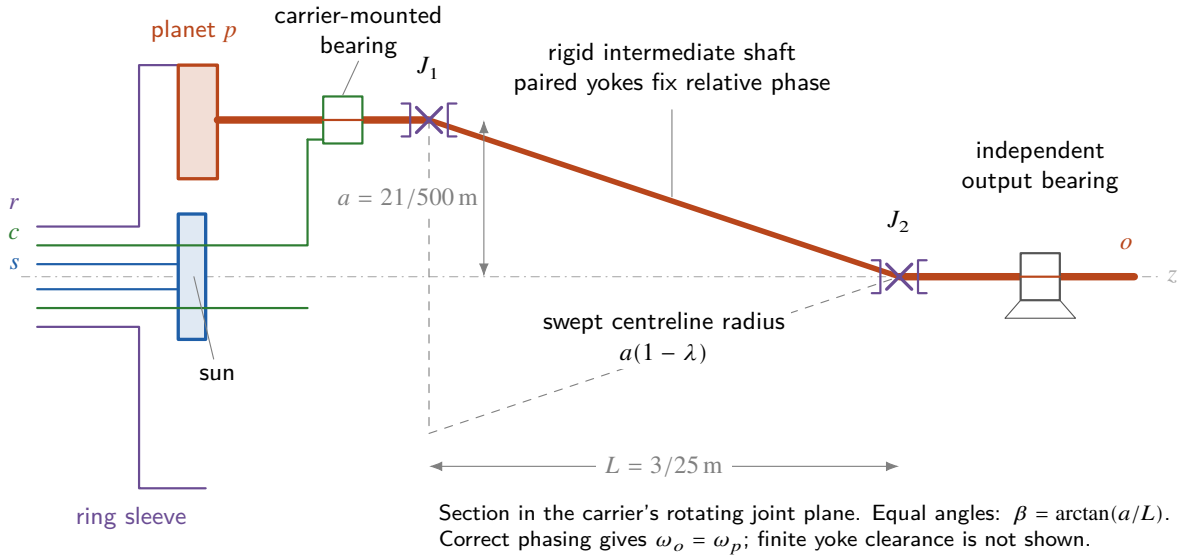
The numerical integration residual throughout this article is exactly zero: no numerical integration is used. The model residual comprises omitted physical laws and ports. It remains unquantified unless a separate, explicit constitutive control calculates a particular contribution.

### 3 Gears: free bodies, stores, and physical lead-outs

#### 3.1 The fourth shaft and its first measurable comparison

The planet body carries a rotating stub inside a carrier-mounted bearing. Two universal joints connect that stub to a separately supported central shaft  $o$ , beside the sun, carrier and ring shafts. The parallel-offset geometry has equal joint angles, with phasing fixed in the carrier's rotating joint plane. The intermediate shaft fixes that relative phase; equal angles and correct phasing are the ordinary double-joint conditions [Belden](#). With continuous joint lift  $H_\beta(x)$  satisfying  $\tan H_\beta(x) = \tan x / \cos \beta$ , the map  $\theta_o = \phi + H_\beta(H_\beta(\theta_p - \phi) + \pi/2) - \pi/2$  reduces to  $\theta_o = \theta_p$ . The later phase and support controls change specified parts of this model.

## Four separate shafts; one rotating planet stub



**Figure 2.** The recurring four-shaft apparatus. The output is connected to the rotating planet stub, not its carrier-fixed support pin. The yokes are phased relative to the rotating joint plane. Dimensions describe centreline geometry; complete joint forces, packaging and physical clearances require independent identification.

For teeth (24, 18, 60), module 1/500 m, offset 21/500 m and joint-centre axial separation 3/25 m, the two mesh equations give

$$(\omega_s, \omega_c, \omega_r, \omega_o) = (u + d, u, u - 2d/5, u - 4d/3). \quad (3)$$

All torques below enter the assembly. For a single steady loaded planet, the sun force is  $F_s = \tau_s/r_s$ , planet spin balance gives  $F_r = F_s - \tau_o/r_p$ , and the tangential pin force is  $-F_s - F_r$ . The ring and carrier moment arms then give

$$\tau_r = \frac{5}{2}\tau_s - \frac{10}{3}\tau_o, \quad \tau_c = -\frac{7}{2}\tau_s + \frac{7}{3}\tau_o. \quad (4)$$

These load reactions were obtained before evaluating power. Hold the ring, maintain  $(\omega_s, \omega_c) = (7/2, 1)$  rad/s and compare the unloaded planet with  $\tau_o = 1 \text{ N m}$ , at measured  $\tau_s = 2 \text{ N m}$ .

Four effective unit inertias give 673/72 J at each endpoint, with orbit counted once in the carrier coefficient. The four separate work integrals sum to zero. This initialized steady interval has no impulse and does not assign a preparation work. The later 7 J comparison uses three receivers or total output torque 3 N m; it is a different load. Equal planet speeds do not determine force sharing. Under equal bilateral contact stiffness, loads (3, 0, 0) and (1, 1, 1) N m have the same aggregate reactions but sun-force vectors (250/3, 0, 0) and (250/9, 250/9, 250/9) N. Individual contacts and pin couples need observations when they enter the boundary.

**Table 1.** The same established motion and sun effort with one attached receiver. Independent control authority and actual effort measurements are requirements of this comparison.

| One-second observation        | Unloaded output | Loaded output |
|-------------------------------|-----------------|---------------|
| Output rate, rad/s            | $-7/3$          | $-7/3$        |
| Carrier torque, N m           | $-7$            | $-14/3$       |
| Ring holding torque, N m      | $5$             | $5/3$         |
| Sun work into assembly, J     | $7$             | $7$           |
| Carrier work into assembly, J | $-7$            | $-14/3$       |
| Ring work into assembly, J    | $0$             | $0$           |
| Output work into assembly, J  | $0$             | $-7/3$        |

The first comparison uses synchronized shaft torque and unwrapped angle; the held-ring transducer checks its reaction although its work is zero. As an illustrative calibration requirement for the two carrier observations, take  $|\widehat{\tau}| \leq 8 \text{ N m}$ ,  $|\widehat{\omega}| \leq 101/100 \text{ rad/s}$ ,  $u_{\tau} \leq 1/200 \text{ N m}$  and  $u_{\omega} \leq 1/1000 \text{ rad/s}$ . The product bound derived below gives  $2611/100000 \text{ J}$  for their difference. A total  $1/10 \text{ J}$  difference budget leaves  $7389/100000 \text{ J}$  for independently bounded timing, setting and other observation errors. This resolves the predicted  $7/3 \text{ J}$  change if those requirements are met. A complete residual also needs every support, drive, thermal and endpoint contribution.

### 3.2 A dimensioned support model with a different coupling

The U-joint angle map leaves its spatial forces undetermined. A mirrored Oldham pair supplies a separate determinate parallel-axis control. Take offset  $a = 21/500 \text{ m}$ , combined middle mass  $m = 1/5 \text{ kg}$ , middle inertia  $I_m = 1/500$  and output inertia  $I_o = 1/250 \text{ kg m}^2$ . Orthogonal slots have half-stroke  $3/50 \text{ m}$ , pad arm  $b = 1/50 \text{ m}$ , bearing separation  $\ell = 3/100 \text{ m}$  and overhang  $h = 1/100 \text{ m}$ . Mirroring puts net pad forces in one plane; locating bearings hold axes parallel. Angular misalignment and axial force are excluded. Bilateral preload is declared as  $100 \text{ N}$ ; contact strength, preload compliance and friction require material data. The declared domain is  $|\omega_c| \leq 1$  and  $|\omega_p| \leq 7/3 \text{ rad/s}$ ; fixed output bearings do zero ideal work.

Put  $u = (\cos \theta, \sin \theta)$ ,  $v = (-\sin \theta, \cos \theta)$ ,  $P = a(\cos \phi, \sin \phi)$  and middle centre  $M = (P \cdot v)v$ , where  $\theta$  is planet orientation and  $\delta = \phi - \theta$ . For constant shaft rates, differentiation and the translational and angular free bodies give

$$K_L = \frac{ma^2}{2} [(\omega_c - \omega_p)^2 \cos^2 \delta + \omega_p^2 \sin^2 \delta] + \frac{I_m + I_o}{2} \omega_p^2, \quad (5)$$

$$F_i = -ma[(\omega_c - \omega_p)^2 + \omega_p^2] \sin \delta v, \quad F_o = -2ma(\omega_c - \omega_p)\omega_p \cos \delta u, \quad (6)$$

$$Q_p = ma^2 \omega_c^2 \cos \delta \sin \delta, \quad Q_c = -ma^2[(\omega_c - \omega_p)^2 + \omega_p^2] \cos \delta \sin \delta. \quad (7)$$

The force sum is  $m\ddot{M}$ . Input couple  $T_i = Q_p - \tau_{\text{rec}}$  follows from the angular equation. Near and far bearing forces are  $-(1 + h/\ell)F_i$  and  $hF_i/\ell$ ; their sum and bending moment close independently. The overhang moment is  $(-hF_{i,y}, hF_{i,x}, 0)$ . The two pad resultants are  $F_n/2 \pm T/(2b)$  for the local resultant  $F_n$  and transmitted torque  $T$ . This bilateral idealization must be checked against physical preload and contact capacity.

At  $(\omega_s, \omega_c, \omega_r, \omega_p) = (7/2, 1, 0, -7/3)$ ,  $\tau_s = 2$ ,  $\tau_{\text{rec}} = 1$  in SI units and initial  $\theta = \phi = 0$ , set  $\tau_o = -T_i$ . The train then requires  $\tau_r = 5 - 10\tau_o/3$ ,  $\tau_c = -7 + 7\tau_o/3$ ; its carrier drive additionally supplies  $Q_c$ .

The whole boundary includes train and linkage, so the two sides of the planet and moving-pin transfers cancel only after their separate integrals.

**Table 2.** Each work is the integral of its specified shaft product. The second interval returns the linkage store; the first does not.

| Interval, s    | $W_s$ , J  | $W_c$ , J                 | $W_r$ , J | $W_{\text{rec}}$ , J | Independent $\Delta K$ , J |
|----------------|------------|---------------------------|-----------|----------------------|----------------------------|
| $[0, 3\pi/40]$ | $21\pi/40$ | $-7\pi/20 - 2499/5000000$ | 0         | $-7\pi/40$           | $-2499/5000000$            |
| $[0, 3\pi/5]$  | $21\pi/5$  | $-14\pi/5$                | 0         | $-7\pi/5$            | 0                          |

The train alone has steady 673/72 J under its declared effective unit inertias; this control treats planet spin as that train coefficient and adds the new middle/output terms in  $K_L$  once. Evaluation at the two ends, rather than an assumed periodic energy, gives the last column. The exact residual is zero and no event occurs. These slot and bearing forces do not determine Cardan-cross forces.

Physical stator mounting also changes a brake law. For a massless viscous receiver with  $D = 1/2 \text{ N m s}$ , shaft rate  $\omega_p = -7/3$  and stator rate  $\omega_f$ , inward powers to the receiver are  $D(\omega_p - \omega_f)\omega_p$  and  $-D(\omega_p - \omega_f)\omega_f$ , with exported heat  $-D(\omega_p - \omega_f)^2$ . On  $[0, 1]$  s the three works are  $(49/18, 0, -49/18)$  J for  $\omega_f = 0$  and  $(35/9, 5/3, -50/9)$  J for  $\omega_f = 1$ . A real reference drive must supply the second mounting.

The zero-cycle support result also depends on the load history. In the massless fixed-phase joint control take  $\cos \beta_1 = 4/5$ ,  $\cos \beta_2 = 3/5$ ,  $\omega_p = 2$ ,  $\omega_c = 1$  and  $\psi = t$  on  $[0, \pi]$  s. For  $k = 3/4$ ,  $\chi = k/(\cos^2 \psi + k^2 \sin^2 \psi)$ . A passive variable torque demand  $\mathcal{T} = 1/\chi$  gives  $P_p = 2$ ,  $P_c = 1/\chi - 1$  and  $P_{\text{rec}} = -(1 + 1/\chi)$ . Since  $\int_0^\pi \chi^{-1} dt = 25\pi/24$ , the works are  $(2\pi, \pi/24, -49\pi/24)$  J. The empty store at both endpoints gives zero residual. The nonzero support work is derived for this load; the constant-load cycle null retains its separate scope. Unequal joint angles require a different axis/support arrangement from the equal-angle parallel-offset baseline.

### 3.3 Locking and release with actual endpoint states

A coaxial sphere inside an annular carrier isolates the locking question without imposing an orbital rolling constraint. Let  $I_b = 2/5$ ,  $I_c = 3/5 \text{ kg m}^2$  and viscous clutch  $D = 1 \text{ N m s}$ . Holding  $\omega_c = 1 \text{ rad/s}$  with the ball initially at rest gives  $\omega_b = 1 - e^{-5t/2}$ . On  $[0, 1]$  s, independently integrating drive  $D(1 - \omega_b)$ , ball transfer  $D(1 - \omega_b)\omega_b$  and heat  $D(1 - \omega_b)^2$  gives

$$W_d = \frac{2}{5}(1 - e^{-5/2}), \quad \Delta K_b = \frac{1}{5}(1 - e^{-5/2})^2, \quad H = \frac{1}{5}(1 - e^{-5}) \quad (\text{J}). \quad (8)$$

The ball store follows independently from its final rate. With a free undriven carrier initially at rate one, the same torque laws instead give  $\omega_c = 3/5 + 2e^{-25t/6}/5$ ,  $\omega_b = 3/5 - 3e^{-25t/6}/5$  and  $H = 3(1 - e^{-25/3})/25$  J at one second. Weighted momentum is conserved, the initial kinetic store is 3/10 J, and both final spin stores must be retained. A finite removal of torque preserves velocity; an impulsive release requires its own contact and transfer law.

A finite actuator model adds strain  $q$ , thermal excess  $H$  and gate voltage  $v_g$ . Declare

$$T_c = Kq + D(\omega_c - \omega_b), \quad \dot{q} = \omega_c - \omega_b, \quad (9)$$

$$I_c \dot{\omega}_c = T_d - T_c, \quad I_b \dot{\omega}_b = T_c - \omega_b/10, \quad T_d = 2(\Omega_d - \omega_c), \quad (10)$$

$$\frac{1}{100} \dot{v}_g = (V_g - v_g)/10, \quad \dot{H} = 2(\Omega_d - \omega_c)^2 + D(\omega_c - \omega_b)^2 + (V_g - v_g)^2/10 - H/2. \quad (11)$$

The independently defined store is  $I_c \omega_c^2/2 + I_b \omega_b^2/2 + Kq^2/2 + H + v_g^2/200$ . External powers are  $T_d \Omega_d$ ,  $\dot{K}q^2/2$ ,  $V_g(V_g - v_g)/10$ ,  $-\omega_b^2/10$  and  $-H/2$ . Multiplying the separate laws gives their sum as the store derivative. The stiffness actuator is physically distinct from the gate supply.

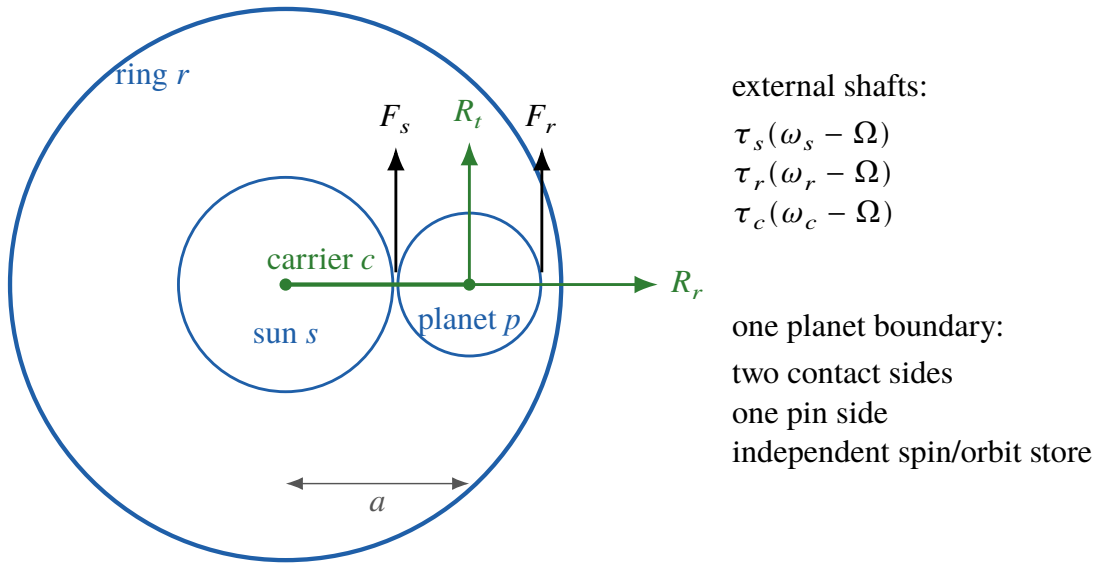
Use one-second stages with smooth cosine ramps  $r(t) = (1 - \cos \pi t)/2$  on each local  $0 \leq t \leq 1$ : prepare  $(\Omega_d, K, D, V_g) = (r, r/5, 1/10, 1)$ ; lock with  $(1, 1/5, 2, 1)$ ; release with  $(1, (1 - r)/5, 2(1 - r), 0)$ ; and reduce the drive with  $(1 - r, 0, 1/2, 0)$ . Start all states at zero. Each actual final store is evaluated from the resulting state; finite damping does not specify an exact reset. The equations define a conditional finite comparison, with no decimal trajectory asserted here. A stiffness actuator's efficiency, real release law and temperature dependence require acquisition. Both shaft channels, spring state, gate and stiffness supplies, heat export and thermal endpoints belong to that comparison.

### 3.4 Simple train and four observation frames

Let the sun, planet, internal ring, and carrier have indices  $s, p, r, c$ . All axes are initially parallel to  $+z$ , with positive right-hand rates. For common module  $m_0$ , radii  $r_j = m_0 Z_j/2$  satisfy  $a = r_s + r_p = r_r - r_p$  and  $Z_r = Z_s + 2Z_p$ . Rolling at each mesh gives

$$\omega_p = \omega_c - \frac{Z_s}{Z_p}(\omega_s - \omega_c) = \omega_c + \frac{Z_r}{Z_p}(\omega_r - \omega_c), \quad Z_s(\omega_s - \omega_c) + Z_r(\omega_r - \omega_c) = 0. \quad (12)$$

This carrier-relative construction is consistent with Culpepper (2002). Three equally spaced planets additionally require  $(Z_s + Z_r)/3$  integral and adequate clearance. Configuration I uses  $(24, 18, 60)$  and  $m_0 = 1/500$  m, so  $(r_s, r_p, r_r, a) = (3/125, 9/500, 3/50, 21/500)$  m and the phase quotient is 28.



**Figure 3.** Schematic pitch contacts and external shaft ports. Only one planet is drawn. Force arrows indicate positive tangential and radial components; their signed values follow the free bodies.

Figure 3 locates the forces and shaft ports. For  $q = 3$  equally loaded planets, tangential forces on one planet are  $F_s, F_r$ , and its pin reaction is  $R_t \mathbf{e}_\theta + R_r \mathbf{e}_r$ . With zero pin couple, Newton–Euler equations,

independently of energy, give

$$F_s = \frac{\tau_s - I_s \dot{\omega}_s}{qr_s}, \quad F_r = F_s + \frac{I_p \dot{\omega}_p}{r_p}, \quad R_t = m_p a \dot{\omega}_c - F_s - F_r, \quad (13)$$

$$\tau_r = I_r \dot{\omega}_r + qr_r F_r, \quad \tau_c = I_c \dot{\omega}_c + qa R_t, \quad R_r = -m_p a \omega_c^2. \quad (14)$$

These radial forces belong to the tangential pitch-contact idealization. Tooth pressure angles would add radial reactions. Constant rates give  $(\tau_s, \tau_r, \tau_c) = \tau_s(1, 5/2, -7/2)$  in configuration I.

The complete train contains gears, carrier, and pins. Only its three shaft ports cross the mechanical boundary in this rolling model. A planet subboundary also has the separately oriented powers

$$P_{sp}^f = F_s r_s (\omega_s - \Omega), \quad P_{rp}^f = F_r r_r (\omega_r - \Omega), \quad P_{cp}^f = R_t a (\omega_c - \Omega). \quad (15)$$

Each opposite mesh or pin side has the opposite product. Here  $\Omega$  is 0,  $\omega_c$ ,  $\omega_s$ , or  $\omega_r$  for the four specified observations. Their coordinate reconstructions do not physically remount any load. The measured division among shaft, mesh and pin transfers must retain each side; a shaft work difference cannot be assigned to a pin [OP-EPI-02].

Define  $J_c = I_c + qm_p a^2$ ,  $J = I_s + I_r + J_c + qI_p$  and

$$K^0 = \frac{1}{2}(I_s \omega_s^2 + I_r \omega_r^2 + J_c \omega_c^2 + qI_p \omega_p^2), \quad H = I_s \omega_s + I_r \omega_r + J_c \omega_c + qI_p \omega_p, \quad E^f = K^0 - \Omega H. \quad (16)$$

The relative kinetic energy is  $K^f = K^0 - \Omega H + J \Omega^2 / 2$ ;  $E^f$  includes effective centrifugal potential  $-J \Omega^2 / 2$ . This effective energy and the physical ground energy are distinct stores. The familiar rotating-velocity law is documented in MIT (2022).

**Theorem 1** (Frame work and its path division). *For the specified rigid train,  $\sum_j \tau_j = \dot{H}$  and*

$$\dot{K}^0 = \sum_j \tau_j \omega_j, \quad \dot{E}^f = \sum_j \tau_j (\omega_j - \Omega) - \dot{\Omega} H, \quad (17)$$

$$W_j^0 - W_j^f = \int \Omega \tau_j dt, \quad \int \Omega \dot{H} dt + \int H \dot{\Omega} dt = [\Omega H]_{t_0}^{t_1}. \quad (18)$$

*Proof.* Sum (14) after multiplying the spin and translation equations by their own rates. The opposite contact products cancel using (12); summing moments uses both equalities defining  $a$ . Differentiating (16), subtracting each shaft product, and integrating the product derivative of  $\Omega H$  gives the remaining identities.  $\square$

The effective-field term separates as Euler power  $-\dot{\Omega}(H - J\Omega)$  and explicit potential-time term  $-J\Omega\dot{\Omega}$ . Each is integrated before combining them. They are coordinate terms, not physical motor or heat supplies. Reconstructing these transient frame balances within independent uncertainty is the physical comparison [OP-EPI-01]. For a genuine rotating support its physical torque–rate work is additional.

For exact rational examples choose  $I_s = I_r = qI_p = J_c = 1 \text{ kg m}^2$ ,  $m_p = 1 \text{ kg}$ , and  $I_c = 248677/250000 \text{ kg m}^2 > 0$ . Flywheels may supply these inertias; no uniform-disc mass law is assumed. Let  $(\omega_s, \omega_r, \omega_c, \omega_p) = (7/2, 0, 1, -7/3) \text{ rad/s}$  and  $\tau_s = 2 \text{ N m}$  on  $[0, 1] \text{ s}$ . Direct state evaluation gives  $H = 13/6$ ,  $K^0 = 673/72$ ,  $E^c = 517/72$  in SI units at **both** endpoints.

**Table 3.** Configuration I in steady operation. Only the first three entries belong to the complete external shaft sum. Each complete residual is zero.

| Separately integrated work, J | Ground | Carrier |
|-------------------------------|--------|---------|
| Sun shaft                     | 7      | 5       |
| Ring shaft                    | 0      | -5      |
| Carrier shaft                 | -7     | 0       |
| One sun contact into planet   | 7/3    | 5/3     |
| One ring contact into planet  | 0      | -5/3    |
| One pin into planet           | -7/3   | 0       |

Preparation instead prescribes  $\omega_j(t) = \omega_{j*}t$  and  $\tau_s(t) = 2t$  on  $[0, 1]$  s from rest. Equations (14) give  $\tau_r = 5t - 595/36$  and  $\tau_c = 673/36 - 7t$ . For any frame  $\Omega = \Omega_*t$ , each shaft integral is

$$W_j^f = (\omega_{j*} - \Omega_*) \left( \frac{a_j}{3} + \frac{b_j}{2} \right), \quad (a_s, a_r, a_c) = (2, 5, -7), \quad (b_s, b_r, b_c) = (0, -595/36, 673/36). \quad (19)$$

The units of  $a_j, b_j$  here incorporate the one-second ramp. Every internal integral follows (15) with the affine forces in (14); explicitly  $\int_0^1 (At + B)t dt = A/3 + B/2$ .

**Table 4.** Preparation works and independent endpoints. Every complete sum equals its store change. The ground and ring frames coincide only for this held-ring path.

| Observation          | $(W_s, W_r, W_c), J$       | Effective-field work, |                 |
|----------------------|----------------------------|-----------------------|-----------------|
|                      |                            | J                     | $E(0), E(1), J$ |
| Ground and held ring | $(7/3, 0, 505/72)$         | 0                     | 0, 673/72       |
| Carrier              | $(5/3, 475/72, 0)$         | -13/12                | 0, 517/72       |
| Sun                  | $(0, 3325/144, -2525/144)$ | -91/24                | 0, 127/72       |

For a distinct history set  $H = (13/6)t$  and  $\Omega = t + bt(1-t)$ . With  $d = \omega_s - \omega_c$  one has  $H = 4\omega_c - (11/15)d$ , so choose  $\omega_c = \Omega$  and  $d = (15/11)(4\Omega - H)$ . Equation (12) determines all other rates; the ring moves during preparation when  $b \neq 0$  and reaches the same held endpoint. The shaft and field differences are  $(13/6)(1/2 + b/6)$  and  $(13/6)(1/2 - b/6)$  J. Thus  $b = 0, 1$  give respectively  $(13/12, 13/12)$  and  $(13/9, 13/18)$  J. The same endpoint total  $13/6$  J follows by integration by parts, not by presuming an equal division or a held ring throughout a nonproportional preparation. Measuring this history-dependent shaft/field division is a separate comparison [OP-EPI-04].

For a locked approach retain  $\omega_c = 1$ , put  $\omega_s = 1 + \epsilon$ , and obtain  $\omega_r = 1 - 2\epsilon/5$ ,  $\omega_p = 1 - 4\epsilon/3$ . At fixed  $\tau_s = 2$  the one-second ground shaft works are  $(2 + 2\epsilon, 5 - 2\epsilon, -7)$  J. At  $\tau_s = 2\epsilon$  they are  $\epsilon$  times that triple. Their exact limits are  $(2, 5, -7)$  and  $(0, 0, 0)$  J. Carrier-frame works vanish at the locked endpoint in both controls. The independent kinetic endpoints are evaluated by (16); a steady interval has identical states at its ends, not necessarily zero energy. Fixed-drive and scaled-drive approaches require their own finite uncertainty bounds [OP-EPI-12].

### 3.5 Rotor, stator, phase, and load

Let  $x = \theta_p - \phi$ , where  $\dot{\phi} = \omega_c$ , and specify  $\theta_o = \phi + F(x)$ . The output rate is  $\omega_o = \omega_c + \chi(\omega_p - \omega_c)$ ,  $\chi = F'(x)$ . For the same rotor path two stator counts obey  $N_o^0 - N_o^c = \Delta\phi/(2\pi)$ . Physical stator remounting under load requires a fresh trajectory, not substitution into the unloaded path.

For a Hooke joint define the continuous angle lift  $H_\beta(x) = \text{atan2}(\sin x, \cos \beta \cos x)$ . The two-joint Z path is  $F = H_{\beta_2}(H_{\beta_1}(x) + \gamma + \pi/2) + \pi/2$ , up to a constant. Equal angles and carrier phasing  $\gamma = 0$  give  $F = x$ ; quarter-turn phasing gives

$$\tan F = \frac{\tan x}{\cos^2 \beta}, \quad \chi = \frac{\cos^2 \beta}{\cos^4 \beta \cos^2 x + \sin^2 x}, \quad F(x + \pi) = F(x) + \pi. \quad (20)$$

An arbitrary finite readout retains  $[F(x) - x]_{t_0}^{t_1}/(2\pi)$ . Ground phasing prescribes  $\gamma = -\phi$  in the full map and introduces its own phase actuator; it supplies a mean carrier-relative count with endpoint ripple. The paired-stator identity does not prohibit differential instruments that report other relative observables.

These maps cover the eleven specified ideal mountings: equal-angle correct and quarter-turn Hooke paths against ground and carrier stators; the kinematic ground-phased Hooke path with ground stator; constant-ratio offset paths against both stators; and bevel ratios one and two against both stators. Oldham's orthogonal sliding coordinates are  $a \cos x, a \sin x$ . A Schmidt path can share the angle map without sharing its force system.

For configuration II the linkage contains input inertia  $I_R$ , orbiting mass  $m_R$ , output inertia  $I_o$ , and massless joint members. An imposed output torque  $\tau_L$  and drag  $d_0 \text{sgn}(\omega_o - \omega_\sigma)$  give

$$\mathcal{T} = I_o \dot{\omega}_o - \tau_L + d_0 \text{sgn}(\omega_o - \omega_\sigma), \quad \tau_d = I_R \dot{\omega}_p + \chi \mathcal{T}, \quad \tau_u = (1 - \chi) \mathcal{T}. \quad (21)$$

The five physical work channels are the separate integrals of  $\tau_d \omega_p, \tau_u \omega_c, \tau_L \omega_o, -d_0 \text{sgn}(\omega_o - \omega_\sigma) \omega_o$ , and  $m_R a^2 \dot{\omega}_c \omega_c$ . Their sum differentiates  $K_L^0 = (I_R \omega_p^2 + m_R a^2 \omega_c^2 + I_o \omega_o^2)/2$ . The stator reaction has power  $+d_0 \text{sgn}(\omega_o - \omega_\sigma) \omega_\sigma$ ; including it gives mechanical conversion  $D = d_0 |\omega_o - \omega_\sigma|$ . Thermal storage changes the larger boundary as derived below.

For constant input rates, constant load, and unchanged drag sign, the Hooke support-cycle integral over  $T_b = \pi/|\omega_p - \omega_c|$  is zero:  $\int (1 - \chi) dt = 0$  and its inertial part is proportional to  $[\chi - \chi^2/2]$  at equal phases. In a massless control set  $(\omega_p, \omega_c) = (3, 1)$ ,  $\tau_L = -1, d_0 = 0$ . On  $[0, \pi/2]$  s the Hooke drive, support, and receiver works are  $(3\pi/2, 0, -3\pi/2)$  J. The bevel map  $F = 2x$  instead gives  $(3\pi, -\pi/2, -5\pi/2)$  J. Both ideal endpoint stores are zero. In the carrier frame the bevel works are  $(2\pi, 0, -2\pi)$  J. A whole-cycle support null cannot be inferred for a partial cycle. Actual Hooke support work is the first measurement [OP-EPI-06]; the different bevel support path needs its own work account [OP-EPI-07].

For unequal angles with correct phasing,  $\tan F = (\cos \beta_2 / \cos \beta_1) \tan x$ . Choose these cosines as  $3/5, 4/5$  respectively, in the order  $\cos \beta_2 = 3/5, \cos \beta_1 = 4/5$ . With  $x = t, \omega_c = 1, \mathcal{T} = 1$ , and  $[0, \pi/4]$  s, the support work is  $\pi/4 - \arctan(3/4) > 0$  J, against zero for equal angles. These deliberately large articulations are mathematical controls; buildability and clearance require separate geometry. Under carrier acceleration, the rotor-pin integral for  $m_R a^2 = 1$  and  $\omega_c = t$  is  $1/2$  J. For an independently moving phase coordinate  $\gamma$ , virtual displacement adds torque  $\tau_\gamma = \mathcal{T} \partial \theta_o / \partial \gamma$ ; its work is  $\int \tau_\gamma \dot{\gamma} dt$ . A unit torque through  $\gamma = t$  supplies 1 J. None is supplied by a count alone. Their combined physical lead-out, carrier acceleration and phase-reference works remain to be determined [OP-EPI-31].

The actual Cardan force and bending system requires its bearing constraints, member stiffnesses, and application-point motions. A finite control with  $F_x = 1$  N,  $\dot{x} = 1$  m/s over one second transfers 1 J; a

fixed support transfers zero at the same force. The same rule applies to a bending couple. It specifies a force/displacement discrimination, not the unknown force in an unspecified double Cardan device. Full transverse forces and bending couples, including their application-point motion, are the acquisition target [OP-EPI-30].

For a stator comparison reproducing the finite scale of Nilre and Herlin (2026), take  $\omega_p = -14\pi/3$ ,  $\omega_c = 2\pi$ ,  $d_0 = 1/50$  and duration  $7/2$  s. The separate relative drag integrals are  $Q_0 = 49\pi/150$  and  $Q_c = 7\pi/15$  J, giving  $Q_c - Q_0 = 7\pi/50$  J. This is a change of physical mounting, not a coordinate dependence of the heat produced by one mounting. The actual loaded readout and its finite endpoint corrections require identification [OP-EPI-05].

### 3.6 Attaching a receiver changes the free bodies

Attach three identical unity-ratio, ground-stator lead-outs to configuration I, with  $\tau_L = +1$  N m on the negative-speed outputs and zero drag. The steady linkage requires  $\tau_d = -1$ ,  $\tau_u = 0$ . The planet equation becomes  $F_r = F_s + (I_p \dot{\omega}_p + \tau_d)/r_p$ ; the carrier equation also includes  $q(\tau_u + m_R a^2 \dot{\omega}_c)$ . The resulting steady torques are  $(2, -5, 0)$  N m. Separate one-second external works are

$$(W_s, W_r, W_c) = (7, 0, 0) \text{ J}, \quad W_{L,j} = -7/3 \text{ J} \quad (j = 1, 2, 3). \quad (22)$$

The shared axle transfers  $+7/3$  J to each linkage and  $-7/3$  J to its planet. Both endpoint stores are  $673/72 + 3[(I_R + I_o)(7/3)^2 + m_R a^2]/2$  J for the stated added inertias. They agree because the actual endpoint rates agree. The carrier work changes from  $-7$  to zero, a  $+7$  J comparison. Adding a receiver to an unchanged unloaded force table would miss that change. The full joint, bearing, phase-driver, and thermal balance still requires those independent physical ports and states. The corresponding complete loaded-train work remains a physical measurement [OP-EPI-33].

## 4 Additional mechanical boundaries

### 4.1 Compound circulation and a distinct two-sun geometry

In configuration III two coaxial members  $A, B$  engage rigidly connected planet sections  $P_A, P_B$  carried by  $C$ . Let  $s_X = -1$  for an external mesh,  $+1$  for an internal mesh, and

$$a = \frac{m_0}{2}(Z_A - s_A Z_{pA}) = \frac{m_0}{2}(Z_B - s_B Z_{pB}), \quad h_X = s_X \frac{Z_{pX}}{Z_X}, \quad R = \frac{h_A}{h_B}. \quad (23)$$

Rolling gives  $\omega_X - \omega_C = h_X(\omega_P - \omega_C)$ . The planet spin equation  $s_A r_{pA} F_A + s_B r_{pB} F_B = 0$ , together with shaft and carrier free bodies, gives  $(\tau_A, \tau_B, \tau_C) = \tau_*(1, -R, R - 1)$ . This is a massless quasistatic model with empty endpoint stores.

Three shaft powers and opposite sides of the two contacts and pin give nine component powers, namely signed copies of  $p_X^f = \tau_X(\omega_X - \Omega)$ . Define  $D_0 = \max_X |\tau_X \omega_X| > 0$ . Then

$$\mathcal{E}_f = \frac{\max_X |p_X^f|}{D_0}, \quad \mathcal{E}_0 = 1, \quad \mathcal{E}_C = \begin{cases} 1/|R - 1|, & A \text{ held,} \\ |R|/|R - 1|, & B \text{ held.} \end{cases} \quad (24)$$

Each work is separately  $p_X^f T$ , with its opposite side integrated as  $-p_X^f T$  before combining. Choose  $\omega_C = 1$ ,  $\tau_* = 1$ ,  $T = 1$  in SI units. Five pitch-compatible tooth sets give the following exact values.

**Table 5.** Compound controls. Ground ratios are one in all cases. The other frames follow by putting  $\Omega = \omega_A$  or  $\omega_B$  in equation (24). Multiple-planet phasing is a separate assembly condition; a single stepped planet suffices for these kinematic controls.

| $(Z_A, Z_{pA}, Z_{pB}, Z_B)$ | $(s_A, s_B)$ | $R$       | $\mathcal{C}_C, A \text{ held}$ | $\mathcal{C}_C, B \text{ held}$ |
|------------------------------|--------------|-----------|---------------------------------|---------------------------------|
| (24, 18, 18, 60)             | (−1, +1)     | −5/2      | 2/7                             | 5/7                             |
| (20, 22, 20, 62)             | (−1, +1)     | −341/100  | 100/441                         | 341/441                         |
| (30, 20, 19, 31)             | (−1, −1)     | 62/57     | 57/5                            | 62/5                            |
| (62, 20, 21, 63)             | (+1, +1)     | 30/31     | 31                              | 30                              |
| (100, 58, 59, 101)           | (+1, +1)     | 2929/2950 | 2950/21                         | 2929/21                         |

At  $R = 1$  with either coaxial member held,  $D_0 = 0$ ; the ratio is undefined. No finite resolution on the denominator establishes an infinite physical ratio. Approaching that geometry at finite tooth numbers is a sequence of specified comparisons. Their frame-dependent power ratios require synchronized torque and speed measurements with a resolved denominator [OP-EPI-08].

A Ravigneaux arrangement has independent long and short planet spins, not the single  $P$  of (23). To exhibit the distinction, use two sun planes and one long/short pair with teeth  $(Z_A, Z_B, Z_R, Z_S, Z_L) = (30, 60, 90, 15, 15)$ . With  $u_j = \omega_j - \omega_C$ , its four independent rolling equations are

$$30u_A + 15u_S = 0, \quad 15u_S + 15u_L = 0, \quad 60u_B + 15u_L = 0, \quad 90u_R - 15u_L = 0. \quad (25)$$

At module 1/30 m the short and long centres can lie on the same radial line at 3/4 and 5/4 m; their separation is 1/2 m, the sum of their pitch radii. Sun planes are axially separated. This is pitch geometry, not a tooth-strength or packaging design. With  $\omega_C = 1$ ,  $\omega_R = 0$ , rates are  $(\omega_A, \omega_B, \omega_S, \omega_L) = (-2, 5/2, 7, -5)$ . The independent reduced constraints are  $u_A + 2u_B = 0$  and  $u_A - 3u_R = 0$ . Their massless contact reactions have multiplier torque vectors  $(1, 2, 0, -3)$  and  $(1, 0, -3, 2)$  on  $(A, B, R, C)$ , as follows by resolving each contact's equal and opposite force. Choosing  $\tau_A = \tau_B = 1$  gives  $\tau_R = -3/2$ ,  $\tau_C = -1/2$ . The separate one-second ground shaft works are  $(-2, 5/2, 0, -1/2)$  J; the carrier works are  $(-3, 3/2, 3/2, 0)$  J. Both sum to zero, and the shaft ratio is 6/5 against its ground denominator 5/2 W. Internal contact powers require these four meshes, not the earlier nine-side set. Empty massless stores close this bounded model only.

Helical thrust or a nonparallel joint adds vector contact and support ports. For instance an imposed axial force 1 N through 1 m over one second adds 1 J, whereas holding that coordinate gives zero. Its existence and sign in a given gear require that gear's contact geometry, compliance, and measured trajectory. The two-sun example supplies no such missing law. Other geometry and contact directions retain their own energy partition [OP-EPI-21].

## 4.2 Contact changes, reversals, and unequal loading

A compliant gap has overlap  $\delta$ , stored energy  $U = k[\max(\delta, 0)]^2/2$ , and, during compression contact,  $F = k\delta + c\dot{\delta}$ . Keep the damper active only on its declared contact branch; separating and tensile branches need their own force law. For the spring/damper subboundary, input is  $F\dot{\delta}$  and heat export is  $-c\dot{\delta}^2$ . At  $\delta = 0$  the spring store is continuous even when damping makes power jump. The control  $\delta = t$ ,  $k = 2$ ,  $c = 1$  on  $[0, 1]$  s has

$$W_{\text{contact}} = \int_0^1 (2t + 1)dt = 2 \text{ J}, \quad W_h = -1 \text{ J}, \quad U(0) = 0, \quad U(1) = 1 \text{ J}. \quad (26)$$

At onset input power changes from 0 to 1 W, while  $\Delta U$  across the instant itself and impulse work are zero. Ramps approaching contact are separate intervals; the actuator imposing  $\delta$  has the opposite work. Actual re-engagement power jumps require the finite contact law and event-side measurements [OP-EPI-10]. For an impulsive alternative take two freely translating effective masses  $m_1, m_2$  and relative incoming speed  $v_- = v_1^- - v_2^- > 0$ . A declared restitution coefficient  $e \in [0, 1]$  gives  $J = (1 + e)\mu v_-$ ,  $\mu = m_1 m_2 / (m_1 + m_2)$ , from impulse momentum and  $v_+ = -e v_-$ . The separate contact-side works under a resolved regularization with no other impulse are

$$W_1 = \frac{m_1}{2} [(v_1^+)^2 - (v_1^-)^2], \quad W_2 = \frac{m_2}{2} [(v_2^+)^2 - (v_2^-)^2], \quad W_1 + W_2 = -\frac{\mu}{2} (1 - e^2) v_-^2. \quad (27)$$

These follow directly by integrating  $m \dot{v}$ , not by multiplying a force impulse by one arbitrarily chosen event velocity. For unit masses,  $(v_1^-, v_2^-) = (1, 0)$  and  $e = 1/2$ , final velocities are  $(1/4, 3/4)$ ,  $J = 3/4$  N s, works  $(-15/32, 9/32)$  J, and kinetic stores  $1/2, 5/16$  J. The contact receives  $3/16$  J. If contact and its thermal store are included, that conversion is internal; if its heat is exported, the external heat work is  $-3/16$  J. Do not also add the two internal contact works to that larger boundary. A finite pulse with the specified restitution can measure these transfers; it cannot realize a zero-time impulse. Physical contact regularization and its energy destination remain independent questions [OP-EPI-22].

For a full compliant train write actual contact coordinates  $\delta_l(q)$ ,  $U = \sum_l k_l \max(\delta_l, 0)^2/2$ , and apply each force through its geometric gradient. The rigid no-slip constraints must be released where that compliance acts. Multiplying the full Newton equations by all body velocities gives  $\dot{K} + \dot{U} = \sum P_{\text{ext}} - \sum_l c_l \dot{\delta}_l^2$  on each active contact interval. In a coaxial observation add  $-\dot{\Omega}H$  to the effective energy  $K + U - \Omega H$ . An explicit full-train control includes all three planets and both contacts on each. Keep configuration  $\Gamma$ 's inertias, hold the sun, ring, and planet spin angles at zero, and prescribe  $\theta_c = t^2/2$  on  $[0, 1]$  s. Define the two linear pitch-compliance coordinates on each planet by  $\delta_s = a\theta_c - r_s\theta_s - r_p\theta_p - g$  and  $\delta_r = a\theta_c - r_r\theta_r + r_p\theta_p - g$ . Let  $g = a/8$ ,  $k = 2/a^2$ , and  $c = 1/a^2$  in SI units. All six contacts engage at  $t = 1/2$  s; during contact they have  $\delta = a(t^2 - 1/4)/2$ ,  $\dot{\delta} = at$ , and  $F = (t^2 + t - 1/4)/a$ . Equal opposite planet spin torques leave  $\omega_p = 0$  without a planet drive. Resolving each contact gradient gives  $\tau_s = -3r_s F$ ,  $\tau_r = -3r_r F$ , and  $\tau_c = 1 + 6aF$  after engagement; before it,  $(\tau_s, \tau_r, \tau_c) = (0, 0, 1)$ . This is an exact lumped linear pitch-contact model, not a claim about finite-deflection tooth geometry.

Separate integration on the gap and contact intervals gives

$$W_c^0 = 99/32, \quad W_s^0 = W_r^0 = 0, \quad \sum_{l=1}^6 W_{h,l} = -7/4, \quad (K + U)(0) = 0, \quad (K + U)(1) = 43/32 \quad \text{J}. \quad (28)$$

Each heat work is  $-7/24$  J; each spring endpoint is  $9/64$  J and final carrier kinetic energy is  $1/2$  J. In carrier observation,  $(W_s^c, W_r^c, W_c^c) = (83/112, 415/224, 0)$  J, field work is  $-1/2$  J, and effective endpoints are  $0, 11/32$  J. At engagement damping makes finite contact powers jump; coordinates, velocities, and stores remain continuous and impulse work is zero. These independently evaluated stores close both balances. Clearance in a lead-out, changing active tooth contacts, unequal loading, and actual force laws remain separate physical questions. Their full accelerating-train remainder is not fixed by this contact control [OP-EPI-23].

With unequal planet fractions  $(1/2, 1/3, 1/6)$ , a steady total 7 J sun-contact transfer divides into  $(7/2, 7/3, 7/6)$  J rather than three  $7/3$  J values. The fractions specify a load-sharing law; they are not inferred from an aggregate balance. A pin couple  $M = 1/10$  N m on a planet with  $\omega_p - \omega_c = -10/3$  rad/s gives pair work  $-1/3$  J in one second. Its two separately integrated sides are  $M\omega_p$  and  $-M\omega_c$ .

Bearing heat or elastic torsion then needs an independently chosen law. Each planet's force and couple channels must therefore be observed. The load-sharing and pin-couple partition is a local measurement [OP-EPI-24].

For a reversal,  $\omega = 1 - 2t$ ,  $\tau = 1$  on  $[0, 1]$  s gives  $W_{[0,1/2]} = 1/4$  J and  $W_{[1/2,1]} = -1/4$  J. Net work is zero while  $\int |P|dt = 1/2$  J. A separate Coulomb drag of unit magnitude converts  $\int |\omega|dt = 1/2$  J; treating its sign as constant would give the wrong zero. A unit inertial rotor has equal endpoint kinetic energies  $1/2$  J on this prescribed reversal, and its inertial actuator torque  $\dot{\omega} = -2$  supplies works  $-1/2, +1/2$  J on the two halves. Any concurrent applied constant torque and brake require their own compensating drive. The examples are separate effort laws, not a claim that torque one alone produces that trajectory. Signed and absolute works through an actual reversal need independently calibrated channels [OP-EPI-28].

### 4.3 Radial motion and vector observation

For a centre at radius  $r$ , planar Newton equations give  $F_r = m(\ddot{r} - r\omega_c^2)$  and  $F_t = m(r\dot{\omega}_c + 2\dot{r}\omega_c)$ . Both  $F_r\dot{r}$  and  $F_t r\omega_c$  are physical work channels. With  $m = 1$ ,  $r = 1 + t$ ,  $\omega_c = 1$  over one second, their works are  $-3/2$  and  $+3$  J. Ground kinetic energy changes from 1 to  $5/2$  J; carrier effective energy changes from 0 to  $-3/2$  J. This radial-guide control requires compliant or variable contact geometry before it can represent an actual radially moving planet.

In any rotating observation,  $\mathbf{F}_{\text{Cor}} = -2m\boldsymbol{\Omega} \times \mathbf{v}^f$  has  $\mathbf{F}_{\text{Cor}} \cdot \mathbf{v}^f = 0$ . At fixed radius,  $\mathbf{v}^f = r(\omega_c - \boldsymbol{\Omega})\mathbf{e}_\theta$ , so the Coriolis force is generally nonzero outside ground and carrier observations. Radial motion adds a tangential component; it does not add net Coriolis power. This distinction follows from the vector product itself. Radial support work and changed contact geometry still need identification [OP-EPI-26].

For observations rotating about a fixed common origin in three dimensions,

$$E^f = K^0 + U - \boldsymbol{\Omega} \cdot \mathbf{H}, \quad P_j^f = \mathbf{F}_j \cdot (\mathbf{v}_j - \boldsymbol{\Omega} \times \mathbf{r}_j) + \mathbf{M}_j \cdot (\boldsymbol{\omega}_j - \boldsymbol{\Omega}), \quad P_{\text{field}} = -\dot{\boldsymbol{\Omega}} \cdot \mathbf{H}. \quad (29)$$

Here  $U$  is rotationally invariant internal potential, and all vectors and time derivatives in the equation are resolved in an inertial basis. The result follows from  $\sum(\mathbf{r} \times \mathbf{F} + \mathbf{M}) = \dot{\mathbf{H}}$  and the kinetic-energy identity, including the full inertia tensor in the relative kinetic and centrifugal terms. A translating origin would add linear-momentum terms and is outside this formula. For a free spherical rotor of unit inertia with  $\boldsymbol{\omega} = (1, 0, 2)$  and observation  $\boldsymbol{\Omega} = (t, 0, 0)$ ,  $K^0 = 5/2$  J at both endpoints, physical work is zero, field work is  $-1$  J, and effective energy changes from  $5/2$  to  $3/2$  J. Dropping the transverse momentum predicts zero field work. A symmetric axial train can have zero transverse momentum; the rotor example distinguishes the vector laws without asserting that every train has this nonzero component. The nonparallel-axis energy comparison requires the full observed vector momentum [OP-EPI-27].

## 5 Windings, returns, and finite dynamic correspondence

### 5.1 Coordinate zero versus an attached branch

In a tapped pair, currents enter winding 1 at  $a$  and winding 2 at  $b$ , with returns  $c$  and  $a$ . Thus

$$(v_1, v_2) = (V_a - V_c, V_b - V_a), \quad (I_a, I_b, I_c) = (i_1 - i_2, i_2, -i_1). \quad (30)$$

A common  $V_j \mapsto V_j + g(t)$  leaves the complete winding voltages unchanged. Its terminal contributions change by  $gI_j$ , whose sum is zero by current conservation. Each changed work is separately  $\int gI_j dt$ ; deleting a return changes that sum.

A physical attachment instead has voltage  $v_b$  across its own resistor  $R_b$  and capacitor  $C_b$ . On a prescribed ramp  $v_b = t$  with  $R_b = C_b = 1$  over one second, its source current is  $i_b = t + 1$ . For the attachment boundary, direct integration gives

$$W_b = \int_0^1 t(t+1)dt = 5/6 \text{ J}, \quad W_h = -\int_0^1 t^2 dt = -1/3 \text{ J}, \quad E_b(0) = 0, \quad E_b(1) = 1/2 \text{ J}. \quad (31)$$

Adding a parallel constant-current receiver of 1 A over the same ramp gives another 1/2 J of receiver delivery and raises source work to 4/3 J. More generally  $W_b = C_b/2 + 1/(3R_b)$  J for the unit ramp; a pure voltage coordinate change gives no added branch work or capacitor store. The driver holding that voltage must be included in the larger circuit; without it the attached branch changes the voltage trajectory. Ground-bond impedance, probe common-mode paths, interwinding capacitance, and chassis capacitance are distinct branches with their own endpoints. Their physical attachment work is the return question [OP-TRF-01].

## 5.2 A regular winding model and its seven ports

Configuration IV has reciprocal inductance

$$\mathbf{L} = \begin{pmatrix} L_1 & M \\ M & \rho^2 L_1 \end{pmatrix}, \quad M = sk\rho L_1, \quad s = \pm 1, \quad 0 \leq k < 1, \quad E_m = \frac{1}{2} i^T \mathbf{L} i, \quad E_C = \frac{1}{2} C v_o^2. \quad (32)$$

Here  $L_1, C, \rho$  are positive. The determinant is  $\rho^2 L_1^2 (1 - k^2) > 0$ . An exact decomposition is  $E_m = kL_1(i_1 + s\rho i_2)^2/2 + (1-k)L_1 i_1^2/2 + (1-k)\rho^2 L_1 i_2^2/2$ . The distinction between finite magnetic storage and the ideal transformer is standard in [Haus and Melcher \(1989\), Section 9.7](#).

Let  $\eta = 1$  for tapped and 0 for isolated connections, whose returns remain physically distinct. A source  $u$  behind  $R_s > 0$ , a core shunt  $G_c$ , and output load/probe conductances  $G_L, G_P$  give, by Kirchhoff laws,

$$v_a = \frac{u - R_s(i_1 - \eta i_2)}{1 + R_s G_c}, \quad I_s = i_1 - \eta i_2 + G_c v_a, \quad (33)$$

$$\mathbf{L} \dot{i} = (v_a, v_o - \eta v_a)^T - \text{diag}(R_1, R_2) \dot{i}, \quad C \dot{v}_o = -i_2 - (G_L + G_P) v_o. \quad (34)$$

The boundary includes windings, output capacitor, and the source, copper, and core resistors. The source, receiver, probe, and immediate heat reservoirs are external. Its seven separate powers are

$$(uI_s, -R_s I_s^2, -R_1 i_1^2, -R_2 i_2^2, -G_c v_a^2, -G_L v_o^2, -G_P v_o^2). \quad (35)$$

Each negative resistor product is its voltage times outward current or, under immediate export,  $-T_b \dot{S}_{\text{out}}$ . Multiplication of (34) by  $i^T$  and  $v_o$  derives  $\dot{E}_m + \dot{E}_C = \sum P_j$ . This does not assume that either store is constant. Equation (2) gives each work from its own quadratic form, and endpoints use (32) on the propagated state.

Use  $L_1 = 1/25$  H,  $\rho = 1$ ,  $k = 19/20$ ,  $s = +1$ ,  $C = 1/500$  F,  $R_s = 1/5$   $\Omega$ ,  $R_1 = R_2 = 1/10$   $\Omega$ ,  $G_c = 1/100$  S. A finite crossed comparison retains all combinations

$$G_L \in \{1/10, 2/5\}, \quad G_P \in \{0, 1/10\}, \quad \omega \in \{0, 2\pi\}, \quad (36)$$

$$x(0) \in \{(0, 0, 0), (0, 0, 1)\}, \quad u(t) = \cos(\omega t) \text{ V}.$$

Conductances are in siemens and states in amperes, amperes, volts. On  $[0, 1]$  s this specifies 16 regular initial-value problems for each chosen connection. Preparation of the charged state is its own interval. For every choice, set  $x = (i_1, i_2, v_o)$  and construct  $A$  directly from (34), adding  $\dot{c} = -\omega s$ ,  $\dot{s} = \omega c$  with  $(c, s) = (1, 0)$ . All seven predictions are the closed forms  $W_j = \mathcal{T}(Q_j, A, y_0, 1)$ , and both endpoint stores are specified by (32); this is an exact finite family, not a numerical trajectory. Changing the load or source at a finite event keeps  $i, v_o$  continuous when a current path remains. Reinitialize the source coordinates on the correct event side and integrate the next interval separately.

A simpler probe branch displays an explicit nonzero contrast. A capacitor  $C = 1$  F initially at 1 V discharges with source disconnected through  $G_L = 1$  S. With no probe its receiver work at  $T = 1$  is  $-(1 - e^{-2})/2$  J and endpoint store  $e^{-2}/2$  J. With  $G_P = 1$  S they become  $-(1 - e^{-4})/4$  J and  $e^{-4}/2$  J; the probe has its own work  $-(1 - e^{-4})/4$  J. These follow by separately integrating  $-G_j v^2$ ,  $v = e^{-(G_L + G_P)t}$ . The winding comparison in (36) still requires its magnetic states and actual source.

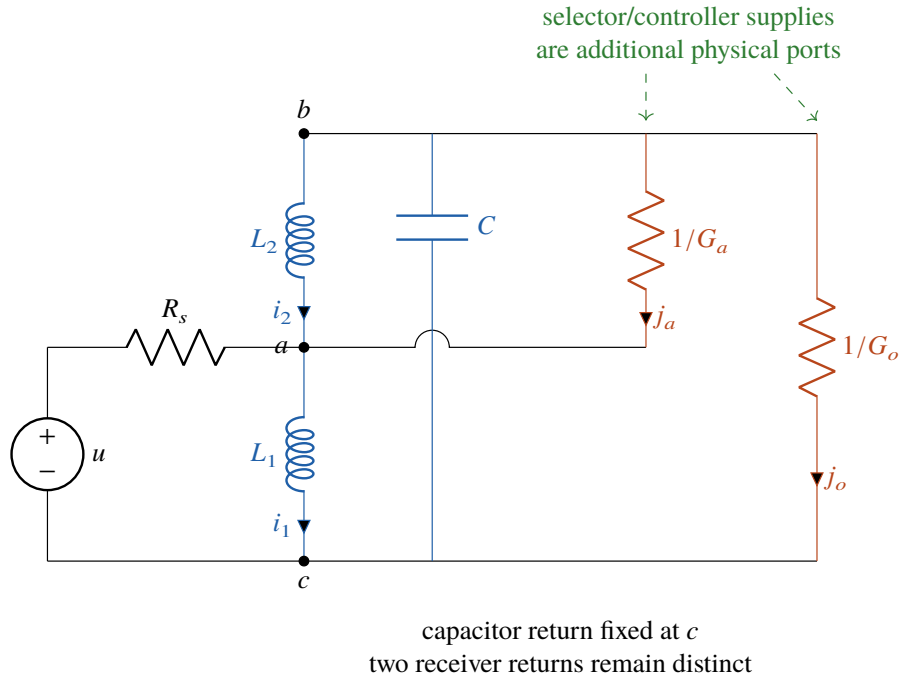
Moving the tapped receiver return requires another current law. Keep the capacitor across  $b-c$  and let  $h = 1$  return the receiver to  $a$ ,  $h = 0$  to  $c$ . With  $G = G_L + G_P$ ,  $v_L = v_b - h v_a$ ,

$$v_a = \frac{u - R_s(i_1 - i_2) + R_s h G v_b}{1 + R_s(G_c + hG)}, \quad I_s = i_1 - i_2 + G_c v_a - h G v_L, \quad C \dot{v}_b = -i_2 - G v_L. \quad (37)$$

The two winding equations use voltages  $(v_a, v_b - v_a)$ . Both receivers have their own separately prepared trajectories and  $-G_L v_L^2$  work. At  $u \neq 0, x = 0$ , return  $a$  has nonzero initial receiver power and return  $c$  zero. At  $u = 0, x = (0, 0, V)$  the return- $a$  voltage is  $V(1 + R_s G_c)/(1 + R_s(G_c + G))$ , smaller in magnitude than  $V$ . Continuity preserves each strict comparison on a finite following interval. There is therefore no preparation-independent ordering of the two works. Their numerical values for a specified duration remain the separate matrix primitives, not a ratio of voltage integrals. The crossed physical load, probe, drive and preparation comparison remains open [OP-TRF-06].

### 5.3 Opening, commutation, and separate constrained limits

Physical commutation can connect two receiver branches,  $G_a$  across  $b-a$  and  $G_o$  across  $b-c$ , during overlap. Replace the last equation of (37) by  $C \dot{v}_b = -i_2 - G_a(v_b - v_a) - G_o v_b$  and use  $v_a = [u - R_s(i_1 - i_2) + R_s G_a v_b]/[1 + R_s(G_c + G_a)]$ . The two receiver powers are separately  $-G_a(v_b - v_a)^2$  and  $-G_o v_b^2$ . A finite piecewise-constant selector, with duration 1/10 s in each of the states  $(G_a, G_o) = (1, 0), (1, 1), (0, 1)$  S, has exact matrix works on all three intervals. An open-gap alternative uses  $(0, 0)$  in its middle interval and needs its own propagated states. Binary  $h$  is not continuously interpolated to represent these connections.



**Figure 4.** Schematic tapped receiver branches during commutation. Winding copper and core-loss elements are suppressed in this connection drawing and retained in the equations. The bridge at the capacitor crossing is not a junction. The two inductors are magnetically coupled.

Figure 4 keeps the two receiver branches distinct. The switch/controller boundary adds  $v_{\text{ctrl}} i_{\text{ctrl}}$ , its own electric and thermal states, and any mechanical actuation. Actual winding opening also adds clamp, arc, and displacement-current paths; a source command of zero with  $R_s$  still connected is a different event. As a finite clamp control, an isolated  $L = 1$  H inductor prepared at 2 A drives a constant opposing clamp voltage 1 V over  $[0, 1]$  s. Its law  $\dot{i} = -1$  gives

$$W_{\text{clamp}} = - \int_0^1 (2 - t) dt = -3/2 \text{ J}, \quad E_L(0) = 2 \text{ J}, \quad E_L(1) = 1/2 \text{ J}. \quad (38)$$

The clamp receiver gets 3/2 J. Releasing at current 1 A into a declared parallel capacitor with zero initial voltage permits a subsequent LC interval; simply deleting that current is not a specified transfer. A switch arc with constant voltage 2 V on the same initial inductor instead reaches zero current in one second and receives 2 J. These are two different finite laws, each with its own  $vi$  integral and endpoint. Singular models must be written before their finite approaches:

**Table 6.** Configuration IV boundary models. The units of the explicit unit controls are SI. An algebraic constraint with incompatible preparation needs a named parasitic or supply path; it cannot receive an assigned impulse from this table.

| Exact condition    | Constrained law and prepared-state issue                                                      | Exact finite contrast                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| $k = 1$            | $q = i_1 + s\rho i_2, L_1 \dot{q} = v_1 - R_1 i_1;$<br>$v_2 - R_2 i_2 = s\rho(v_1 - R_1 i_1)$ | $L_1 = \rho = 1$ , zero copper and<br>$v_1 = v_2 = 1$ give $q = t$ ,<br>$E_m(1) = 1/2$ J; a zero-store<br>ideal transformer gives 0 |
| $C = 0$            | $i_2 + Gv_o = 0$ ; no independent capacitor<br>state                                          | $C = \epsilon, v_o = 1$ gives $E_C = \epsilon/2$ ;<br>$v_o = \epsilon^{-1/2}$ gives $1/2$ J                                         |
| $R_s = 0$          | $v_a = u$ ; source current follows KCL                                                        | A capacitor ramp $v = t$ has<br>$W_s = C/2 + R_s C^2$ and heat<br>$-R_s C^2$ J on one second                                        |
| Exact output short | $v_o = 0$ ; nonzero prepared voltage is<br>incompatible without an edge law                   | A resistor across $C = 1, V = 1$<br>for $R \ln 2$ seconds exports $3/8$<br>J and leaves $1/8$ J                                     |
| $\rho = 0$         | $L_2 = M = 0$ ; winding 2 becomes<br>$v_2 = R_2 i_2$ , or $v_2 = 0$ if also $R_2 = 0$         | With $L_1 = 1, i_1 = 0, i_2 = 1/\rho$<br>the finite pair retains<br>$E_m = 1/2$ J                                                   |

For the perfect-coupling ramp, imposing  $i_2 = 0$  gives  $i_1 = t$ ; the separately integrated primary input is  $1/2$  J and secondary input zero. Thus the example includes its effort allocation as well as  $q$ . For  $C = 0$  or zero turns ratio, bounded-state approaches and scaled prepared states have different stores. These examples give no universal joint limit.

Opening an energized physical winding requires its switch, arc and clamp laws [OP-TRF-02]. Perfect coupling, vanishing capacitance and other constrained limits are separate models with separate admissible preparations [OP-TRF-04].

#### 5.4 Ideal terminal matching and finite independent stores

Two ideal transformers with common secondary nodes  $P, C$  satisfy  $V_X - V_C = h_X(V_P - V_C)$  and  $j_X = -h_X i_X$  for  $X = A, B$ . KCL gives  $j_A + j_B = 0$  and  $i_C = -(i_A + i_B)$ , hence  $(i_A, i_B, i_C) = i_A(1, -R, R - 1)$ . With a dimensional scale  $\alpha$ , the rotational mobility mapping is

$$V = \alpha\omega, \quad I = \tau/\alpha, \quad VI = \tau\omega, \quad C = J/\alpha^2. \quad (39)$$

No factor  $2\pi$  is omitted if rates were originally in revolutions per second. The nine mechanical powers equal signed terminal products  $(V_X - \alpha\Omega)i_X$ ; complete winding powers instead use  $V_X - V_C$ . Consequently the maximum complete-winding ratio is  $\mathcal{C}_C$  and that of the three external plus four winding ports is  $\max(1, \mathcal{C}_C)$ . For  $R = -341/100$  with  $B$  held and observation fixed to  $A$ , the mapped ratio is  $341/100$  but the complete-port ratio is one. Their exact difference is  $-241/100$ . For  $R = 2929/2950$  with  $A$  held, the winding and carrier ratios are  $2950/21$  while the ground terminal ratio is one. These compare explicitly different observables; each corresponding signed work remains identical under the ideal map. Constant voltage on an ideal transformer is not a claim about indefinitely sustainable excitation of a finite core.

A dynamic circuit needs node capacitors and elastic counterparts to magnetic states. For incidence matrix  $B$ , positive diagonal  $C$ , positive definite  $L$ , and nonnegative  $R$ , independently specify the two models

$$C\dot{v} = I_{\text{ext}} - Bi, \quad Li = B^T v - Ri, \quad (40)$$

$$J\dot{\omega} = \tau_{\text{ext}} - Bf, \quad \dot{z} = B^T \omega - Mf, \quad f = Kz, \quad (41)$$

$$J = \alpha^2 C, \quad K = \alpha^2 L^{-1}, \quad M = R/\alpha^2. \quad (42)$$

Substitution of  $v = \alpha\omega$ ,  $i = f/\alpha$ ,  $z = Li/\alpha$  proves a power-preserving correspondence for compatible initial states. Capacitor energy maps to inertia, magnetic energy to elastic energy  $z^T Kz/2$ , and resistor loss to  $f^T Mf$ . Each source, receiver, and loss is independently integrated with (2); the stores are independently evaluated. For a compound pair use  $L_X = L_0 \begin{pmatrix} h_X^2 & kh_X \\ kh_X & 1 \end{pmatrix}$  and incidence columns  $e_A - e_C, e_P - e_C, e_B - e_C, e_P - e_C$ . A rational finite control has  $L_0 = 1$ ,  $k = 1/2$ , every retained node capacitance one, each resistance one, zero initial state, a unit current source at  $A$ , and a unit conductance receiver at  $B$  over one second. Together with any specified geometry in configuration III these coefficients fully determine every closed matrix work. A rigid gear and a three-terminal winding pair do not acquire these extra degrees of freedom by a voltage rename; the dynamic correspondence is to the declared elastic model.

A physical accelerating electrical reference needs supplied compensation. For a capacitor between  $A$  and  $R$ , write  $w = V_A - V_R$ ,  $C = J/\alpha^2$ ,  $V_R = \alpha\Omega$ , and

$$C\dot{w} = I_d + I_e, \quad I_d = \tau/\alpha, \quad I_e = -C\dot{V}_R, \quad (P_d, P_e) = (wI_d, wI_e). \quad (43)$$

This realizes  $w = \alpha(\omega - \Omega)$  only with compatible initial data and its two floating-source supply ports. For  $C = J = \alpha = 1$ ,  $\tau = 0$ ,  $\Omega = 2t$ , zero initial states,  $w = -2t$ ,  $I_e = -2$ :  $W_d = 0$ ,  $W_e = 2$  J,  $E_C(0) = 0$ ,  $E_C(1) = 2$  J. Disabling compensation gives  $W_e = 0$ ,  $w = 0$ , and  $E_C(1) = 0$ . A common coordinate shift on the unchanged physical capacitors also leaves their physical energy unchanged. It therefore misses the independent relative-kinetic quantity  $-\Omega H + J\Omega^2/2$ .

A separate reference driver  $C_d = 1/200$  F,  $R_d = 3/4 \Omega$  has  $I_r = C_d \dot{V}_R = 1/100$  A and command  $u_r = 2t + 3/400$  V. Its independently integrated works are  $W_r = 403/40000$  J and heat  $-3/40000$  J, with stores 0,  $1/100$  J. Floating-source return currents cancel at  $R$  for this cell; their nonzero powers still require supplies.

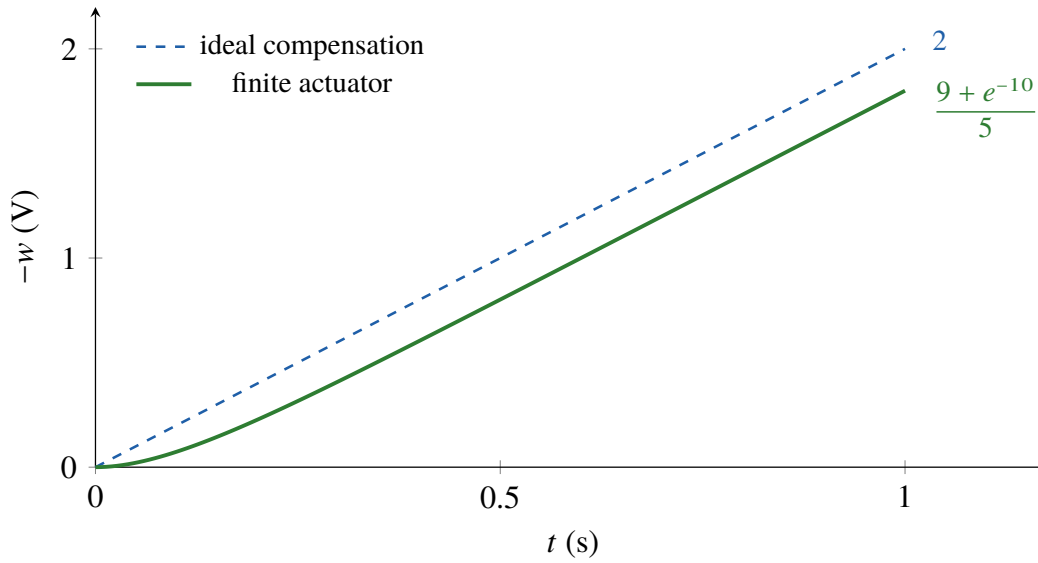
For a finite actuator with current response time  $\tau_a > 0$ , prescribe  $\tau_a \dot{j} + j = -Ca$ ,  $C\dot{w} = j$ , and actual actuator effort  $e = w + R_a j + L_a \dot{j}$ , with zero initial states and  $V_R = at$ . The exact solution and separate works on  $[0, T]$  are

$$j = -Ca(1 - e^{-t/\tau_a}), \quad w = -a[t - \tau_a(1 - e^{-t/\tau_a})], \quad (44)$$

$$W_C = \frac{1}{2}Ca^2[T - \tau_a(1 - b)]^2, \quad Q_a = R_a C^2 a^2 [T - 2\tau_a(1 - b) + \frac{1}{2}\tau_a(1 - b^2)], \quad (45)$$

$$W_{a,s} = W_C + Q_a + \frac{1}{2}L_a C^2 a^2 (1 - b)^2, \quad b = e^{-T/\tau_a}. \quad (46)$$

The three terms in  $W_{a,s} = \int ejdt$  are obtained by integrating  $wj$ ,  $R_a j^2$ , and  $L_a \dot{j}j$  separately. Actuator output work is  $-W_C$ , heat work  $-Q_a$ , and its final independent store is  $L_a j(T)^2/2$ . For  $C = 1$ ,  $a = 2$ ,  $T = 1$ ,  $\tau_a = 1/10$ ,  $R_a = L_a = 1$  in SI units,  $E_C(1) = (9 + e^{-10})^2/50$  J. Its difference from 2 J is strictly negative because  $0 < \tau_a(1 - e^{-T/\tau_a}) < T$ . Separate actuator and driver balances have zero exact residual even when their state correspondence fails.



**Figure 5.** Exact reference-cell response from equation (46), with  $C = 1$  F,  $a = 2$  V/s, and  $\tau_a = 1/10$  s. The plotted quantities are the stated closed forms, with their exact one-second endpoints.

Figure 5 shows the exact finite tracking departure. Finite sensors, converter losses, command limits, and a finite rail need additional constitutive states; their calibration is not supplied by (42). Existing bounded smooth-model correspondence and event results do not establish unrestricted hardware duality. A nonzero reference jump  $\Delta V$  imposed in time  $\delta$  through fixed  $R_d, C_d > 0$  has

$$Q_d = R_d C_d^2 \int_0^\delta \dot{V}_R^2 dt \geq \frac{R_d C_d^2 (\Delta V)^2}{\delta}. \quad (47)$$

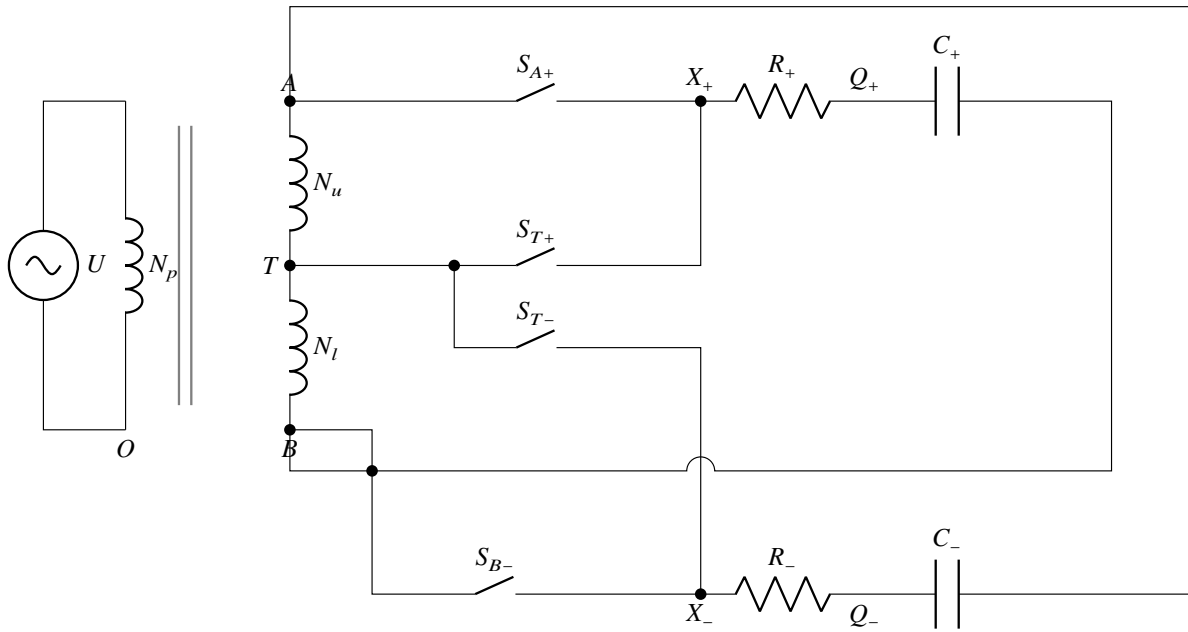
Cauchy–Schwarz proves the bound, with equality for a linear ramp. It diverges as  $\delta \rightarrow 0$  for this driver; finite slow comparisons do not realize that event. With  $L_0 = \epsilon^{-1}$  and  $k = 1 - \epsilon^2$ , the pair’s common and differential magnetic eigenvalues scale differently. A prepared common current proportional to  $\sqrt{\epsilon}$  retains finite energy while the current vanishes. For example  $(hi_1, i_2) = (\sqrt{\epsilon}/2, \sqrt{\epsilon}/2)$  gives  $E_m = (2 - \epsilon^2)/4$  J, tending to  $1/2$  J. Zero common preparation gives zero instead. Thus convergence of selected port works or small differential voltage does not determine convergence of all winding states and stores. Fixed-effort clamps and unchanged passive source/load circuits are different limiting initial-value problems.

The ideal terminal map leaves physical inertial, magnetic and controller states to be identified before claiming a loaded dynamic correspondence [OP-TRF-09]. Complete physical-port ratios also require the actual return and supply ports [OP-TRF-11]. The finite-reference actuator’s separately predicted departure is a bounded model result; its physical supply and tracking accuracy remain measurable [OP-TRF-12].

## 6 A tapped secondary with two switched receiving branches

Murray’s SERPS Figure 11 supplies a concrete switched-return architecture: an isolated primary, a secondary tap, and two resistor–capacitor branches selected on opposite waveform polarities Murray (2017). The following finite component values are declared model choices. Figure 6 names every persistent winding section and branch. The primary return  $O$  is electrically separate from secondary reference  $B$ . A selector changes an attachment; it does not discard an energized winding section.

**Isolated primary, persistent secondary sections, two switched RC branches**



Cell orientations:  $v_+ = V_{Q_+} - V_B$ ,  $v_- = V_{Q_-} - V_A$ .

Selector edges, copper, leakage, parasitics, clamps and gate supplies follow the finite equations.

**Figure 6.** Schematic of the isolated primary and two-section secondary with four selectors. The positive RC branch ends at B; the negative branch ends at A. Each switch is a finite conductance in the model, with separately supplied gate state. Winding polarities, leakage and mutual coupling are specified by the inductance matrix. The dashed magnetic connection is not a conductor between primary and secondary.

The four switches connect  $A - X_+$ ,  $T - X_+$ ,  $T - X_-$  and  $B - X_-$ . Receiver resistors join  $X_+$  to  $Q_+$  and  $X_-$  to  $Q_-$ . The positive cell is  $Q_+ - B$  and the negative cell is  $Q_- - A$ . For a one-second sinusoidal drive, the nominal commands are:

**Table 7.** The connected state and capacitor voltage determine actual current direction; the clock alone does not establish it.

| Quarter                          | Selected connection | Intended branch operation                                  |
|----------------------------------|---------------------|------------------------------------------------------------|
| Positive rising                  | $A - X_+$           | Charge through full secondary                              |
| Positive falling                 | $T - X_+$           | Return through lower section                               |
| Negative increasing<br>magnitude | $B - X_-$           | Charge through full secondary with<br>opposite orientation |
| Negative falling<br>magnitude    | $T - X_-$           | Return through upper section                               |

For an ideal positive charging interval,  $v_s = Ri + v_C$  and  $C\dot{v}_C = i$ . On return,  $v_C = Rj + fv_s$  and  $C\dot{v}_C = -j$ , where  $j$  enters the winding and  $f$  is the tap fraction. The condition  $fv_s < v_C < v_s$  permits both directions under the respective connections. For  $R = C = 1$  in SI units, a constant connected voltage  $v_s$  gives  $v_C = v_s + (v_0 - v_s)e^{-t}$  on  $[0, T]$ . With  $a_0 = v_0 - v_s$ , separate integration gives

$$W_s = -v_s a_0 (1 - e^{-T}), \quad Q_R = \frac{a_0^2}{2} (1 - e^{-2T}), \quad (48)$$

$$\Delta E_C = \frac{1}{2} \{ [v_s + a_0 e^{-T}]^2 - v_0^2 \}. \quad (49)$$

Units are joules when the displayed constants use volts, ohms, farads and seconds. Substitution verifies  $W_s - Q_R = \Delta E_C$ . The negative branch has its own orientation and prepared voltage. Its full and tapped connected voltages are  $-v_s$  and  $-(1-f)v_s$ ; the positive branch uses  $v_s$  and  $fv_s$ . For a constant selected voltage  $v_\alpha$ , actual source return requires  $v_\alpha (v_\alpha - v_C) < 0$ . Selecting a tap does not enforce that inequality. The same exponential solution applies separately to either sign and either fraction. For  $v_\alpha = 0$ , a prepared cell still heats the resistor while source work is zero; for  $v_0 = v_\alpha$ , all currents and works vanish.  $v_0 = v_s$  gives zero current; zero preparation can instead give nonzero charging current. Finite leakage prevents assigning an instantaneous winding current reversal to a physical tap event.

### 6.1 Persistent states and finite commutation

Let  $v = (V_A, V_T, V_{X+}, V_{Q+}, V_{X-}, V_{Q-})$  relative to  $B$ . Use three winding currents in primary, upper-secondary and lower-secondary order, and incidence columns  $B_w = (0, e_A - e_T, e_T)$ . With turns vector  $N = (1, \sigma(1-f), \sigma f)$ , define

$$L = (1 - \kappa) \text{diag}(N_f^2) + \kappa N N^T, \quad 0 < f < 1, \quad 0 \leq \kappa < 1, \quad (50)$$

with unit henry scale. The tap fraction  $f$  and magnetic coupling  $\kappa$  are independent. Positive leakage makes  $L$  positive definite. For  $f = \kappa = 1/2$ ,  $\sigma = 1$ ,  $L = \begin{pmatrix} 1 & 1/4 & 1/4 \\ 1/4 & 1/4 & 1/8 \\ 1/4 & 1/8 & 1/4 \end{pmatrix}$  H. Each receiving resistor is  $1 \Omega$ , each cell  $1/10$  F, and each node has parasitic  $1/50$  F to  $B$ :

$$C_n = \frac{1}{50} I + \frac{1}{10} e_{Q+} e_{Q+}^T + \frac{1}{10} (e_{Q-} - e_A)(e_{Q-} - e_A)^T. \quad (51)$$

All selector paths have  $R_{\text{on}} = 1/20 \, \Omega$  and  $R_{\text{off}} = 10000 \, \Omega$ . With source resistance  $1/5 \, \Omega$  and each winding resistance  $1/20 \, \Omega$ , the independent circuit equations are

$$L\dot{i} = B_w^T v + e_p U - R_t i, \quad C_n \dot{v} = -B_w i - G(t)v - i_{\text{cl}}, \quad R_t = \text{diag}(1/4, 1/20, 1/20). \quad (52)$$

Every resistor or switch with incidence  $a_j$  contributes  $g_j a_j a_j^T$  to  $G$ . Clamps across  $A - B$ ,  $T - B$ , and each cell use  $i_{\text{cl},j} = 5 \, \text{sgn}(w_j) \max(|w_j| - 2, 0)$  in amperes. Each gate is a  $1/10000 \, \text{F}$  capacitor supplied through  $20 \, \Omega$ . Prescribed selector commands and the gate circuit are separate model laws; actual semiconductor coupling and actuator cost need identification.

For finite edges use  $s(\xi) = 3\xi^2 - 2\xi^3$  on  $0 \leq \xi \leq 1$  to interpolate conductance between its off and on values. Specify opening before closing for a gap, or closing before opening for overlap. Each path has its own width and edge endpoints. One initialized commutation uses  $i = (1/10, 1/5, -1/10) \, \text{A}$  and  $V_{Q+} = V_{Q-} = 3 \, \text{V}$ , with other nodes zero; the clamp law therefore participates. The separate zero-state history drives two one-second periods with  $U = \sin(2\pi t) \, \text{V}$ , then sets source and gates to zero and connects  $1/5 \, \text{S}$  node bleeders for one second. Its actual final states remain. Finite positive  $L, C_n$  give continuous states and zero impulse at these events, even when a commanded coefficient or power jumps.

The reactive store is  $(i^T L i + v^T C_n v)/2$  plus all gate stores. For a boundary including thermal matter  $H$ , source  $U i_p$  and gate-supply products enter; receiver powers  $-(V_{X+} - V_{Q+})^2$  and  $-(V_{X-} - V_{Q-})^2$  and bath  $-H/2$  leave. Copper, source resistor, selector, clamp and gate-resistor products enter  $\dot{H}$  as internal conversion. An electrical-only boundary instead exports those losses and excludes  $H$ . Both endpoints are evaluated independently under the chosen convention.

On a fixed linear interval the exact primitive (2) computes each work independently. A smooth edge uses its fundamental matrix  $\Phi(t, t_0)$ , defined by  $\dot{\Phi} = A(t)\Phi$ ,  $\Phi(t_0, t_0) = I$ , with the forcing convolution. The separate integral  $\int y_0^T \Phi^T Q_j(t) \Phi y_0 \, dt$  defines that edge's work. Where a clamp changes branch, its guard and event sides must also be retained. This conditional exact formulation preserves the finite-event question without assigning a numerical trajectory or a unique zero-width limit. Removing leakage or parasitics imposes additional constraints and can exclude the chosen initial state.

## 6.2 A parallel/series bank is another connection graph

The patent's Figure 19 specifies a parallel/series network block without resolving its individual cell selectors Murray (2017). A declared finite realization uses cells  $X_1 - Y_1$  and  $X_2 - Y_2$ , charge selector  $A - X_1$ , injection selector  $T - X_1$ , parallel links  $X_1 - X_2$  and  $Y_1 - Y_2$ , and series link  $Y_1 - X_2$ . Its  $1 \, \Omega$  receiver joins  $Y_2 - B$ . This receiver attachment differs from the later hidden-state experiment's  $X_1 - Y_2$  attachment.

Take  $C_1 = 1/10 \, \text{F}$ ,  $C_2 = 1/10$  or  $3/20 \, \text{F}$ , with the same node parasitics, finite selector resistances, windings, clamps and gate laws as above. Command charge and both parallel links during the first and third quarters, injection and series during the second and fourth. The series-to-parallel return at half a second retains the actual cell voltages. Zero preparation and a separately initialized unequal-cell case with  $(v_1, v_2) = (1, 1/2) \, \text{V}$  require their own source, switch, receiver and endpoint expressions. Those expressions follow from the new incidence matrices and finite-edge maps; the initialized case supplies no preparation work by itself.

An isolated pair gives a useful exact control. For  $(C_1, C_2) = (1, 2) \, \text{F}$ , initial  $(v_1, v_2) = (1, 0) \, \text{V}$  and exchange resistance  $1 \, \Omega$ ,  $v_1 = 1/3 + 2e^{-3t/2}/3$  and  $v_2 = 1/3 - e^{-3t/2}/3$ . On  $[0, 1] \, \text{s}$ , the independent heat integral is  $(1 - e^{-3})/3 \, \text{J}$  and the capacitor endpoints are  $1/2$  and  $1/6 + e^{-3}/3 \, \text{J}$ . Positive resistance gives continuous voltages without an impulse. Its isolated single-path zero-resistance heat limit is unique; several competing paths need their own partition laws. This result does not determine the transformer-

connected bank's work.

### 6.3 Clocked cycles, magnetic limits, and physical service

For the linear fixed-clock graph, compose interval maps to obtain  $x(T) = Mx(0) + b$ . A periodic state satisfies  $(I - M)x_* = b$ . Existence, uniqueness and attraction are different tests: respectively  $b \in \text{range}(I - M)$ , invertibility of  $I - M$ , and  $\rho(M) < 1$ . With  $M = I$ ,  $b = 0$  every state is fixed;  $M = I$ ,  $b \neq 0$  has none;  $M = 2I$  has a unique unstable fixed state. The remaining physical question is whether a declared clocked realization stays within magnetic and controller limits [OP-TRF-13].

**Theorem 2** (Clocked contraction). *For the above network with inactive clamps, fixed positive-definite  $L, C_n$ , strictly positive winding resistance and uniformly grounded positive conductance, each periodic forcing and clock has one attracting periodic electrical state.*

*Proof.* The difference of two forced solutions obeys the homogeneous laws. Multiplication by its own currents and voltages gives  $\dot{E} = -i^T R_t i - v^T G(t) v$ . The declared positive resistances and connected off paths imply  $\text{diag}(R_t, G(t)) \geq \eta \text{diag}(L, C_n)$  for some  $\eta > 0$  on the compact clock period. Hence  $\dot{E} \leq -2\eta E$  and the period map contracts its energy norm by at most  $e^{-\eta T} < 1$ . The affine map has one attracting fixed point. The same estimate covers continuous finite edges.  $\square$

Gate RC states have their own affine fixed points. A thermal law  $\dot{H} = D(t) - H/2$  has the period relation  $H(T) = e^{-T/2}H(0) + \int_0^T e^{-(T-t)/2}D(t) dt$ . An insulated thermal state with positive conversion does not repeat merely because voltages do. Repeated flux endpoints also leave continuous peak flux and material admissibility to be checked.

A state-triggered controller needs its own map. One example switches at  $V_{Q+} = 1/100$  V upward and  $1/200$  V downward. For a guard with normal  $n$ , continuous state and vector fields  $f^-, f^+$ , its transverse-event variation contains  $S = I + (f^+ - f^-)n^T / (n^T f^-)$ . This formula requires  $n^T f^- \neq 0$ ; grazing, simultaneous guards and changed event order require separate treatment. A clocked contraction theorem does not supply full-wave hybrid attraction.

Two material extensions illustrate the needed constitutive information. With positive leakage  $D_L$  and inductance scale  $L_m = 1$  H, let  $\lambda = D_L i + L_m \kappa I_s \tanh(N^T i / I_s) N$ . Integrating  $i^T d\lambda$  gives

$$E_m = \frac{1}{2} i^T D_L i + L_m \kappa \left[ I_s q \tanh(q / I_s) - I_s^2 \log \cosh(q / I_s) \right], \quad q = N^T i. \quad (53)$$

Here  $q$  is a current coordinate. At  $I_s = 1/2$  A, a 4 V sinusoidal drive defines a finite amplitude comparison against the fresh linear model. This reversible saturating law has no remanence or hysteresis state. A separate temperature law  $R_w = (1/20)(1 + H/(100 \text{ J})) \Omega$ , for the stated thermal scaling, changes the trajectory through heating. Neither extension inherits the constant-matrix theorem without a new proof. Compare complete service only after fixing its definition: for example total receiver heat on  $[0, 3]$  s, both zero-state preparations, source  $\sin(2\pi t)$  on  $[0, 2]$  and zero afterward, and the actual finite bleeder reset. A full-secondary baseline retains both secondary sections and selects its load on a declared descending branch of the service equation before cost comparison. If several branches exist, their selection is part of the contract. Retain each endpoint, even if the selected service is equal. Gross source draw and return, net source work, gate work and useful delivery are separate integrals. A source model supplies one finite ordering; an exact general efficiency ordering is not established by the formulation here. Matched reset states or their explicit costs are additional requirements for a closed-cycle comparison.

#### 6.4 The generator boundary

Enclosing the generator makes capacitor-to-generator electrical return internal. Prime-mover shaft work, field excitation and controller supplies remain external, with rotor, magnetic, field and thermal stores at both ends. The physical prime-mover comparison requires the actual flux/coenergy law and independently established motion and excitation [OP-TRF-14].

A bounded ideal converter control shows the signs. Choose  $\omega = 2\pi$  rad/s,  $e = \sin(2\pi t)$  V and  $i = \pm \sin(2\pi t) + \frac{1}{2} \cos(2\pi t)$  A on  $[0, 2]$  s, with converter torque  $ei/\omega$ . Let rotor inertia be  $1/100$  kg m<sup>2</sup>, drag  $D = 1/1000$  N m s, and a separate unit-inductance unit-resistance field carry 1 A under 1 V. Separate products give electrical delivery  $\pm 1$  J, shaft input  $\pm 1 + \pi^2/125$  J, field input 2 J, and drag/field heat exports  $-\pi^2/125, -2$  J. Rotor and field stores remain  $\pi^2/50$  and  $1/2$  J at both independently evaluated ends. The imposed reciprocal converter and decoupled field are explicit assumptions. Their positive or negative shaft input does not determine a physical machine's complete-cycle ordering.

Finite disconnected preparation is also explicit. A unit  $RL$  field driven by constant voltage  $U$  from zero has  $i = U(1 - e^{-t})$ ,  $W_s = U^2[T - 1 + e^{-T}]$ , resistor heat  $U^2[T - 2(1 - e^{-T}) + (1 - e^{-2T})/2]$ , and store  $U^2(1 - e^{-T})^2/2$ . Setting its source to zero through the resistor gives  $i = i_0 e^{-t}$ , heat  $i_0^2(1 - e^{-2T})/2$  and a nonzero finite endpoint. A rotor accelerated by constant torque  $J\omega_*/T$  with zero drag receives  $J\omega_*^2/2$  by its own torque–rate integral. These separate paths clarify what remains to be specified when actual field and armature dynamics are coupled.

#### 6.5 State and observation correspondence

For the fixed linear charging topology, the selected outputs  $(i_p, V_A, V_T, V_{Q+}, V_{Q-} - V_A)$  and known component laws give an observability matrix  $[H; HA; \dots; HA^8]$  of rank nine for the nine electrical states. This can be seen directly with inactive clamps. The measured last output and  $V_A$  give  $V_{Q-}$ . The two cell-node laws give

$$V_{X+} = V_{Q+} + \frac{3}{25} \dot{V}_{Q+}, \quad V_{X-} = V_{Q-} + \frac{3}{25} \dot{V}_{Q-} - \frac{1}{10} \dot{V}_A, \quad (54)$$

with the coefficients in seconds for the stated resistor and capacitance values. All six node voltages are now known. Put  $r = C_n \dot{v} + Gv$ ; the  $A, T$  node laws then give  $i_u = -r_A$  and  $i_l = i_u - r_T$ . These two components need only the already observed node derivatives. Together with measured  $i_p$ , they recover all nine states, proving the rank statement. This is structural reconstruction from ideal traces. Temperature-independent elements leave  $H_{\text{thermal}}(0) = 0$  or 1 J electrically indistinguishable while their thermal difference at time  $T$  is  $e^{-T/2}$  J. A fixed-topology rank test gives neither a noise bound nor a switched observation certificate. Individual cell and material observations still belong to the full endpoint account. Even the nonnegative cell pairs (1, 1) and (0, 2) V share a 2 V series terminal with unit-cell stores of 1 and 2 J.

The analogous mechanical ambiguity is already present in unit-inertia states  $(\omega_c, \omega_p) = (0, 1)$  and (10, 11) rad/s: their relative rate is one, but their kinetic stores are  $1/2$  and  $221/2$  J. Absolute unwrapped motion and signed effort must be observed together. An encoder count is  $2\pi/N$  radians for  $N$  counts per turn; pulse totals without reversal direction lose signed work. For a 1 F rail at 1000 V, a positive  $1/1000$  V endpoint error changes the inferred energy by  $2000001/2000000$  J, from the exact expression  $Cv\delta v + C(\delta v)^2/2$ . Small relative voltage error therefore does not guarantee a small absolute energy error. A full low-energy rig target of  $1/100$  J needs independently bounded channels, endpoint covariance, event timing and bandwidth; it is not established by naming the instruments.

A finite mechanical counterpart can be constructed with  $J = \alpha^2 C_n$ ,  $z = \lambda/\alpha$ ,  $K = \alpha^2 L^{-1}$  and  $i = Kz/\alpha$ . Six rotors supply node inertias; two differential flywheels supply the capacitor terms between

nodes. Three positive modal springs realize  $K$ . At  $\alpha = 1$ ,  $K = \begin{pmatrix} 3/2 & -1 & -1 \\ -1 & 6 & -2 \\ -1 & -2 & 6 \end{pmatrix}$  and  $LK = I$ . Its independent equations are  $J\dot{\omega} = -B_w Kz - G\omega - \tau_{cl}$  and  $\dot{z} = B_w^T \omega - R_t Kz + b_{\text{anchor}}$ . An anchored drive supplies the primary forcing; every dashpot, gate rotor and thermal port must map as well. Substitution proves state, power and constitutive-store correspondence for the declared laws. The rigid planetary train has two independent speeds, whereas this circuit has six independent capacitive node coordinates. A bijection with that unchanged rigid train is impossible. The added rotors, differentials, springs and actuators are distinct physical hardware.

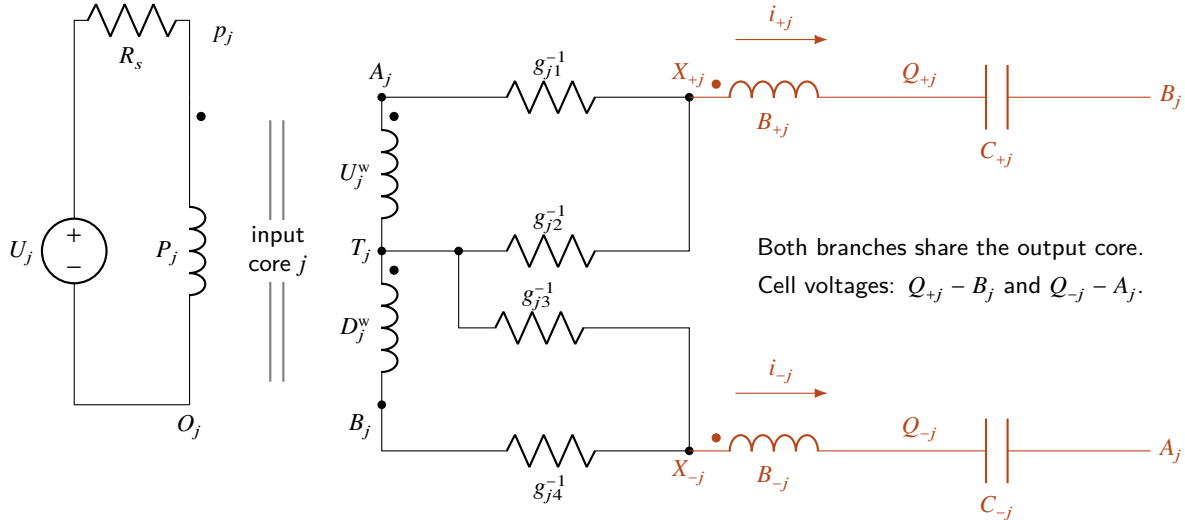
## 7 A shared output, supplied timing and finite observation

Multiple switched branches can deliver to one magnetic receiver, but their currents must solve the newly connected circuit. The following model keeps six isolated input cores and a seventh output core. Each input channel has two polarity branches; all twelve branch windings and one output winding share the output core. This is a declared extension of the switched-return concept in Murray's patent [Murray \(2017\)](#). Its six-channel topology, finite values and supplied controls are specified here. The patent's output combination examples do not identify this graph or its material laws.

### 7.1 The complete finite electrical fixture

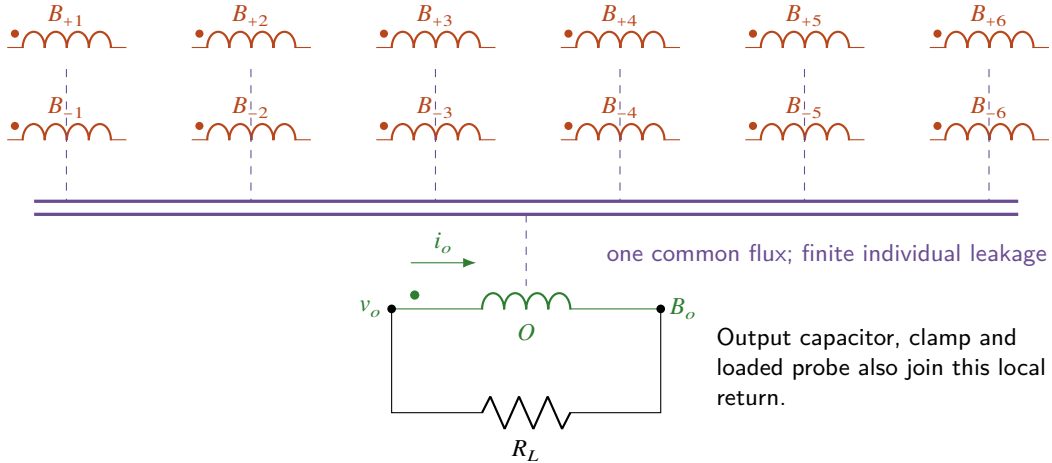
For  $j = 1, \dots, 6$ , the primary reference  $O_j$ , secondary reference  $B_j$ , four gate references  $D_{jk}$  and receiver reference  $B_o$  are physically separate. A reference elimination in the equations does not bond them. There is no postulated chassis or interwinding capacitance. Those additional physical paths would change the model. Figure 7 defines the main connections; the following tables include every remaining electrical path.

**A. One channel,  $j = 1, \dots, 6$ : isolated input and two persistent branches**



Repeated  $A_j, B_j$  labels denote the same channel conductors. Returns with different names are unconnected. Finite selectors are shown as resistive paths.

**B. Twelve separate branch windings, one shared output core, one receiver**



Panel B repeats the branch windings of panel A; it adds no conductors between channels. Dashed links show magnetic coupling. The tables give copper, gates, parasitics, clamps and all seven probes.

**Figure 7.** Schematic of one indexed channel and the shared output. Instantiate the channel six times with separate primary and secondary returns. Both branch windings in every channel couple to the single output core; dashed magnetic connections are not conductors.  $U_j^w$  and  $D_j^w$  label the upper and lower secondary windings. Dots and arrows specify persistent winding orientations. The component tables specify every finite selector, gate, clamp, parasitic, bleeder and loaded probe omitted from the main paths.

Let  $p_j$  be primary-node voltage relative to  $O_j$  and  $(A_j, T_j, X_{+j}, Q_{+j}, X_{-j}, Q_{-j})$  the six secondary voltages relative to  $B_j$ . Winding currents are positive along the following ordered terminal pairs. The negative cell keeps voltage  $Q_{-j} - A_j$  throughout; its physical orientation is not changed by a selector.

**Table 8.** One channel's complete paths, repeated with separate returns.

| Element                           | Ordered terminals or connection                                    | Declared value                        |
|-----------------------------------|--------------------------------------------------------------------|---------------------------------------|
| Primary, upper and lower windings | $p_j - O_j, A_j - T_j, T_j - B_j$                                  | Copper 1 $\Omega$ each                |
| Two output-core branch windings   | $X_{+j} - Q_{+j}, X_{-j} - Q_{-j}$                                 | Copper 1 $\Omega$ each                |
| Four selectors                    | $A_j - X_{+j}, T_j - X_{+j}, T_j - X_{-j}, B_j - X_{-j}$           | Conductance $g_{jk}$ below            |
| Positive and negative cells       | $Q_{+j} - B_j, Q_{-j} - A_j$                                       | 1 F each                              |
| Secondary parasitics and bleeders | Each of the six secondary nodes to $B_j$                           | 1 F and 1/10 S each                   |
| Source and its resistor           | $U_j$ relative to $O_j$ , through $R_s$ to $p_j$                   | $U_j = 1$ V, $R_s = 1$ $\Omega$       |
| Primary-node capacitor            | $p_j - O_j$                                                        | 1 F                                   |
| Four secondary clamps             | Across $A_j - B_j, T_j - B_j$ and each cell                        | Threshold 5 V, slope 1 S              |
| Each independent gate             | Command $c_{jk}$ through $R_g$ to $z_{jk}$ , capacitor to $D_{jk}$ | $R_g = 1$ $\Omega$ , $C_g = 1/1000$ F |

The output winding is oriented  $v_o - B_o$ , with current  $i_o$  into its positive terminal and copper 1  $\Omega$ . A 1  $\Omega$  receiver, 1 F capacitor and the same clamp law each join  $v_o$  to  $B_o$ . Seven physical probes load  $p_1, \dots, p_6, v_o$ : a 10  $\Omega$  resistor joins each node to a probe node  $y_j$  or  $y_o$ , with 1 F from that probe node to the corresponding local return. There is no probe supply. Each probe has its own resistor heat and capacitor store.

Order currents as six primary/upper/lower triples, then the twelve branch currents, then  $i_o$ . Let  $n = (1, 1/2, 1/2)^T$  and  $s = \mathbf{1}_{13}$ . The inductance blocks and joint magnetic store are

$$L_j = (I_3 + nn^T/4) \text{ H}, \quad L_o = (I_{13} + ss^T/4) \text{ H}, \quad L = \text{diag}(L_1, \dots, L_6, L_o), \quad E_m = \frac{1}{2} i^T L i. \quad (55)$$

Each core has positive unit leakage and positive common inverse reluctance; the fractional entries describe effective turn ratios. This construction proves positivity within a lumped model, not a manufactured geometry or material rating. The input tap fraction 1/2 is separate from its coupling. There are 31 persistent winding currents.

Order the 50 voltage coordinates as the six secondary sextuples, six primary nodes,  $v_o$ , six source probes and the output probe. Let  $e_a$  select a named node and let  $a_e = e_a - e_b$  select an oriented terminal pair, with the eliminated local return represented by zero. Each physical capacitor contributes  $C_e a_e a_e^T$ . Thus

$$C = \left[ I_{50} + \sum_{j=1}^6 \{ e_{Q_{+j}} e_{Q_{+j}}^T + (e_{Q_{-j}} - e_{A_j})(e_{Q_{-j}} - e_{A_j})^T \} \right] \text{ F}. \quad (56)$$

The identity includes secondary, primary, output and probe capacitances; the two additional terms per channel are its physical cells. In particular  $L \geq I_{31}$  H and  $C \geq I_{50}$  F.

Let  $B$  collect the oriented winding-incidence columns in the first table and the output column  $e_o$ . Let  $G_0$  contain all bleeder, receiver and probe conductance terms  $g_e a_e a_e^T$ , plus the six primary diagonals  $e_{p_j} e_{p_j}^T / R_s$ . If  $a_{jk}$  are the four selector incidences,  $G(z) = G_0 + \sum_{j,k} g_{jk} a_{jk} a_{jk}^T$  and  $d(U) = \sum_j e_{p_j} U_j / R_s$ . With clamp incidences  $a_c$ , set  $h(w) = (1 \text{ S}) \operatorname{sgn}(w) \max(|w| - 5 \text{ V}, 0)$ . Kirchhoff and component laws then give

$$Li = B^T v - R_w i, \quad R_w = I_{31} \Omega, \quad (57)$$

$$C\dot{v} = -Bi - G(z)v + d(U) - \sum_c a_c h(a_c^T v), \quad (58)$$

$$C_g \dot{z}_{jk} = (c_{jk} - z_{jk}) / R_g, \quad g_{jk} = \frac{1}{10} \text{ S} + \frac{9}{10} \text{ S } z_{jk} / (1 \text{ V}). \quad (59)$$

The current at source  $j$  is  $I_j = (U_j - p_j) / R_s$ , including its capacitor and probe branches; it need not equal the primary winding current. These definitions specify the full graph without discarding an energized winding at a tap change.

Set  $T_* = 1/200 \text{ s}$  and  $t_e = 1/800 \text{ s}$ . Every channel's command changes from  $(1, 0, 0, 0) \text{ V}$  to  $(0, 1, 0, 0) \text{ V}$  at  $t_e$ . Gates start at the former values. With  $\tau_g = R_g C_g = 1/1000 \text{ s}$ , after the command  $z_1 = e^{-(t-t_e)/\tau_g} \text{ V}$ ,  $z_2 = (1 - e^{-(t-t_e)/\tau_g}) \text{ V}$ ,  $z_3 = z_4 = 0$ . All selector conductances remain between  $1/10$  and  $1 \text{ S}$ . The off paths remain finite; this is a simultaneous overlapping RC transition, not ideal opening, six phase-shifted histories or a bank reconnection. The gate/channel relation is a declared controlled law; identifying an actual switch or cam remains a separate task.

All clamp laws are continuous and globally Lipschitz. The known gates, invertible  $L$ ,  $C$ , and affine growth bound give a unique global electrical solution. Currents, capacitor voltages and gates are continuous at  $t_e$ ; command and power can jump, but no impulse or state reset occurs.

For the boundary retaining internal thermal energy  $H$ , let  $\dot{H} = D - H/\tau_h$ ,  $\tau_h = 2 \text{ s}$ . Here  $D$  sums winding, source-resistor, selector, bleeder, clamp, probe and gate-resistor heating. For example, source heating is  $(U_j - p_j)^2 / R_s$  and clamp heating is  $(a_c^T v) h(a_c^T v) \geq 0$ . Receiver heat is outside. Integrate each external product independently, splitting at  $t_e$ :

$$W_j = \int_0^{T_*} U_j I_j dt, \quad W_{gjk} = \int_0^{T_*} c_{jk} (c_{jk} - z_{jk}) / R_g dt, \quad (60)$$

$$W_L = \int_0^{T_*} v_o^2 / R_L dt, \quad Q_b = \int_0^{T_*} H / \tau_h dt, \quad (61)$$

$$E(t) = \frac{1}{2} i^T L i + \frac{1}{2} v^T C v + \frac{1}{2} C_g \sum z_{jk}^2 + H(t), \quad r_E = E(T_*) - E(0) - \sum_j W_j - \sum_{j,k} W_{gjk} + W_L + Q_b. \quad (62)$$

Multiplying the independently specified electrical laws gives their reactive-store derivative. At each source use  $p_j I_j = U_j I_j - R_s I_j^2$ , and add the gate and caloric equations. This proves  $r_E = 0$  for the declared model. Each winding's terminal work  $\int (b_w^T v) i_w dt$  and its opposite node transfer are retained before internal cancellation. Mutual magnetic energy is not assigned uniquely to individual windings. An electrical-only boundary instead excludes  $H$  and exports the separately integrated  $D$ ; it must not also export the same retained heat a second time.

## 7.2 A prepared transition with continuous current margins

Does this connected transition admit simultaneous source return and receiver delivery within specified component limits throughout a nonzero interval? The following prepared family answers that bounded

question [OP-TRF-18]. In every channel the nominal initial data are

$$(i_p, i_u, i_d) = (-1, 1/5, 1/10) \text{ A}, \quad (i_+, i_-) = (1, 1) \text{ A}, \quad p_j = 2 \text{ V}, \\ (A, T, X_+, Q_+, X_-, Q_-) = (1, 1/2, 4/5, 7/10, 1/5, 1/10) \text{ V}, \quad i_o = -1 \text{ A}, \quad v_o = 1 \text{ V}. \quad (63)$$

All probes and  $H$  start at zero; gates have the already specified values. The admitted family varies every plant electrical coordinate independently by at most  $1/1000$  A or V, keeping probes, gates and  $H$  fixed. Preparation work is outside this operation interval.

**Theorem 3** (Finite shared-output witness). *Every preparation in this family has  $I_j < -1/2$  A,  $i_{\pm j} > 1/2$  A and  $v_o > 1/2$  V throughout  $[0, T_*]$ . Every component current has magnitude below 2 A, every component voltage below 3 V, every winding linkage below 10 Wb turn and every common-core flux below 7 Wb. The receiver work exceeds  $1/1000$  J.*

*Proof.* Normalize currents by 1 A, voltages by 1 V and time by 1 s. In the unclamped domain the 81-coordinate electrical vector  $x$  obeys  $\dot{x} = A(t)x + b$ . The incidence calculation in Appendix A gives  $\|A(t)\|_\infty \leq 61/10$  and  $\|b\|_\infty \leq 1$  over the entire gate range. Use the looser bounds  $K = 32$ ,  $\beta = 2$ ,  $\rho = 1/1000$ ,  $\|x_0\|_\infty = 2$  and  $T = 1/200$ . The integral equation bounds the displacement by

$$\|x(t) - x_0\|_\infty \leq \rho + \frac{[K(2 + \rho) + \beta]T}{1 - KT} = \frac{331}{840} =: d_*. \quad (64)$$

This follows from  $\int_0^t e^{Ks} ds \leq T/(1 - KT)$ , not an expansion of the trajectory. Every relevant terminal difference is below  $2 + 2d_* = 1171/420 < 3 < 5$  V. A first clamp crossing is therefore impossible, which justifies continuation over the whole interval. Since the nominal source node is 2 V, branch currents are 1 A and output is 1 V, the signed margins are  $1 - d_* = 509/840$  in the respective units. Hence

$$W_L \geq \frac{1}{200} \left( \frac{509}{840} \right)^2 \text{ J} = \frac{259081}{141120000} \text{ J} > \frac{1}{1000} \text{ J}. \quad (65)$$

Winding currents are below  $1 + d_* < 2$  A. The largest absolute sum of inductance coefficients is  $17/4$  H, giving linkage below  $19907/3360$  Wb turn. The output common flux is  $\Phi_o = (1/4 \text{ H}) s^T i_o^{\text{block}}$  in the stated turn scale, and is bounded by  $15223/3360$  Wb; input-core bounds are smaller. These are declared model limits, not material saturation ratings.

Capacitor currents require the derivative equation, not only the state box. Reusing the sharper  $K = 61/10$ ,  $\beta = 1$  gives displacement  $\eta = 134/1939$ . Appendix A bounds every physical capacitor's  $C_e d_e^T \dot{v}$ ; the largest is  $187351/96950 \text{ A} < 2$  A. Its separate resistor, selector, probe, source, receiver and gate bounds are also below 2 A. The changing full-winding selector has  $A_j - X_{+j} \geq (1/5 - 2\eta) \text{ V} = 599/9695 \text{ V}$ , so its current stays at least  $599/96950 \text{ A} > 0$  during the edge. All bounds hold at both event sides and at every intervening time.  $\square$

The nominal initial store is independently

$$E_m(0) = \frac{40507}{1600} \text{ J}, \quad E_C(0) = \frac{2369}{100} \text{ J}, \quad E_g(0) = \frac{3}{1000} \text{ J}, \quad E(0) = \frac{392079}{8000} \text{ J}. \quad (66)$$

The receiver and magnetic states are already energized. For  $D_* = T_* - t_e = 3/800 \text{ s}$ , independent integration of the six rising gate sources, six falling gate resistors and the remaining zero gates gives

$$\sum W_{gjk} = \frac{3}{500} (1 - e^{-15/4}) \text{ J}, \quad Q_g = \frac{3}{500} (1 - e^{-15/2}) \text{ J}. \quad (67)$$

Each gate endpoint is evaluated from its exponential voltage, and their store change equals the difference of these two independently integrated terms. No residual defines either term.

For the electrical endpoints, the fundamental matrix of the connected equations gives  $x(t) = F(t, 0)x(0) + \int_0^t F(t, s)b(s)ds$ , composed across the command with the identity state map. Substitute this state separately in every integral in (62) and in its quadratic endpoint store; the thermal endpoint follows its own convolution. These exact expressions define the remaining works without assigning numerical trajectories. In particular the proved source and receiver margins, the gate integral and  $Q_b \geq 0$  imply

$$E(T_*) - E(0) \leq \left[ \frac{3}{500}(1 - e^{-15/4}) - \frac{6 \cdot 509}{840 \cdot 200} - \frac{259081}{141120000} \right] J < 0. \quad (68)$$

The endpoint store remains defined by the physical state in (62); this inequality follows from the independently derived dynamics and port bounds. It is not an inferred preparation cost. The exact identities and enclosures involve no numerical integration error; the other finite-history integrals are left unevaluated and physical model uncertainty remains unmeasured.

As a separate coupling control, set output mutual inductances to zero while retaining their diagonal self-inductances and the same initial currents, voltages, sources, receiver and gates. Its initial store becomes 284079/8000 J, instead of 392079/8000 J. The receiver winding is then magnetically independent and initially energized; its current must be solved from the changed equations. This is neither a separate loaded-transformer bank nor an equal-store efficiency comparison. With all plant states and sources zero, plant and receiver work remain zero, while initialized gates and their supplies still exchange energy. Neither control supplies a full-wave cycle. Staggered commands, longer intervals, singular component limits, other preparations and material behavior require their own analyses.

### 7.3 Six machines and an independently supplied timing shaft

Can the supplied assembly return its complete state after a timing-shaft revolution in  $T \in [1/2, 2]$  s? For the insulated laws below, that question has a negative answer [OP-TRF-19]. It is a different contract from the preceding fixed-clock finite witness.

Replace each ideal source by a field-excited armature, retaining its  $R_s = 1 \Omega$  resistor and primary-node/probe capacitances. The armature current  $a_j$  leaves the generator and enters  $p_j$  through  $R_s$ ;  $V_{gj} = p_j + R_s a_j$ . Six rotors share one rigid shaft with angle  $\phi$ , speed  $\omega$  and total inertia  $J = 1 \text{ kg m}^2$ . Define  $\chi_j = \phi + (j - 1)\pi/3$ . Field current  $f_j$  enters the generator's positive field terminal from an independent  $V_f = 1 \text{ V}$  supply. With  $L_a = L_f = 1 \text{ H}$ ,  $M = 1/10 \text{ H}$  and  $R_a = R_f = 1 \Omega$ , declare the reciprocal laws

$$E_{gj} = \frac{1}{2}L_a a_j^2 + \frac{1}{2}L_f f_j^2 + M \cos \chi_j a_j f_j, \quad (69)$$

$$\begin{pmatrix} L_a & M \cos \chi_j \\ M \cos \chi_j & L_f \end{pmatrix} \begin{pmatrix} \dot{a}_j \\ \dot{f}_j \end{pmatrix} = \begin{pmatrix} M \omega \sin \chi_j f_j - (R_a + R_s) a_j - p_j \\ V_f - R_f f_j + M \omega \sin \chi_j a_j \end{pmatrix}, \quad (70)$$

$$\dot{\phi} = \omega, \quad J \dot{\omega} = \tau_{\text{pm}} - b \omega - \sum_j M \sin \chi_j a_j f_j, \quad (71)$$

$$\tau_{\text{pm}} = 8 \text{ N m}, \quad b = \frac{1}{10} \text{ N m s/rad}. \quad (72)$$

The magnetic matrix has eigenvalues at least 9/10 H. The primary-node equation now receives  $a_j$ , not the former ideal-source current  $(U_j - p_j)/R_s$ . Remove that source forcing and diagonal when forming its nodal law. Fixed electrical offsets add no independent rotor angles. Direct differentiation of the independently

declared magnetic store and substitution of the winding laws gives

$$\dot{E}_{gj} = -V_{gj}a_j + V_{fj}f_j - R_a a_j^2 - R_f f_j^2 + M \omega \sin \chi_j a_j f_j. \quad (73)$$

The opposite rotor conversion follows by multiplying its torque equation by  $\omega$ . Integrate both sides of each generator/fixture and rotor/generator interface before cancellation. Source-resistor heat is now  $R_s a_j^2$ . The timing motor has angle  $\theta$ , speed  $\Omega$ , inertia  $J_t = 1 \text{ kg m}^2$ , coil  $L_t = 1 \text{ H}$ ,  $R_t = 1 \Omega$ , torque constant  $k_t = 1 \text{ N m/A}$  and the corresponding back-emf constant. Four followers, indexed  $k = 0, 1, 2, 3$ , have  $x_k = h \cos(\theta - k\pi/2)$ ,  $h = 1/100 \text{ m}$ ,  $K_k = (k + 1) \text{ N/m}$  and  $d_k = 1 \text{ N s/m}$ . With  $x'_k = dx_k/d\theta$  and  $b_t = 1/10 \text{ N m s/rad}$ , the supplied laws are

$$L_t \dot{i}_t = u - R_t i_t - k_t \Omega, \quad \dot{\theta} = \Omega, \quad (74)$$

$$J_t \dot{\Omega} = k_t i_t - b_t \Omega - \sum_k K_k x_k x'_k - \sum_k d_k (x'_k)^2 \Omega, \quad (75)$$

$$c_{jk} = \frac{1}{2} [1 + \cos(\theta - k\pi/2)] \text{ V}. \quad (76)$$

The compatible SI values of the motor constants make the electrical and mechanical conversion products equal. Cam springs store  $\sum_k K_k x_k^2/2$ ; follower heat is  $\sum_k d_k (x'_k \Omega)^2$ . Thus the mechanical command has actual spring and damping reactions. The cosine commands supply the same finite gates as above. No physical capacitance depends on cam position in this declared model; a variable-capacitance actuator requires its own conjugate power and constitutive store.

For relative phase  $\psi = \theta - \phi$  with target zero, two actual  $1 \Omega$ ,  $1/100 \text{ F}$  sensing branches have voltages  $q_\phi, q_\omega$  driven by the independent transducer ports  $v_\phi = \sin(\phi - \theta) \text{ V}$  and  $v_\omega = (\omega - \Omega) (1 \text{ V s/rad})$ . They obey  $C_q \dot{q}_v = (v_v - q_v)/R_q$ . The command  $U_c = 8 \text{ V} + q_\phi + q_\omega$  drives a  $R_u = 1 \Omega$ ,  $C_u = 1/100 \text{ F}$  node supplying the motor:

$$C_u \dot{u} = (U_c - u)/R_u - i_t. \quad (77)$$

Each transducer work is  $\int v_v (v_v - q_v)/R_q dt$ ; the controller supply work is  $\int U_c (U_c - u)/R_u dt$ . Their finite states, resistor heats and capacitor endpoints are retained. These ideal transducer and regulated-supply laws have explicit external ports; sensor backreaction and regulation losses beyond those ports remain unidentified.

For this assembly replace the thermal bath by insulation:  $\dot{H} = D_m$ , where  $D_m$  contains all fixture losses except external receiver heat, armature/field copper, both bearings, followers, timing coil, controller and sensing resistors. Coefficients are temperature-independent. The full 126-coordinate state includes the previous electrical, gate and thermal states, twelve armature/field currents, both shaft angles and speeds, timing-coil current and three sensing/supply voltages. Cam positions are functions of  $\theta$ , with their stores retained.

The external signed powers are  $\tau_{\text{pm}} \omega$ , each  $V_{fj} f_j$ , all 24 gate-supply products, the two transducer products and the controller supply product; receiver export is  $v_o^2/R_L$ . There is no bath port. Add  $\sum E_{gj}$ ,  $J \omega^2/2$ ,  $J_t \Omega^2/2$ ,  $L_t i_t^2/2$ , the cam springs and the three controller/sensor capacitors to the fixture's independent store. The winding, shaft, motor, spring and caloric laws prove its signed balance after each external work is integrated separately. Electrical return into an enclosed generator is internal, never another external source contribution.

Take the section  $\theta = 0 \pmod{2\pi}$  with  $\Omega > 0$ , returning after one positive timing revolution. Both shaft speeds must lie in  $[\pi, 4\pi] \text{ rad/s}$ , period in  $[1/2, 2] \text{ s}$ , winding/field currents below  $10 \text{ A}$  in magni-

tude and node voltages below 10 V. Angles return modulo  $2\pi$ ; every electrical, speed, relative-phase, controller and thermal state must also return. The caloric law and the timing-bearing product give, for every admitted history,

$$H(T) - H(0) \geq b_t \int_0^T \Omega^2 dt \geq \frac{b_t}{T} \left( \int_0^T \Omega dt \right)^2 = \frac{4\pi^2 b_t}{T} \geq \frac{\pi^2}{5} \text{ J} > 0. \quad (78)$$

The middle step is Cauchy–Schwarz and uses the separately integrated bearing power. Thus no full-state return fixed point, and hence no attracting full-state cycle, exists in this domain. No initial thermal preparation removes the positive drift. Two trajectories differing only in initial  $H$  keep that difference exactly under these laws. This proof does not infer instability from a unit multiplier.

For a transverse return, the differential of the flow-to-section map is  $[I - fn^T/(n^T f)]DF_T$ , restricted to section tangents, with the thermal integral's derivative retained. Here  $n$  is the section normal and  $f$  the full vector field at return. This factor includes the change of return time; holding the clock fixed would omit it. Smooth cams give no selector reset. At a transverse clamp crossing the vector field is continuous, so its saltation matrix is the identity although its variational Jacobian changes. Grazing, tied crossings and nontransverse returns need separate analysis. There is no fixed point at which to assign cycle multipliers.

The admitted current bounds imply input linkage at most 15 Wb turn, output linkage 85/2 Wb turn, generator linkage 11 Wb turn and output common flux 65/2 Wb at every time. These bounds on the declared linear model do not identify a material's saturation or remanence. The obstruction also holds in its larger linear continuation. A bath law or permission for thermal drift changes the full-state question. Electrical/shaft attraction, heat-rejecting operation, physical material limits and complete-service performance retain their separate scopes.

A distinct finite initialized control on  $[0, 1/200]$  s uses the earlier fixture preparation,  $a_j = 0, f_j = 1 \text{ A}$ ,  $\phi = \theta = 0$ ,  $\omega = \Omega = 2\pi \text{ rad/s}$ ,  $i_t = 8 \text{ A}$ ,  $u = 8 \text{ V}$ ,  $q_\phi = q_\omega = 0$  and  $H = 0$ . Its independent initial store is  $(3373203/40000 + 4\pi^2) \text{ J}$ . Its actual final state and each external power integral are defined by (72)–(77); no evaluated trajectory or work ordering is assigned here. This control supplies neither an initialized cycle nor an attraction result. Positive magnetic matrices and locally Lipschitz laws give uniqueness up to exit from a bounded operating region, which suffices for the preceding return obstruction.

#### 7.4 A complete electrical history with insufficient energy accuracy

Can complete electrical-port records identify initial electrical energy uniformly within 1/250 J for the following two known preparations? The specified probes and error set exclude that target [OP-LR23-03]. Return to the electrical fixture and its original fixed clock; the mechanical assembly and its cycle question are absent.

The first preparation is (63). The second adds 1/10 V to  $Q_{+1}$  and subtracts it from  $Q_{+2}$ . All other states, inputs, probes and gates are identical;  $H(0) = 0$  is independently known. This pair is separate from the smaller feasibility neighborhood. Every individual cell and parasitic, joint magnetic state, gate and probe capacitor contributes to the initial electrical store. The two changed nodes each have a cell and a unit parasitic, so direct evaluation yields

$$E_0 = \frac{392079}{8000} \text{ J}, \quad E_1 = \frac{392239}{8000} \text{ J}, \quad E_1 - E_0 = \frac{1}{50} \text{ J}. \quad (79)$$

The common baseline cancels the linear cross terms. With preparation radius 1/10, the same continuation bound gives maximum terminal difference  $127/42 \text{ V} < 5 \text{ V}$ , so both trajectories remain unclamped. The

smaller family's 3 V component bound is not invoked.

Observe all six exact source voltages and reported source currents  $\hat{I}_j = (U_j - y_j)/R_s$ , the receiver channels  $y_o, y_o/R_L$ , all 24 gate command/current pairs and the event time on  $[0, T_*]$ . The actual passive probe satisfies  $(10\text{ s})\dot{y}_j = p_j - y_j$ ,  $y_j(0) = 0$ ; the receiver probe obeys the same law with  $p_j$  replaced by  $v_o$ . Thus, for constant  $U_j$ , the reported current is a filtered current with known initial value, not instantaneous  $I_j(0)$ . The two receiver channels are redundant under the known load law.

Each reported source-current record permits any additive error function bounded pointwise by  $1/10\text{ A}$ . There is no independence assumption in time or across channels. Output, supply, driver, event-time, gain, offset, coefficient and initial-sensor errors are fixed at zero for this mathematical comparison. The 10 s response and loading are component choices, not measured calibration. Heat export is integrated in the balance but is not an observed channel here. "Complete electrical ports" therefore does not mean calorimetric observation or instantaneous unfiltered records.

**Theorem 4** (Uniform energy-error obstruction). *For this two-state experiment the smallest possible worst-case absolute initial-energy error is  $1/100\text{ J}$ . In particular the requested  $1/250\text{ J}$  target is impossible.*

*Proof.* The connected electrical difference obeys the homogeneous linear equations through both the plateau and finite edge. Its independent quadratic form and derivative are

$$D = \frac{1}{2}\delta i^T L \delta i + \frac{1}{2}\delta v^T C \delta v, \quad \dot{D} = -\delta i^T R_w \delta i - \delta v^T G(z) \delta v \leq 0, \quad D(0) = \frac{1}{50}\text{ J}. \quad (80)$$

Each primary node has a unit capacitor, so  $|\delta p_j| \leq \sqrt{2D(0)/(1\text{ F})} = 1/5\text{ V}$ . The passive filter has zero initial difference and a nonnegative impulse response of integral at most one on any finite interval. Hence  $|\delta y_j| \leq 1/5\text{ V}$  and  $|\delta \hat{I}_j| \leq 1/5\text{ A}$  throughout.

Swapping channels 1 and 2 commutes with  $L, C, B$  in their corresponding coordinates, all permanent connections and each synchronous selector coefficient. It also fixes the input forcing. The initial difference changes sign under this permutation; uniqueness therefore keeps it antisymmetric. The output winding, receiver and its probe are invariant coordinates, so their differences are identically zero throughout. Gate histories are identical. This is a property of the connected flow across the entire finite edge, not a fixed-topology rank assertion or a static cell sum.

Choose the midpoint of the two reported source-current histories and the common output and gate records. Each of the two explanations needs error at most  $1/10\text{ A}$ , so both are admitted. For its consistent-state set  $\mathcal{S}(y)$ , this record has exact energy width  $1/50\text{ J}$ . Every estimate must err by at least half that width for one of its endpoints. For any other admitted record the two-state set is either a singleton or the same pair. Its energy-interval midpoint attains worst-case error at most  $1/100\text{ J}$ , proving equality.  $\square$

Exact source traces can distinguish more than this finite-error record. A longer history, tighter errors, additional cell/flux probes, observed heat or unknown coefficients defines another experiment, with its own physical loading and consistent-state set. The earlier successful binary cell-probe result retains its own graph and error assumptions. Neither result implies universal state/sign reconstruction or a missing physical energy transfer.

## 7.5 Finite comparisons for the shared assembly

For an identified implementation of the finite transition, hold the input waveforms, command time and individual initial capacitor/winding states, then observe simultaneous complete source, receiver and gate terminal products. The predicted sign margins are  $509/840\text{ A}$  and  $\text{V}$ . To establish the stricter  $1/2$  thresholds from those model values, total current or voltage error must be below  $89/840$  in the corresponding

units; for receiver service it must be below  $(259081/141120000 - 1/1000)$  J. Actual observations instead use their measured lower bounds. Independently justified model error is part of those bounds, not calibrated from a small balance residual. The finite 1 ms gate response requires verified timing and bandwidth; synchronous voltage/current acquisition alone does not establish that adequacy.

Observe cell and magnetic endpoints independently, including the initially energized receiver, all probes, gates and thermal state. Winding terminal integrals and an identified flux/material law are needed to reconstruct magnetic state; unit model flux limits are not physical core ratings. Resolve actual clamp, switch, gap and return paths. A source return with positive receiver work is compatible with the proved decreasing prepared store; a complete measured remainder requires every boundary term in (62) and propagated uncertainty.

For the machine assembly, synchronized torque and encoder observations at the common prime-mover and timing shafts resolve their effort–rate products. Field, sensor, controller and gate supplies are separate electrical observations. Thermal endpoints and exported heat distinguish insulated drift from a cooled return. Under the stated insulated bearing law the lower bound is  $\pi^2/5$  J per admitted revolution. A finite thermal uncertainty smaller than a proposed separation is necessary to test it. A cooled or temperature-dependent apparatus needs its own law; that comparison leaves the proved insulated obstruction resolved.

For energy identification, the existing error set cannot meet the stated target regardless of estimator choice. An added observation can discriminate the known pair if its two predicted responses differ by more than twice its independently bounded error. Establish those responses from the new loaded graph, including probe work and disturbance, before using that criterion. Alternatively, bound the energy width of its entire consistent state set by  $2\epsilon_E$ . This condition concerns the energy target; full state reconstruction is stronger.

A switched/bypass performance comparison additionally fixes useful receiver service and actual preparation, operation, relaxation and reset paths. Integrate prime-mover and every auxiliary supply separately; match endpoints or retain their differences and explicit reset costs. The finite witness, insulated obstruction and observation theorem do not supply that ordering. Ordinary synchronized electrical, torque/encoder and thermal instruments provide the relevant observables, but their calibration, loading and bandwidth must resolve the particular difference. No apparatus data or unmeasured physical remainder are assigned by these model results.

## 8 Signed states and a reconnection experiment

### 8.1 Four quantities that an instrument must distinguish

An oriented charge, its absolute value, its energy and its work have different units and different laws. Fix each capacitor's physical plate orientation before defining  $q_j = C_j v_j$ . Reversing a sensor changes its reported sign; reconnecting plates changes the incidence matrix and can change the conserved quantities. On a fixed isolated two-cell exchange, KCL gives  $\dot{q}_1 = -i$ ,  $\dot{q}_2 = i$ , so  $Q = q_1 + q_2$  is constant. For  $C_* = C_1 C_2 / (C_1 + C_2)$  and  $d = v_1 - v_2$ ,

$$E_C = \frac{Q^2}{2(C_1 + C_2)} + \frac{1}{2} C_* d^2, \quad A_q = |q_1| + |q_2|. \quad (81)$$

Neither expression is a signed port work. With a resistor  $R$ ,  $d = d_0 e^{-t/(RC_*)}$  and its separately integrated heat is  $C_* d_0^2 (1 - e^{-2T/(RC_*)})/2$ . The absolute charge sum is nonincreasing: when the signs agree its derivative is zero; when they oppose,  $\dot{A}_q = -2|i|$  until a charge reaches zero. This sign argument depends

on the resistor law and the declared orientations.

An inductor permits a different exact result. With initial charges  $(Q, 0)$  and zero current,

$$q_1 = \frac{Q(C_1 + C_2 \cos \omega t)}{C_1 + C_2}, \quad q_2 = \frac{QC_2(1 - \cos \omega t)}{C_1 + C_2}, \quad i = \frac{QC_2 \omega \sin \omega t}{C_1 + C_2}, \quad \omega^2 = \frac{1}{LC_*}. \quad (82)$$

For  $(C_1, C_2, L) = (1, 19, 20/19)$  in SI units and  $Q = 10$  C, the states at 0 and  $\pi$  s are  $(10, 0, 0)$  and  $(-9, 19, 0)$ . Thus  $Q = 10$  C,  $A_q$  grows from 10 to 28 C, and total reactive energy is 50 J at both ends. Integration of  $-v_1 i$ ,  $v_2 i$  and  $Li \dot{i}$  separately gives  $-19/2$ ,  $19/2$  and 0 J. No external work enters this isolated boundary. The inductor's nonzero intermediate store is essential.

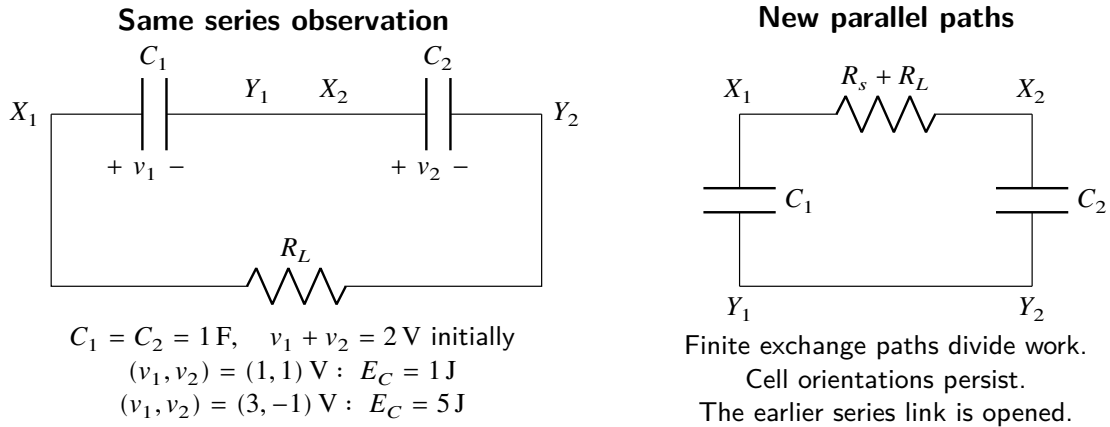
A driven lossy winding has no corresponding universal winding-only linear invariant. If  $a^T i$  were conserved for all inputs and states in  $Li = bU - Ri$ , it would require both  $a^T L^{-1}b = 0$  and  $a^T L^{-1}R = 0$ . Invertible positive  $L, R$  force  $a = 0$ . At every topology change the applicable invariant must be derived anew.

## 8.2 The same terminal history can hide four joules

Connect two equal cells in series through a receiver  $R_L$ . For cell voltages  $v_1, v_2$ , set  $s = v_1 + v_2$  and  $d = v_1 - v_2$ . The common current gives

$$\dot{s} = -\frac{2s}{R_L C}, \quad \dot{d} = 0, \quad E_C = \frac{C}{4}(s^2 + d^2), \quad W_L = -\frac{Cs_0^2}{4}(1 - e^{-4T/(R_L C)}). \quad (83)$$

With  $C = 1$  F, preparations  $(1, 1)$  and  $(3, -1)$  V both give  $s_0 = 2$  V and the same entire terminal history. Their independently evaluated initial stores are 1 and 5 J; the unobserved difference  $d = 0$  or 4 V retains a 4 J energy separation at every time. Preparation II contains a reversed cell.



**Figure 8.** Two equal series cells can have the same terminal voltage and different internal energy. A later parallel connection exposes the difference mode only through its declared finite paths; the individual cell orientations remain fixed.

Reconnecting corresponding plates through a finite resistance can expose that difference mode. If a receiver and selector resistance are in series, the receiver receives the fraction  $R_L / (R_L + R_s)$  of the converted difference energy, with finite-time factor  $1 - e^{-2T/((R_L + R_s)C_*)}$ . The selector receives the complementary fraction. Residual cell energy remains at finite  $T$ . This calculation specifies neither the preparation work

nor the useful delivery in a different switched graph.

### 8.3 A finite switched observation map

The bounded reconnection result concerns identification of two known preparations by a loaded cell probe [OP-TRF-15]. Its exact finite formulation below states the component laws and separation inequalities required for that binary certificate. Recovering arbitrary cell signs, energies or temperatures is a different inverse problem.

Use nodes  $(A, T, X_1, Y_1, X_2, Y_2)$  relative to  $B$ , unit-farad cells across  $X_1 - Y_1$  and  $X_2 - Y_2$ , and  $1/50$  F parasitics at every node. A  $1 \Omega$  receiver joins  $X_1 - Y_2$ . Selectors join  $A - X_1$ ,  $T - X_1$ ,  $X_1 - X_2$ ,  $Y_1 - Y_2$ ,  $Y_1 - X_2$  and the probe path  $X_1 - Y_1$ . Use the winding matrix (50) with  $f = \kappa = 1/2$ , each winding resistance  $1/20 \Omega$ , source resistance  $1/5 \Omega$ , and zero source voltage during operation. The five ordinary selectors have on/off resistances  $1/20$  and  $10000 \Omega$ ; the probe has  $10$  and  $10^6 \Omega$ . Cell and terminal clamps have slope  $5 \text{ S}$  above  $4 \text{ V}$  magnitude. Gate capacitance and resistance remain  $1/10000 \text{ F}$  and  $20 \Omega$ .

The initial node vectors are

$$v_0^I = (0, 0, 2, 1, 1, 0), \quad v_0^{II} = (0, 0, 2, -1, -1, 0) \quad \text{V.} \quad (84)$$

Both terminal voltages  $V_{X_1} - V_{Y_2}$  are  $2 \text{ V}$ ; both parasitic stores are  $3/50 \text{ J}$ , independently of the  $1$  or  $5 \text{ J}$  cell store. Initial winding currents are zero. The series selector is on until  $t = 1/4 \text{ s}$ . During the next  $1/100 \text{ s}$ , smooth conductance edges open it and close both parallel selectors and the probe. At  $t = 1 \text{ s}$ , source and gates are turned off and  $1/5 \text{ S}$  node bleeders operate until  $t = 2 \text{ s}$ . All final states remain in the account. Finite off conductance means that the two earlier terminal histories need not agree exactly as they do in (83).

Let  $x$  collect winding and node states, and append the probe filter  $\dot{z} = (V_{X_1} - V_{Y_1} - z)/(1/50 \text{ s})$  with zero initial filter state. On each prescribed clamp branch the augmented linear or affine equations have a fundamental matrix. Their ordered composition defines the switched observation  $z(t) = \mathcal{O}_t(x_0)$ . It includes the probe conductance in the physical state equations; filtering is not a license to omit probe work.

For an exact conditional certificate, calculate  $z_j = \mathcal{O}_{7/20}(x_0^j)$  and form the two uncertainty intervals  $I_j = [z_j - u_z, z_j + u_z]$ ,  $u_z = 1/20 \text{ V}$ . If  $|z_{II} - z_I| > 2u_z$  and the precommutation terminal difference is bounded by  $1/10 \text{ V}$  throughout its window, these readings distinguish the two specified preparations. Since the two candidate cell energies are known exactly, correct binary classification meets a  $1/100 \text{ J}$  energy target in the exact-coefficient model. Those inequalities, including continuous-time preterminal bounds, are the certificate conditions for the finite model; no sampled voltage is an all-state observability proof. Physical acquisition must bound component drift, probe loading, gain, bandwidth, polarity, timing and preparation uncertainty before adopting these decision intervals. Unknown analog preparations require a new uncertainty analysis rather than a two-label classifier.

### 8.4 Paired work includes the chargers and the retained states

The same receiver and clock define a bounded paired-work comparison [OP-TRF-16]. More prepared energy alone fixes no sign for the difference of that receiver's work. The work is determined by the loaded trajectory and every connected conversion path.

Preparation is physical. Disconnect windings and selectors and charge the six-node capacitance matrix  $C_b$  from zero during  $[-1, 0] \text{ s}$  using six independent  $1 \Omega$  Thevenin sources. For constant source vector  $u$ ,  $C_b \dot{v} = u - v$ , whence  $v(0) = (I - e^{-C_b^{-1}})u$ . The positive matrix  $C_b$  makes the charge map invertible,

so each vector in (84) has its own specified finite  $u$ . Each source work is  $\int u_j(u_j - v_j)dt$ ; each resistor exports  $-\int (u_j - v_j)^2 dt$ . The endpoint is independently  $v(0)^T C_b v(0)/2$ .

During operation evaluate separately, for each preparation,

$$W_L^j = -\int_0^2 (V_{X_1}^j - V_{Y_2}^j)^2 dt, \quad W_{\text{probe}}^j = -\int_0^2 g_{\text{probe}}(t) (V_{X_1}^j - V_{Y_1}^j)^2 dt. \quad (85)$$

Source, selector, clamp, gate and bath products have their own integrals. With thermal matter retained, all internal resistive conversions enter  $\dot{H} = D - H/2$ ; the bath work is  $-\int H/2 dt$ . The total endpoint includes winding, cell, parasitic, gate and thermal stores. For each event retain the same continuous state on both sides and both one-sided powers. Finite positive parameters give zero impulse.

On linear intervals the independent quadratic primitive (2) evaluates these works; finite edges use their ordered fundamental matrix and separate quadratic integrals. The sign of  $W_L^{\text{II}} - W_L^{\text{I}}$  follows from those expressions, not from the 4 J initial cell difference. A physical comparison requires an uncertainty below its predicted separation and retains actual reset endpoints. It establishes neither equal useful service nor a closed-cycle generator ranking.

### 8.5 An all-time sign and magnitude certificate

A finite topology change can be treated without inferring sign events from sampled states. Retain the LC path in (82) with  $(C_1, C_2, L) = (1, 19, 20/19)$  and add a  $1 \Omega$  shunt across cell 1 only on  $[a, b] = [\pi/2, \pi/2 + 1/10]$  s. On the full window  $[0, 2]$  s, write its conductance as  $g = 0$  or  $1$  S. The equations and independent energy are

$$\dot{q}_1 = -i - gq_1, \quad \dot{q}_2 = i, \quad i = \frac{19q_1 - q_2}{20}, \quad E = \frac{q_1^2}{2} + \frac{q_2^2}{38} + \frac{10i^2}{19}. \quad (86)$$

Here  $\dot{E} = -gq_1^2$  and  $\dot{Q} = -gq_1$ . The old conserved charge is therefore not conserved during the shunt. For initial state  $(10, 0, 0)$ , Cauchy–Schwarz gives, at every time,  $A_q^2 \leq 40E_C \leq 40E \leq 2000 C^2$ . The identically zero initial state remains zero. This is a conservative bound for these preparations and components, not a universal multiplier.

Before  $a$ , (82) gives  $(q_1, q_2, i)(a) = (1/2, 19/2, 19/2)$  in SI units.  $q_2 = 0$  has its initial tangency at  $t = 0$ ; both charges are positive on  $(0, a]$ . Energy bounds give  $|i| < 10$  and  $|q_2| < 44$ . On the shunt interval, putting  $\tau = t - a$ , the integrating-factor solution of  $\dot{q}_1 + q_1 = -i$  gives  $|q_1| \leq 1/2 + 10\tau \leq 3/2$ . Also  $17/2 \leq q_2 \leq 21/2$  and  $|i| \leq 39/20$ , hence  $i \geq 1861/200 > 9$ . Thus  $\dot{q}_1 < 0$  throughout this interval. While  $q_1 \geq 0$ , it satisfies  $-21/2 < \dot{q}_1 \leq -9$ . Consequently its unique zero obeys

$$\frac{\pi}{2} + \frac{1}{21} < t_* < \frac{\pi}{2} + \frac{1}{16}. \quad (87)$$

Before that zero  $\dot{A}_q = -q_1 < 0$ ; after it,  $\dot{A}_q = 2i + q_1 > 0$ . The zero is a magnitude cusp with distinct one-sided derivatives, not a differentiable minimum.

After  $b$ , the energy bound also gives  $|q_1| \leq 10$  and  $|i| < 117/10$ . Since  $2 - b < 2/5$ ,  $i > 1861/200 - 117/25 = 37/8 > 0$  to the final time. Therefore  $q_1$  stays negative,  $q_2$  stays positive, and there are no further charge zeros. At  $a, b$  states are continuous; the appropriate one-sided vector fields give the switched derivatives. This covers every time, both preparations, the tangency, persistent zeros, single crossing, cusp and both switch sides.

The independent shunt work is  $W_h = - \int_a^b q_1^2 dt$ . On each constant- $g$  interval set  $x = (q_1, q_2, i)^T$  and  $x(t) = e^{A_g(t-t_0)} x(t_0)$  from (86); the quadratic primitive evaluates  $W_h$  separately from the two endpoint energies. Multiplying the three independent equations verifies their equality  $E(2) - E(0) = W_h$ . If the resistor is enclosed, its heat store replaces that export and total retained energy is constant. The finite shunt interval does not specify an actual switch's ignition, parasitic or sensor law.

## 9 Other prepared states and switching paths

### 9.1 A passive two-store circuit has several different limits

Configuration V first uses a parallel  $R_C, C$  section in series with a parallel  $R_L, L$  section. Its terminal impedance is

$$Z(s) = \frac{R_C}{1 + sR_C C} + \frac{sLR_L}{R_L + sL}. \quad (88)$$

KCL in each section and KVL in their series connection derive this formula. At  $s = 1 \text{ s}^{-1}$  with untargeted components equal to one in SI units,  $C = \epsilon$ ,  $R_C = 1$  gives  $Z = 1/(1 + \epsilon) + 1/2$ ;  $C = R_C = \epsilon$  in their respective units gives  $Z = \epsilon/(1 + \epsilon^2) + 1/2$ . At  $\epsilon = 1/10$  these are  $31/22$  and  $121/202 \Omega$ ; the limits are  $3/2$  and  $1/2 \Omega$ . This Laplace-domain comparison is a terminal-law statement, not a work measurement or a physical component with infinite or zero value.

The exact zero/infinite topologies have distinct hypotheses. A zero capacitor removes its capacitive branch, retaining  $R_C$ ; infinite capacitance constrains its voltage to a constant only for bounded current and a declared prepared voltage. A zero inductor enforces zero branch voltage, while infinite inductance constrains current for bounded voltage. Zero resistance shorts its section; infinite resistance removes that resistor. Incompatible prepared charge or flux needs a physical regularization, not substitution into (88).

Close the terminal by a zero-voltage source, retaining finite positive components. For capacitor voltage  $v$  and inductor current  $i$ , KCL gives

$$C\dot{v} = -(R_C^{-1} + R_L^{-1})v + i, \quad L\dot{i} = -v, \quad E = \frac{1}{2}Cv^2 + \frac{1}{2}Li^2. \quad (89)$$

The complete store boundary excludes the two thermal reservoirs and the terminal source. Its powers are  $0$ ,  $-v^2/R_C$ , and  $-v^2/R_L$ . For  $C = L = R_C = R_L = 1$ , prepared  $(v, i) = (1, 0)$  gives  $v = (1 - t)e^{-t}$ ,  $i = -te^{-t}$ . On  $[0, 1] \text{ s}$ ,

$$W_C^h = W_L^h = -\frac{1 - e^{-2}}{4} \text{ J}, \quad E(0) = \frac{1}{2} \text{ J}, \quad E(1) = \frac{e^{-2}}{2} \text{ J}. \quad (90)$$

The sum of absolute physical heat works is  $(1 - e^{-2})/2 \text{ J}$ ; the zero-state control gives zero for every work and store. This prepared experiment retains the original two-section topology. Other nondimensional coscalings require their own positive component paths, compatible endpoints, and individually resolved source and heat works.

Two explicitly different clamp regularizations illustrate why zero-state transfer functions do not fix prepared switching work. A unit capacitor at  $1 \text{ V}$  discharged through resistance  $R = \epsilon$  for  $T = \epsilon \ln 2$  exports  $3/8 \text{ J}$  and retains  $1/8 \text{ J}$ . A lossless auxiliary inductor  $L = \epsilon^2$  attached instead gives  $v = \cos(t/\epsilon)$ ,  $i = \sin(t/\epsilon)/\epsilon$ . At  $T = \pi\epsilon/2$  capacitor energy is zero, inductor energy  $1/2 \text{ J}$ , and each capacitor/inductor interface work is respectively  $-1/2, +1/2 \text{ J}$ ; external heat work is zero. Both follow exact finite equations and the two endpoint laws. Neither is the unique instantaneous fate of the initial charge. The finite component and time choices, their peak currents, and all additional switch or clamp paths remain part of the

experimental claim.

These finite paths do not select a unique prepared singular limit [OP-LR21-01]. Smooth zero-state forcing also leaves the work of an actual prepared-state switch unspecified [OP-LR21-02].

## 9.2 Hidden prepared energy and its observability

A transformer-coupled series  $R, C$  branch in parallel with  $G_0$  has

$$Y(s) = G_0 + \frac{n^2 s C}{1 + s R C}, \quad v_C(t) = V_0 e^{-t/(RC)}, \quad I_{\text{terminal}}(t) = -\frac{n V_0}{R} e^{-t/(RC)} \quad (91)$$

on a zero-voltage terminal preparation release, with the indicated current positive into the passive input. The internal resistor current has magnitude  $|V_0| e^{-t/(RC)} / R$ . At  $n = 0$  the transformer is interpreted as the decoupled boundary topology, not an invertible zero-ratio transformer. There is no terminal hidden-mode current, but  $E_C = C V_0^2 e^{-2t/(RC)} / 2$ .

Choose  $C = 1/1000000$  F,  $R = 100 \Omega$ ,  $V_0 = 1$  V,  $T = RC \ln 2 = \ln 2 / 10000$  s. The source work is zero, initial store  $1/2000000$  J, final store  $1/8000000$  J, and heat work  $-3/8000000$  J by direct integration of  $-v_C^2 / R$ . An empty mode predicts zero internal current and zero heat; a prepared hidden mode predicts initial internal current  $1/100$  A and the stated heat. A branch voltage/current observation resolves the difference even when the terminal observation cannot. Calorimetric resolution at that energy would need its own demonstrated uncertainty; branch electrical measurement avoids presuming it.

For a known single nonzero  $n$ , a certified terminal current bound  $u_I$  at release implies

$$|V_0| \leq R u_I / |n|, \quad E_C(0) \leq \frac{C R^2 u_I^2}{2 n^2}. \quad (92)$$

For  $n = 1/100$ ,  $u_I = 1/1000000$  A, and the preceding  $R, C$ , this upper bound is  $1/20000000000$  J. It grows without bound as sensitivity  $n$  approaches zero. Known gain, loading, and bandwidth are hypotheses of the bound. A noisy or nonideal transformer requires their independently bounded transfer law and additional stores.

Multiple modes introduce a further obstruction: two equal  $R, C, n$  branches prepared at  $+V, -V$  have cancelling terminal release currents but total energy  $C V^2 > 0$ . With the preceding  $C, R, n$  and  $V = 1$  V this is  $1/1000000$  J, exceeding the single-mode bound despite exact terminal cancellation. All their initial opposite branch currents are observable separately. More generally, if boundary observations are  $y(t) = H e^{A t} x_0$ , an energy bound from them exists on a preparation subspace only if no nonzero energy-bearing direction lies in its observation nullspace, with a quantitatively bounded inverse on the retained subspace. Internal branch measurements remove this nullspace in the two-mode example. More modes, nearly coincident time constants, nonideal couplings, and correlated noise require their own finite observability and error bounds; a single terminal noise threshold cannot establish them.

Near-cancelled or unobserved modes require branch-state observations with their own uncertainty [OP-LR23-01]. Exact decoupling leaves prepared hidden energy and any later reconnection as separate physical questions [OP-LR23-02].

## 9.3 Common current is a physical preparation

Take two lossless inductors  $L_1 = 1$  H,  $L_2 = 19$  H connected through a capacitor  $C = 20/19$  F. The oriented equations are  $L_1 \dot{i}_1 = -v$ ,  $L_2 \dot{i}_2 = v$ ,  $C \dot{v} = i_1 - i_2$ . For initial  $(i_1, i_2, v) = (10 + u, u, 0)$ , define

$c = i_1 + 19i_2 = 10 + 20u$  and  $d = i_1 - i_2 = 10 \cos t$ . Then

$$i_1 = \frac{c + 19d}{20}, \quad i_2 = \frac{c - d}{20}, \quad v = \frac{19}{2} \sin t, \quad E = \frac{c^2}{40} + \frac{19d^2}{40} + \frac{Cv^2}{2} = 50 + 10u + 10u^2 \text{ J.} \quad (93)$$

The equality follows by substitution of the state, not an assumed constant store. The operating boundary has no external ports; the capacitor and two coil works integrate separately as changes of their explicit quadratics. On  $[0, \pi]$  s,  $(W_1, W_2, W_C) = (-19c/20, +19c/20, 0)$  J, so their sum is zero. For  $u = 0$ , this pair is  $(-19/2, +19/2)$  J; for  $u = -1/2$  it is  $(0, 0)$ ; for  $u = -1$  it is  $(+19/2, -19/2)$ .

For actual preparation, disconnect the operating capacitor and independently ramp both coil currents from zero as  $i_1 = (10 + u)t$ ,  $i_2 = ut$  on  $[0, 1]$  s. With ideal lossless windings the two imposed source voltages are  $10 + u$  and  $19u$  V. Their independently integrated works are  $(10 + u)^2/2$  and  $19u^2/2$  J. Add winding resistance  $R_j$  and each source work additionally contains  $R_j I_{j*}^2/3$ , with separate negative heat work of that magnitude. A reversed controlled ramp supplies the explicit reset; passive relaxation is another path.

Over  $-1 \leq u \leq 1/2$ , completing the square gives  $E = 95/2 + 10(u + 1/2)^2$  J, hence the stated range  $[95/2, 115/2]$  J. At  $u = 0$  initial fluxes are  $(10, 0)$  Wb and final fluxes  $(-9, 19)$  Wb; at  $u = -1/2$  they are  $(19/2, -19/2)$  and  $(-19/2, 19/2)$  Wb. Thus physical sign changes and absolute sums vary with preparation while the differential trajectory is the same. A differential pickup that rejects  $c$  sees the same drive but retains its own backaction, load, and stores; its full coupled model is needed for the loaded comparison. Translating a reported flux origin alone changes none of the preparation works above.

The common-current preparation and pickup backaction therefore need their own source and load measurements [OP-LR42-02].

#### 9.4 Commuting endpoints with noncommuting work partitions

Two capacitors  $C_1, C_2$  exchange charge through one of two finite conducting channels  $A, B$ . The store boundary contains both capacitors; the active channel's heat reservoir is external. With  $d = v_1 - v_2$ ,  $\bar{v} = (C_1 v_1 + C_2 v_2)/(C_1 + C_2)$ , and  $C_e = C_1 C_2/(C_1 + C_2)$ , KCL gives

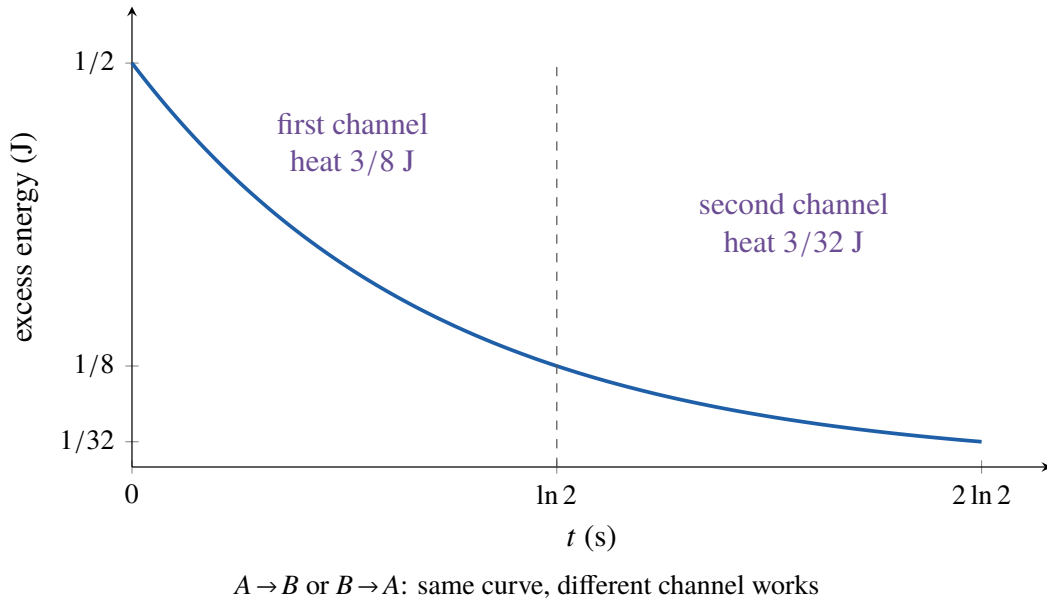
$$\dot{d} = -\frac{G}{C_e} d, \quad \dot{\bar{v}} = 0, \quad E = \frac{1}{2}(C_1 + C_2)\bar{v}^2 + \frac{1}{2}C_e d^2, \quad W_h = -\int_0^T G d^2 dt = -\frac{1}{2}C_e d_0^2(1 - e^{-2GT/C_e}). \quad (94)$$

Each stage is a finite  $d \mapsto d/2$  map when  $T = C_e \ln 2/G$ . The two maps commute. Their heat integrals retain the incoming  $d$  and therefore the stage order.

For  $C_1 = C_2 = 2$  F,  $(v_1, v_2) = (1, 0)$  V,  $G_A = G_B = 1$  S, each stage lasts  $\ln 2$  s. Independently evaluated total stores are 1,  $5/8$ , and  $17/32$  J before, between, and after the stages. Final voltages are  $(5/8, 3/8)$  V in both orders.

**Table 9.** Equal endpoints do not determine the channel partition. Both controls have zero residual and no impulse at either finite switching edge.

| Channel work into capacitor boundary, J | A then B | B then A |
|-----------------------------------------|----------|----------|
| $W_A$                                   | $-3/8$   | $-3/32$  |
| $W_B$                                   | $-3/32$  | $-3/8$   |
| Total work                              | $-15/32$ | $-15/32$ |
| Final store                             | $17/32$  | $17/32$  |



**Figure 9.** Exact excess-energy decay during two finite charge-sharing stages. The continuous state is the same for both orders, while the channel receiving the larger heat transfer changes. The plotted curve is equation (94).

Figure 9 separates the unchanged state curve from the changed heat recipient. Unequal-capacitance controls retain the common store. With  $(C_1, C_2) = (1, 2)$  F,  $C_e = 2/3$  F, the same voltage preparation and half-difference maps give channel works  $-1/4, -1/16$  J and endpoints  $1/2, 3/16$  J. With  $(C_1, C_2) = (2, 1)$  F those works agree but endpoints are  $1, 11/16$  J. Changing preparation to  $(1, -1)$  V in the equal-capacitance case multiplies the excess store and both channel works by four, with common store zero. Thus orientation and capacitance ratios must accompany a work claim.

Actual arc ignition, restrike, clamp release, winding capacitance, and switch supply laws are not determined by (94). A finite released winding/capacitor model can instead be written as  $L\dot{i} = -v$ ,  $C_p\dot{v} = i - \sum_l G_l(v, t)v$  with branch powers  $-G_l(v, t)v^2$ . Each proposed switch, arc, and clamp law is a separate  $G_l$ , with its own heat or receiver destination. Constant positive coefficients give exact matrix primitives; specified threshold switching uses event-side states and separate intervals. A constant-current or constant-voltage clamp uses its actual effort–flow law instead of this conductance representation.

For a connected perfectly coupled pair with  $L = aa^T$ ,  $a = (\sqrt{L_1}, \sqrt{L_2})$ , and capacitor incidence  $b = (1, -1)^T$ , the equations  $L\dot{i} = bv$ ,  $C\dot{v} = -b^T i$  impose  $n^T b v = 0$  for  $n = (-\sqrt{L_2}, \sqrt{L_1})$ . At  $L_2/L_1 = 19$  this requires  $v = 0$  and  $i_1 = i_2$  on a smooth solution. The prepared state  $(10, 0, 0)$  is incompatible; a putative terminal voltage impulse also has  $n^T b \int v dt = 0$  and cannot select the null-current jump. Named parasitic paths can select finite releases, but different paths retain their different works. Other orientations require their own nullspace calculation.

Finite values  $C_p = 1/100, 1/1000, 1/10000$  F and durations  $1/50, 1/200$  s are separate realizable comparisons, conditional on their bandwidth and current envelopes. For explicit wider ratios, retain  $(v_1, v_2) = (1, 0)$  V and both half-difference stages, but use  $(C_1, C_2) = (1/2000, 1)$  F or  $(1, 1/2000)$  F. The final stores are respectively  $21/1334000$  and  $10667/21344$  J; both export  $5/21344$  J in total, from initial stores  $1/4000$  and  $1/2$  J. Each stage lasts  $\ln 2/2001$  s at conductance  $1$  S. These finite ratios  $1/2000$  and  $2000$  probe beyond the preceding range, conditional on their own measurement bounds; they do not

certify a continuous limiting surface. At fixed prepared voltage,  $C_p \rightarrow 0$  removes energy; at fixed nonzero charge,  $E = q^2/(2C_p)$  diverges. At fixed  $R, C$  a prescribed voltage jump obeys (47); an RC charge-sharing path instead has finite limiting conversion but unbounded peak current as resistance vanishes. The exact zero-time and zero-component models remain mathematical questions separate from these finite tests.

Actual release, ignition, restrike and parasitic paths need independently identified laws before their channel partition is known [OP-LR48-03]. Finite ratios and finite edges test specified approaches, while other orientations and continuous limiting surfaces remain unproved [OP-LR48-05].

### 9.5 Different connections need not commute

Another scalar control has two independent 1 F cells initially at  $(1, -1)$  V. Each selectable channel contains a separate 1  $\Omega$  shunt across each cell, so its two laws are  $\dot{v}_j = -v_j$ . Operating a channel for  $\log 2$  s halves both voltages. Two successive channels therefore end at  $(1/4, -1/4)$  V and 1/16 J from an initial 1 J. The independent two-resistor integrals give heats 3/4 then 3/16 J; reversing the channel order interchanges their recipients. These are shunts across individual cells; the earlier exchange resistor instead has time constant  $C_e/G$ . Stating the connections prevents a factor-of-two ambiguity in the duration of a half-difference map.

The scalar difference-mode maps above commute. With three stores a connection change can alter the final state as well as the recipient of heat. Take three 1 F cells with initial charges  $q = (1, 0, -1)^T$  C. Channel A joins cells 1 and 2 through 1  $\Omega$ ; channel B joins 2 and 3. Each operates for  $(\log 2)/2$  s. For incidence  $a$ , KCL is  $\dot{q} = -aa^T q$  and its heat is  $\int (a^T q)^2 dt$ . The two exact maps are

$$M_A = \begin{pmatrix} 3/4 & 1/4 & 0 \\ 1/4 & 3/4 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad M_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 3/4 & 1/4 \\ 0 & 1/4 & 3/4 \end{pmatrix}. \quad (95)$$

Their ordered products and separately integrated channel heats give:

**Table 10.** Each final store is 133/256 J, evaluated from  $q^T q/2$ ; each total heat is 123/256 J, from an initial 1 J.

| Order    | Final charges, C       | Heat in A, J | Heat in B, J |
|----------|------------------------|--------------|--------------|
| A then B | $(3/4, -1/16, -11/16)$ | 3/16         | 75/256       |
| B then A | $(11/16, 1/16, -3/4)$  | 75/256       | 3/16         |

Indeed  $M_B M_A - M_A M_B \neq 0$ . Equal energy here follows from the symmetric preparation; it does not restore equal final state. Finite resistor maps do not reset the cells exactly to zero. This bounded noncommuting extension leaves the physical partition under other connections and switch laws open [OP-LR48-02].

### 9.6 A damped electrical and mechanical preparation

A reciprocal port map extends beyond lossless oscillators. For a series circuit  $\dot{q} = i$ ,  $Li = V - q/C - Ri$  and a mass–spring–dashpot  $\dot{x} = v$ ,  $m\dot{v} = F - kx - bv$ , set

$$q = \beta x, \quad i = \beta v, \quad V = F/\beta, \quad L = m/\beta^2, \quad 1/C = k/\beta^2, \quad R = b/\beta^2. \quad (96)$$

Then  $Vi = Fv$ ,  $Ri^2 = bv^2$ ,  $q^2/(2C) = kx^2/2$  and  $Li^2/2 = mv^2/2$ . Substitution proves the state and port correspondence without assigning a balance residual as a missing work. Actual collision/contact laws and a manufactured realization remain unidentified [OP-LR50-01].

For  $L = C = m = k = 1$ ,  $R = b = 1/5$  and  $\beta = 1$  in SI units, start at zero and apply effort +1 on  $[0, 1]$ , 0 on  $[1, 2]$ , then  $-1$  on  $[2, 3]$  s. The common state matrix is  $A = \begin{pmatrix} 0 & 1 \\ -1 & -1/5 \end{pmatrix}$ . On each interval the exact forced exponential determines its endpoint; the independent integrals  $\int ui \, dt$  and  $-\int i^2/5 \, dt$  are given by (2) after adjoining a constant state. The two physical models therefore have identical signed works and stores on each finite interval. The third endpoint is retained; opposite drive is not an assumed exact reset.

A physical common-motion preparation also has a source. Over  $t \in [-1, 0]$  s, put  $s = t + 1$  and prescribe  $x = (3/10)(s^3 - s^2)$  m,  $v = \dot{x}$ ,  $F = \ddot{x} + \dot{x}/5 + x$  in the same unit model. Integration of the force and dashpot products gives

$$W_F = \frac{237}{5000} \text{ J}, \quad W_b = -\frac{3}{1250} \text{ J}, \quad E(-1) = 0, \quad E(0) = \frac{9}{200} \text{ J}. \quad (97)$$

The endpoints  $(x, v) = (0, 0), (0, 3/10)$  are evaluated independently. Changing an observer coordinate alone performs none of this preparation.

## 10 Material, thermal, sensor, and supply boundaries

### 10.1 Conversion is different from heat already exported

For a fixed physical mesh, opposing sliding forces convert  $D = \mu N |v_{\text{slip}}|$ . Both contact points receive the same observation velocity subtraction, so  $v_{\text{slip}}$  and  $D$  are invariant. For  $\mu = 1/2$ ,  $N = 2$  N,  $|v_{\text{slip}}| = 1$  m/s over one second, the mechanical conversion is 1 J in every observation; at  $\mu = 0$  it is zero. Bearings use relative angular speed, for example  $D_b = b(\omega - \omega_b)^2$ , while a declared quadratic drag effort gives windage conversion  $D_w = c|\omega - \omega_a|^3$ . Lubricant churning needs its own measured constitutive law and moving-fluid boundary. None follows from the ideal rolling equation.

A thermal body with heat capacity  $C_T$ , excess temperature  $\theta$  above a fixed ambient  $T_a > 0$ , and conductance  $H_T$  obeys

$$C_T \dot{\theta} = D - H_T \theta, \quad E_T = C_T \theta, \quad P_{\text{amb}} = -H_T \theta = -T_b \dot{S}_{\text{out}}. \quad (98)$$

For constant  $D$  and initial  $\theta = 0$ ,  $\theta = (D/H_T)(1 - e^{-H_T t/C_T})$ . Choose  $D = 1$  W,  $C_T = H_T = 2$  in SI units on one second. The independently integrated internal conversion, external heat, and store change are +1,  $-e^{-1}$ , and  $1 - e^{-1}$  J. Immediate export would predict  $-1$  J and zero thermal change. In the combined mechanical/thermal boundary the conversion cancels between the two subsystems; only ambient heat crosses the combined boundary. For repeated identical heating intervals the exact map is  $\theta_{n+1} = e^{-H_T T/C_T} \theta_n + (D/H_T)(1 - e^{-H_T T/C_T})$ . No equality of initial and final temperatures is assumed. If copper resistance varies with temperature, specify  $R(T)$  independently and integrate  $R(T)i^2$  on the actual path; a fixed-temperature law cannot identify that feedback.

Coulomb conversion is invariant under a change of observation for the same physical contact; the actual friction law still requires identification [OP-EPI-13]. Bearing drag, windage, churning and retained heat require their own ports and endpoint observations [OP-EPI-25].

### 10.2 Magnetic material states and finite settling

A normalized but dimensionally scaled magnetic mode in configuration VI has flux  $\phi$ , material state  $m$ , and energy

$$E_m = \frac{\phi^2}{2} + \frac{a\phi^4}{4} + \frac{h(\phi - m)^2}{2}, \quad i = \frac{\partial E_m}{\partial \phi} = \phi + a\phi^3 + h(\phi - m), \quad \dot{\phi} = v - Ri, \quad \dot{m} = \frac{h(\phi - m)}{\zeta}. \quad (99)$$

Positive  $a, h, \zeta$  carry the units making each term an energy or the stated electrical variable; numerical controls set their SI coefficients to one where specified. Differentiation from these independent laws gives  $\dot{E}_m = vi - Ri^2 - h^2(\phi - m)^2/\zeta$ . Thus electrical input, copper heat, and material heat are distinct integrals. Positive  $a$  gives a saturation-type increasing reluctance; a single lag is one admissible rate-dependent model, not an identified magnetic material.

For a reversible ramp,  $\phi = t, h = 0, R = 1/5$  on  $[0, 1]$  s,  $i = t + at^3, v = 1 + Ri$ , and initial energy zero. With  $a = 0$  the separate source and copper works are  $17/30$  and  $-1/15$  J, giving final energy  $1/2$  J. With  $a = 1$  they are  $3/4 + 92/525$  and  $-92/525$  J, giving  $3/4$  J. The extra magnetic input is  $1/4$  J, calculated from  $\int i\phi dt$  for each path. This does not specify hysteretic loss.

For an exact periodic lag control take  $a = h = \zeta = 1, R = 1/5, \phi = \sin t, m = (\sin t - \cos t)/2$  and impose  $v = \cos t + Ri$  on  $[0, 2\pi]$  s. Direct integration gives

$$W_s = 63\pi/40 \text{ J}, \quad W_{\text{copper}} = -43\pi/40 \text{ J}, \quad W_{\text{material}} = -\pi/2 \text{ J}, \quad E_m(0) = E_m(2\pi) = 1/8 \text{ J}. \quad (100)$$

For the reversible  $h = 0$  control on the same imposed  $\phi$ , magnetic cycle input and material heat are both zero; its source still supplies its own copper integral. The lag control's initial  $m = -1/2$  is a prepared periodic state. Its preparation work is not assigned from  $1/8$  J.

Finite settling can be checked exactly. Set  $m = (\sin t - \cos t)/2 + \delta e^{-t}$  on the same interval. Material conversion becomes  $\pi/2 - \delta(1 - e^{-2\pi}) + \delta^2(1 - e^{-4\pi})/2$ ; magnetic electrical input is  $\pi/2 - \delta(1 - e^{-2\pi})/2$ . The endpoint stores, evaluated from the material state, are  $(1/2 - \delta)^2/2$  and  $(1/2 - \delta e^{-2\pi})^2/2$  J. Their change is  $\delta(1 - e^{-2\pi})/2 - \delta^2(1 - e^{-4\pi})/2$  J. Measure  $\int i(v - Ri)dt$ , flux linkage, thermal transfer, and endpoint material indicators over successively specified finite periods. Saturation, remanence and hysteresis laws require independent identification; a loop that has not returned to its initial material state is not an exactly periodic heat measurement.

Finite magnetic constitutive behavior must be identified beyond the linear winding law [OP-TRF-03]. Remanence, hysteresis and temperature-dependent states in an actual material retain a separate cycle-work question [OP-LR57-01].

### 10.3 Finite sensors and biased electromechanical actuation

A sensor capacitor driven by a held unit voltage through  $R = 1 \Omega, C = 1 \text{ F}$  from zero has  $v = 1 - e^{-t}$  and source current  $e^{-t}$ . On  $[0, \ln 2]$  s, the source work is  $1/2$  J, resistor heat work  $-3/8$  J, and capacitor end-point store  $1/8$  J. Their separate products are  $1 \cdot e^{-t}$  and  $-e^{-2t}$ . A nonloading ideal observation assigns all three zero; this finite sensor has both delay and supplied energy. If its resistor is inside a thermal boundary, use (98) instead of exporting its conversion twice. Real probe input capacitance, transducer bias, electronics supply, and calibrated transfer function belong to the sensor model. Noise, delayed or nonlinear observers, and multiple feedback channels are additional states or error processes, not consequences of this scalar control.

For the electromechanical plant, independently specify

$$Li = v_s - Ri - g_e v_m, \quad \dot{x} = v_m, \quad m\dot{v}_m = g_m i - cv_m - kx + F_s. \quad (101)$$

Its store is  $Li^2/2 + mv_m^2/2 + kx^2/2$ . Multiplication by  $i$  and  $v_m$  identifies the separate controller transfer

$$P_a = (g_m - g_e)iv_m, \quad \dot{E}_{\text{plant}} = v_s i + F_s v_m - Ri^2 - cv_m^2 + P_a. \quad (102)$$

The controller receives  $-P_a$  on the opposite boundary. Reciprocal coupling has  $g_m = g_e$ ; unequal or

reversed backaction requires a separately supplied physical realization, not a negative store.

For an exact finite supply control use  $L = m = 1$ ,  $R = c = k = 0$ ,  $g_e = 1$ ,  $g_m = 2$ ,  $i = v_m = 1$ ,  $v_s = 1$ ,  $F_s = -2$  on one second. The imposed motion satisfies (101), its store is 1 J at both endpoints, and separate works are  $W_e = 1$ ,  $W_m = -2$ ,  $W_a = 1$  J. Let a declared bidirectional converter have efficiency  $1/2$  in delivery, so that delivering  $P_a = 1$  W draws 2 W and converts 1 W to heat. A bias coil  $L_b = 1/2$  H,  $R_b = 1 \Omega$ , held at  $i_b = 1$  A requires another 1 W and has store  $1/4$  J at both endpoints. The source-to-controller work is 3 J; output, converter heat, and bias heat works are each  $-1$  J. The controller therefore has zero store change by its independent states.

Realize the finite source by a rail  $C_s = 2$  F,  $V_s(t) = \sqrt{4 - 3t}$  V, with ideal regulated draw  $I_s = 3/V_s$  A. Its store changes from 4 to 1 J, and its separately integrated output is  $-\int V_s I_s dt = -3$  J. The combined plant, controller, and rail endpoints are  $21/4, 9/4$  J; external electrical input, receiver output, and the two heat works sum to  $1 - 2 - 1 - 1 = -3$  J. A remaining voltage above the declared cutoff 1 V permits this interval; beyond its endpoint the regulator law must change. The regulated draw and conversion law are assumptions requiring their own actuator and supply measurements.

Source-off does not imply passive decay of every controller store. For a separate return control, a mechanical actuator delivers 1 W to a converter with recovery efficiency  $1/2$  for one second. The rail receives  $1/2$  J, converter heat receives  $1/2$  J, and the actuator receives work  $-1$  J. If  $C_s = 2$  F and  $V_s(0) = 1$  V, then  $V_s(t) = \sqrt{1 + t/2}$  and rail current is  $1/(2V_s)$ ; direct integration gives the stated positive rail work. This is an explicitly supplied return path, not a prediction for an unidentified bias coil. The actual bias law can make coupling coefficients depend on  $i_b$  and must be solved with the plant. A bounded algebraic controller or a positive aggregate store cannot certify its separate coil, rail, thermal, or whole-apparatus work accuracy.

Finite sensors and thermal endpoints require measured loading, supply work and retained states [OP-LR57-07]. The realized actuator and bias supply must support each forward or returned controller transfer [OP-LR59-02].

#### 10.4 Complete cycles and what a prescribed path establishes

For every actual apparatus keep stages  $\ell$  distinct: preparation, operation, switching, relaxation, reset, and any hold interval. The measured statement is

$$r_{\text{cycle}} = E_{\text{final}} - E_{\text{initial}} - \sum_{\ell} \sum_j \int_{t_{\ell,0}}^{t_{\ell,1}} e_j f_j dt. \quad (103)$$

A periodic thermal state, restored battery state, or emptied capacitor is an endpoint observation, not a premise. For a source-free regular linear relaxation,  $x(T) = e^{AT}x(0)$  and the exponential is invertible. A nonzero state cannot become exactly zero at finite  $T$  by that law alone. For a capacitor  $C$ ,  $V_0$  relaxed through  $R$  for  $T = RC \ln 2$ , separate heat work is  $-3CV_0^2/8$  and endpoint store  $CV_0^2/8$ .

A completely specified elementary 2 J receiver cycle uses a unit capacitor. On preparation  $[0, 2]$  s impose  $v = t$  with source current 1 A; its work is +2 J and store rises from 0 to 2 J. On operation  $[2, 6]$  s use an ideal controlled receiver with  $v = (6 - t)/2$ , receiving current  $1/2$  A. Its work into the capacitor boundary is  $-2$  J and store falls to zero. On reset  $[6, 10]$  and hold  $[10, 12]$  s,  $v = i = 0$  and all works are zero. Currents change finitely at the switches; capacitor voltage is continuous. This is a prescribed, lossless current-control model. A one-half-efficient physical preparation converter instead draws 4 J to deliver the same 2 J preparation; its independent heat work is  $-2$  J. Both predict the same receiver transfer and different measured source work.

A finite reset compares further paths. Discharging a prepared unit-energy store to zero through a resistor sends 1 J to heat in a complete asymptotic reset, with finite endpoints retained on every finite interval. A lossless resonant auxiliary receives that energy in a finite quarter-period and must return it during a specified later preparation to close its own cycle. A one-half-efficient regenerative converter delivering the same 1 J removal to a battery stores 1/2 J and converts 1/2 J to heat; drawing that 1/2 J back through the same efficiency returns only 1/4 J to the plant. Battery terminal work, chemical state, and temperature must be independently specified; those efficiencies are illustrative laws, not battery measurements.

For the broader physical cycle questions, retain the different apparatus. A differential three-coil pickup, rectification before or after combining pickup signals, and a tuned receiver each have separate preparation and reset connections. Finite paths for those systems do not establish cycles for a local/distant-field winding arrangement, a moving or relaxing magnetic material, or a spark-gap resonator. For the field arrangement integrate every winding source and reset branch and include field energy endpoints. For motion/material apparatus add force-velocity and material/thermal states. For a spark-gap resonator retain primary, secondary, gap, clamp, and parasitic capacitor ports on each event side. A 2 J receiver target can be held for each comparison, but its required source work is  $2J + Q_{\text{ext}} + \Delta E - W_{\text{other}}$ , where each term is independently observed or predicted from its specified law; this identity is not a way to assign an unmeasured port. No unspecified source, arc, battery, or controller law has a definite efficiency or cycle work here.

A complete transformer cycle requires all finite stages and observed reset states [OP-TRF-05]. Prescribing a path does not establish a physical supply's work or feasibility [OP-LR49-01]. The broader field, pickup, moving-material and spark-gap apparatus retain their own missing laws [OP-LR49-02].

## 11 What must be measured

### 11.1 Independent uncertainty for work and endpoints

Record conjugate effort and flow on the same physical interval, including all event sides. If true channels differ from recorded  $\hat{e}, \hat{f}$  by at most  $u_e, u_f$ , an exact bound is

$$|W - \widehat{W}| \leq \int_{t_0}^{t_1} (|\hat{e}|u_f + |\hat{f}|u_e + u_e u_f) dt + U_{\text{time}} + U_{\text{band}} + U_{\text{read}}. \quad (104)$$

Here the additional bounds cover integration endpoints and relative timing, unresolved signal bandwidth, and reading reconstruction, including noise and quantization. They are established independently of a small energy residual. For a finite endpoint error  $\delta t$  and bound  $|P| \leq P_{\text{max}}$ , the omitted endpoint work is at most  $P_{\text{max}}|\delta t|$ . If a flow channel has  $|\dot{f}| \leq B_f$ , a relative timing offset bounded by  $u_t$  contributes flow error at most  $B_f u_t$  by the mean-value theorem. Neither bound applies across an unresolved ideal impulse; use the finite physical pulse and its actual envelopes.

For a capacitor with recorded positive  $\widehat{C}$ ,  $|C - \widehat{C}| \leq u_C$ , and  $|V - \widehat{V}| \leq u_V$ , the exact endpoint bound is

$$U_E \leq \frac{u_C}{2} (|\widehat{V}| + u_V)^2 + \widehat{C} \left( |\widehat{V}|u_V + \frac{u_V^2}{2} \right). \quad (105)$$

Use the quadratic-form analogue for coupled magnetic or mechanical stores, including covariance or interval dependence of their parameters and states. The mutual term is not two independent inductors. Thermal endpoints require heat capacity, temperature distribution, and heat-transfer uncertainties; mate-

rial endpoints require the stated internal-state law. A residual bound is

$$U_r = U_{E_0} + U_{E_1} + \sum_j U_{W_j}, \quad r_E \in [\hat{r}_E - U_r, \hat{r}_E + U_r]. \quad (106)$$

This conservative interval needs no independence assumption. Correlation can sharpen it only when established by calibration and propagated jointly. For probabilistic intervals, specify the joint error law and coverage; for certified intervals, enclose its support and nonlinear products. General metrological treatment of inputs and correlations is described by JCGM (2008); the finite product and endpoint bounds here are derived directly and discard no second-order term.

Phase and polarity errors illustrate the size of the problem. For peak signals  $v = \cos t$ ,  $i = \cos(t + \phi)$  over  $[0, 2\pi]$  s,

$$W = \pi \cos \phi \text{ J}, \quad \widehat{W} = \sigma(1 + a)(1 + b)\pi \cos(\phi + \delta) \text{ J}. \quad (107)$$

Here  $\sigma = \pm 1$  is channel polarity,  $a, b$  are gains, and  $\delta$  is relative phase including timing at the stated angular frequency. At  $\phi = \pi/2$ ,  $\delta = \pi/6$ , unit gains and correct polarity,  $W = 0$ ,  $\widehat{W} = -\pi/2 \text{ J}$ . At  $\phi = 0$ ,  $a = 3/100$ ,  $b = -3/100$ ,  $\delta = 0$ , the gain error is  $-9\pi/10000 \text{ J}$ . Reversing only current changes  $+\pi$  to  $-\pi \text{ J}$ . Reversing both channels does not do so. These predictions follow by direct trigonometric integration. A sensor voltage error  $1/100 \text{ V}$  at a true  $100 \text{ V}$  on a  $1 \text{ F}$  capacitor gives endpoint error  $20001/20000 \text{ J}$ . Similar-looking endpoint voltages do not establish cancellation of their calibration errors.

A low-pass channel's transfer, loading, time alignment, noise spectrum, and quantization must be calibrated over the signals being integrated. In a multi-port assembly, moving a probe across a shunt can include a different branch: the extra current is a physical port-placement error with correlated voltage and store consequences. Synchronizing instruments alone does not bound it. A sensor that loads the apparatus changes its trajectory and belongs in its constitutive comparison.

The calibrated polarity, gain, phase, timing, bandwidth and endpoint uncertainties determine the signed residual that can be resolved [OP-LR63-02].

## 11.2 Finite comparisons and decision requirements

In the following tables, every pair gives different predictions for at least one accessible quantity under its stated control. A zero in the first column may be a null hypothesis about an omitted contribution, not an established physical law. An exact alternative applies only with its explicitly stated constitutive assumptions. For a pair separated by  $\Delta > 0$ , require a combined prediction-setting and observation uncertainty  $U < \Delta/2$ ;  $U \leq \Delta/4$  is a concrete design target. This requirement is applied to the individual comparison, never inferred from instrument names. If the required bound is unavailable, that comparison remains unresolved. A finite outcome can reject one of these readings on its interval; it does not close the same question for every material, geometry, or drive.

For an unknown physical remainder, the complete-model reading is  $r_E = 0$ ; a surviving-discrepancy reading is  $r_E = \delta$  with  $|\delta| > U_r$ , where sign and magnitude must be measured. These are distinct boundary predictions but there is no justified fixed nonzero value before measurement. To validate the observation chain at a known scale, introduce an independently metered  $1/4 \text{ J}$  positive input. Omitting it predicts  $r_E = +1/4 \text{ J}$ ; including it predicts zero for the otherwise exact control. Requiring  $U_r < 1/8 \text{ J}$  distinguishes that omission. This calibration control is not a prediction of a physical energy anomaly.

The loaded carrier comparison has a practical conditional budget. Suppose both torque records have  $|\widehat{\tau}| \leq 8 \text{ N m}$  and  $|\widehat{\omega}| \leq 4 \text{ rad/s}$ , with respective pointwise errors  $1/100 \text{ N m}$  and  $1/100 \text{ rad/s}$ . On two one-second observations the product part of (104) is at most  $1201/5000 \text{ J}$ . A total comparison bound of

1 J leaves 3799/5000 J for the other contributions and separates the 7 J attachment difference. It does not establish a 1 J full-boundary residual uncertainty.

For the reference actuator,  $\hat{C} = 1 \text{ F}$ ,  $u_C = 1/100 \text{ F}$ ,  $|\hat{w}| \leq 2 \text{ V}$ , and  $u_V = 1/100 \text{ V}$  give  $U_E \leq 80501/2000000 \text{ J}$  at each endpoint. Two final-state observations have total 80501/1000000 J before setting and timing uncertainty. A 1/10 J comparison budget leaves 19499/1000000 J for those effects. It resolves  $[(9 + e^{-10})^2 - 100]/50 \text{ J}$  from zero: since  $e^{-10} < 1/100$ , its magnitude exceeds 188199/500000 J. For full work closure also observe actuator, sensor, driver, rail, and thermal endpoints or their external supply products.

### 11.3 Mechanical comparisons

Hold the declared rates or paths with separately metered actuators and observe torque and angle, force and displacement, and each endpoint motion together. The table preserves distinct contact, support, and whole-system boundaries.

**Table 11.** Distinct mechanical measurement targets. The deliberate extra-torque and extra-work alternatives are explicit falsification controls, not assigned properties of unknown joints or friction laws.

| Question and held control                   | Reading A                                     | Reading B and decisive observation                                                                                                                                |
|---------------------------------------------|-----------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stator mounting, same negative-speed motion | No change in conversion: 0                    | $Q_c - Q_0 = 7\pi/50 \text{ J}$ ; integrate drag torque against each relative angle, retaining phase endpoints                                                    |
| Attached receiver, configuration I          | Unloaded carrier work $-7 \text{ J}$          | Loaded 0 J; measure sun, ring, carrier, and all three receiver works on the same motion                                                                           |
| Transient frame closure                     | Carrier shafts alone sum to 595/72 J          | Effective change 517/72 J with field work $-13/12 \text{ J}$ ; reconstruct all four observations from synchronized motions and inertias                           |
| Location of frame difference                | Held-ring work remains 0 in both observations | Carrier-ring work $-5 \text{ J}$ in steady operation; integrate its independently measured reaction torque                                                        |
| History partition, same endpoints           | Shaft difference 13/12 J at $b = 0$           | 13/9 J at $b = 1$ ; drive the distinct intermediate ring path and observe both field and shaft shares                                                             |
| Hooke support over a complete cycle         | Torque-path $W_u = 0$                         | A specified extra unit support torque adds $\pi/2 \text{ J}$ over the control cycle; resolve its torque and support angle rather than assign it from the residual |
| Ratio-two bevel support                     | Held support flow gives 0                     | Moving support gives $-\pi/2 \text{ J}$ in ground; carrier reconstruction gives zero on the same cycle                                                            |
| Compound observation, close-ratio geometry  | Ground $\mathcal{C}_0 = 1$                    | Carrier 2950/21; measure each signed shaft and contact side and bound the nonzero ground denominator                                                              |

| Question and held control                     | Reading A                                                                        | Reading B and decisive observation                                                                                                                          |
|-----------------------------------------------|----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Other geometry, two-sun control               | Ground shaft ratio 1                                                             | Carrier shaft ratio 6/5; instrument four shafts and retain all four meshes; helical or bevel thrust requires its own force/motion observations              |
| Compliant engagement                          | An inferred instantaneous 1 J jump from a power change                           | Actual instantaneous store jump 0 despite a 1 W power jump; resolve finite contact work 2 J, heat 1 J, and spring endpoint 1 J                              |
| Restitution control                           | Elastic $e = 1$ : kinetic loss 0                                                 | $e = 1/2$ : loss 3/16 J; observe incoming/outgoing motion and resolved contact work, with adequate pulse bandwidth                                          |
| Compliance with acceleration                  | Omitting the field leaves carrier final accounting at 27/32 J                    | Including $-1/2$ J field work gives 11/32 J effective store in the six-contact train; observe all contact states and event sides                            |
| Unequal planets and pin couple                | Equal first-contact work 7/3 J; zero pin-pair work                               | First contact 7/2 J; the separate one-pin control gives $-1/3$ J; measure individual forces and pin torques                                                 |
| Locked approach                               | Drive-scaled limiting works (0, 0, 0)                                            | Fixed-drive (2, 5, $-7$ ) J; record drive and relative motion separately at each finite approach                                                            |
| Mesh friction invariance                      | Frame heat difference 0                                                          | A postulated frame-dependent extra 1/4 J differs from zero; compare reconstructed relative-slip products on the same contact, not remounted friction bodies |
| Bearing/windage/churning with a thermal store | Immediate export 1 J, retained heat 0                                            | Unit thermal control exports $e^{-1}$ J and retains $1 - e^{-1}$ J; observe temperatures and ambient heat as well as mechanical conversion                  |
| Radial motion                                 | Omitted radial work 0                                                            | Radial guide work $-3/2$ J; measure radial force and displacement with tangential support work +3 J                                                         |
| Nonparallel observation                       | Scalar-only field work 0                                                         | Vector control $-1$ J; reconstruct all angular-momentum components and observer motion                                                                      |
| Within-interval reversal                      | Net signed work 0 taken as zero traffic                                          | Absolute transfer 1/2 J and unit-drag heat 1/2 J; integrate both sides of the flow zero                                                                     |
| Cardan transverse and bending work            | Fixed application point: 0                                                       | Unit force through unit displacement: 1 J; determine actual bearing constraints and all force/couple conjugate motions                                      |
| Unequal angles, acceleration, phase actuation | Equal-angle partial support, steady rotor pin, and absent phase drive: (0, 0, 0) | Separate controls give $(\pi/4 - \arctan(3/4), 1/2, 1)$ J; observe each support and phase supply on its own interval                                        |

| Question and held control | Reading A                    | Reading B and decisive observation                                                                                                                              |
|---------------------------|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Physical gear closure     | Complete physical residual 0 | $\delta$ outside its independent interval bound; validate the chain with the $1/4$ J input, then retain any actual signed discrepancy with all completed checks |

A physical gear remainder has unknown magnitude and sign until the complete boundary is measured within those independent uncertainties [OP-EPI-34].

#### 11.4 Electrical and correspondence comparisons

Use synchronized complete voltage/current pairs for source, receiver, probe, reference driver, switch, and heat-conversion channels. Independently observe winding currents, capacitor voltages, flux or identified material states, and thermal endpoints. A specified finite waveform must fit the calibrated channel bandwidth without clipping; opening pulses require a separate bound.

**Table 12.** Electrical comparisons. Finite-model success includes every port and both state endpoints on each interval; neither an ideal ratio nor a small aggregate remainder supplies a missing dynamic map.

| Question and control                             | Reading A                                                      | Reading B and decisive observation                                                                                                                              |
|--------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Physical reference attachment                    | Common coordinate shift: added work 0                          | Unit RC attachment requires $5/6$ J input, retains $1/2$ J; observe its actual return current and voltage                                                       |
| Complete transformer cycle                       | Passive finite relaxation called reset: final store 0          | At $RC \ln 2$ , final store $CV_0^2/8$ ; observe preparation, receiver, switch, supplies, and all remaining states separately                                   |
| Nonlinear magnetic/thermal state                 | Linear unit ramp: magnetic input $1/2$ J                       | Saturated ramp: $3/4$ J; observe $i(v - Ri)$ and actual material/thermal endpoints                                                                              |
| Energized winding opening and return commutation | Unit-voltage clamp receives $3/2$ J                            | Two-volt clamp receives 2 J; observe each clamp and switch port; the tapped overlap/gap alternatives use their separately defined matrix works                  |
| Finite reference realization                     | Ideal receiver endpoint 2 J                                    | Finite actuator $(9 + e^{-10})^2/50$ J; compare source products, tracking, actuator stores, and finite rail state                                               |
| Constrained winding limits                       | Zero-store ideal transformer at the unit ramp: 0               | Perfect-coupling finite-magnetizing model: $1/2$ J; retain the separate capacitance, source-resistance, short, and zero-ratio controls in the constrained table |
| Crossed load, probe, frequency, and preparation  | Prepared RC receiver gets $(1 - e^{-2})/2$ J without the probe | With probe it gets $(1 - e^{-4})/4$ J; use all 16 specified winding settings for interaction claims and each exact work primitive on its own state              |

| Question and control                          | Reading A                                           | Reading B and decisive observation                                                                                                            |
|-----------------------------------------------|-----------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Physical transformer closure                  | Complete physical residual 0                        | $\delta$ outside its independently bounded interval; the known $1/4$ J input validates detection separately from physical closure             |
| Loaded dynamic mapping                        | Coordinate-shift-only capacitor change 0            | Supplied compensation gives 2 J, matching the relative kinetic control; observe the extra ports and state mapping, not solely terminal counts |
| Ideal compound terminal versus physical ratio | Complete-port ratio 1 for $R = -341/100$ , $B$ held | A-frame mapped ratio $341/100$ ; observe actual winding voltages and reconstructed terminal contributions separately                          |

The corresponding physical transformer remainder requires simultaneous port and endpoint observations, including switch, probe and supply effects [OP-TRF-07].

### 11.5 Preparation, switching, material, supply, and calibration comparisons

These comparisons retain the original apparatus classes. A general work law can be shared; a measured outcome for one device cannot be transferred to a different winding, field, impact, pickup, or controller arrangement.

**Table 13.** Prepared and supplied controls. The source-off rail comparison, hidden-mode control, and material loop belong to their explicitly stated systems; they do not certify wider apparatus or unmeasured constitutive laws.

| Question and control                             | Reading A                                                                | Reading B and decisive observation                                                                                                            |
|--------------------------------------------------|--------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Prepared singular-component paths                | Zero-state two-section circuit: heat 0                                   | Prepared state in equation (89): heat $(1 - e^{-2})/2$ J; cross finite component scalings and observe both independent states                 |
| Switching work beyond smooth zero-state transfer | Resistive regularization exports $3/8$ J by its declared endpoint        | Reactive quarter-period transfer exports zero heat and leaves $1/2$ J in the auxiliary; integrate all ports on each finite path               |
| Nearly cancelled, multiple, or nonideal modes    | One known mode: at most $1/20000000000$ J under the stated current bound | Two modes at $\pm 1$ V: $1/1000000$ J despite zero terminal signal; resolve internal branches, coupling uncertainty, and observation rank     |
| Exact hidden mode                                | Empty branch: initial energy and release heat 0                          | Prepared microfarad branch: $1/2000000$ J and $3/8000000$ J; observe branch current/voltage and retained endpoint                             |
| Physical common-mode preparation                 | $u = -1/2$ : input $95/2$ J and no half-period coil exchange             | $u = 1/2$ : input $115/2$ J and coil works $(-19, +19)$ J; observe preparation sources and flux signs, retaining differential load backaction |

| Question and control                                            | Reading A                                                       | Reading B and decisive observation                                                                                                                                    |
|-----------------------------------------------------------------|-----------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Same switching endpoint, different partition                    | <i>A</i> then <i>B</i> : heat pair $(3/8, 3/32)$ J              | <i>B</i> then <i>A</i> : $(3/32, 3/8)$ J; measure channels separately, not only total heat or final voltages                                                          |
| Actual switch/gap and release laws                              | Unit clamp: $3/2$ J receiver transfer                           | Two-volt clamp: 2 J; distinguish ignition, restrike, overlap, gap, and release states through their branch products and endpoint capacitances                         |
| Finite capacitance ratios/orientations and limiting surfaces    | $(C_1, C_2) = (1, 2)$ F: final store $3/16$ J                   | Reversed ratio $(2, 1)$ F: $11/16$ J; extend to declared finite smaller parasitics and wider ratios with their actual bandwidth; no finite outcome realizes zero time |
| Controlled cycle versus physical supply                         | Lossless 2 J preparation requires 2 J source input              | Half-efficient preparation requires 4 J; observe supplies, receiver, conversion and thermal endpoints on the complete schedule                                        |
| Missing preparation/reset and converter laws in other apparatus | Ideal recovery of a 1 J removal stores 1 J in the receiver      | Half-efficient recovery stores $1/2$ J; observe every actual source, field/mechanical, gap, battery, and reset path for that apparatus                                |
| Identified hysteretic material and settling                     | Reversible magnetic cycle input 0                               | Periodic lag control $\pi/2$ J; measure internal-state closure and copper-subtracted magnetic input over declared periods                                             |
| Realized thermal and sensor ports                               | Nonloading ideal sensor input and stored energy 0               | Finite sensor input $1/2$ J and store $1/8$ J; record sensor supply, loading, temperature, and delay before extending to other feedback structures                    |
| Finite electromechanical bias supply                            | Treating unequal backaction as supplied at zero work misses 1 J | Plant receives 1 J actuator work, controller draws 3 J including its declared losses; source-off recovery control raises rail energy $1/2$ J                          |
| Channel-specific uncertainty                                    | True quadrature work 0                                          | Relative phase error gives $-\pi/2$ J; inject known polarity/gain/timing changes and calibrate joint port-placement and endpoint errors separately                    |

A comparison closes its bounded question only when the actual domain, connections, event law, source settings, endpoint states, and uncertainty meet the stated conditions. A measured mismatch is retained with sign, magnitude, interval, and completed polarity, loading, timing, endpoint, and model checks. Missing constitutive identification or inadequate resolution leaves the question open. A finite family of successful comparisons does not establish universal absence of a discrepancy.

## 12 Relation to earlier work and limits of the mathematics

The carrier-relative gear construction in [Culpepper \(2002\)](#) and the rotating-coordinate mechanics in [MIT \(2022\)](#) support the starting kinematics. The individual work differences here follow from explicit free

bodies and exact integrals. They do not turn a rotation count into a work measurement. The magnetic terminal relations in Haus and Melcher (1989) distinguish finite inductive storage from an ideal algebraic winding relation. The physical return, clamp, and supply comparisons require further circuit laws given here. The metrological framework of JCGM (2008) motivates recording input quantities and correlations; it does not provide a numerical uncertainty for this apparatus.

The common ground with Nilre and Herlin (2026) is the separation of complete physical ports, observation-dependent contributions, and independent state endpoints. The additional controls expose contact, topology, preparation, hidden-state, material, and calibration assumptions needed for finite comparisons. Every theorem above is conditional on its equations. The simple rolling and axial linkage results do not settle actual Cardan transverse reactions. The Ravigneaux pitch example does not settle helical thrust or every nonparallel contact. Finite compliant and restitution controls do not identify a real impact law. Ideal terminal correspondence and the elastic dynamic map have different state spaces and different closure domains.

Exact polynomial or trigonometric controls can locate every power zero algebraically. A root-free interval, a tangent zero, a persistent identity, and equality of maximum absolute powers have different roles. A complete event description for one specified smooth constitutive model does not supply nonlinear switch laws for another. A same-sign narrow transfer may also carry substantial work without any interior power zero. The power integrals and their bandwidth requirement remain independent of event counts.

Physical magnetic identification, finite-source realization, and full cycle closure remain empirical questions. Bounded results for prepared paths and smooth-model correspondences leave their wider domains intact. The numerical integration residual here is exactly zero, and each complete ideal control has  $r_E = 0$  by independently calculated works and endpoints. Physical model residuals remain unmeasured. They include omitted joint, contact, bearing, windage, churning, switch, arc, magnetic, dielectric, thermal, sensor, controller, and supply behavior. No aggregate model identity assigns the magnitude or destination of an unexplained transfer.

### 13 Conclusion

The fourth shaft turns a readout question into a loaded-machine test. In the ideal one-receiver control, the output takes  $7/3$  J and carrier work changes by the same amount while the held ring carries torque without ground work. The free bodies determine that result. Actual U-joints require measured spatial reactions and application-point motions. A dimensioned Oldham model, a finite clutch and specified stator loads establish their own transfers and show which assumptions change.

The switched electrical apparatus has equally specific connections. Full-to-tap return changes the circuit; its winding, cell, gate and thermal states persist through the event. A clocked contraction theorem leaves hybrid guards and material admissibility to their own analysis. Generator comparisons require prime-mover, field and controller work over matched service with actual preparation and reset.

Two series banks can share an entire terminal history while retaining 1 or 5 J. Reconnection can distinguish those known states under a specified observation map, but receiver delivery follows its separate integral and need not track the hidden-energy ordering. The LC/shunt certificate establishes every charge-sign event for its declared window. Three-store reset changes the final state with order; the damped correspondence gives a physical preparation whose source work is independently calculated. These are bounded results with explicit state and topology domains.

The shared-output extension has a continuous prepared source-return witness with positive receiver work and a decreasing independently defined store. Its supplied machine admits no full-state cycle under

the stated insulated thermal law. In the separate electrical fixture, loaded complete-port records leave a  $1/100$  J optimal worst-case initial-energy error over two known states. Full-wave operation, heat rejection, richer sensors and complete-service performance each require their own model and comparison.

The remaining experiments require measured laws, resolved work differences and independent end-point uncertainties. A small algebraic residual supplies none of those observations. Any positive or negative physical remainder retains its measured value, conditions and uncertainty until its cause is identified.

## A Component bounds for the shared output

This appendix supplies the finite arithmetic behind (64), using precisely the graph and coordinate order of (59). Currents, voltages and time are normalized by 1 A, 1 V and 1 s; the following matrix coefficients are dimensionless. Their rates correspond to inverse seconds in physical units. The inverse inductance blocks are

$$L_j^{-1} = I_3 - 2nn^T/11, \quad L_o^{-1} = I_{13} - ss^T/17. \quad (108)$$

Each inverse-capacitance block on  $(A_j, Q_{-j})$  is  $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} / 3$ ; each  $Q_{+j}$  diagonal is  $1/2$ ; all other coordinates have inverse 1. These follow by direct multiplication, so there is no numerical inverse. The input-block determinant is  $11/8$  and output-block determinant  $17/4$ ; the positive identity-plus-outer-product forms separately prove full rank.

With  $G_0$  excluding selectors, write

$$A_0 = \begin{pmatrix} -L^{-1} & L^{-1}B^T \\ -C^{-1}B & -C^{-1}G_0 \end{pmatrix}, \quad b = \begin{pmatrix} 0 \\ C^{-1}d(U) \end{pmatrix}, \quad \bar{A} = |A_0| + \sum_{j,k} \begin{pmatrix} 0 & 0 \\ 0 & |C^{-1}a_{jk}a_{jk}^T| \end{pmatrix}. \quad (109)$$

Absolute values act entrywise. The selector upper conductance is one in this normalization, so  $|A(t)| \leq \bar{A}$  throughout the finite edge, including independently varying channel gates. Summing its coefficients for each coordinate gives the following bounds, identical across channels.

**Table 14.** Exact infinity-norm majorant. Its maximum is  $61/10$  and  $\|b\|_\infty = 1$ .

| Coordinate                  | Sum of majorant coefficients |
|-----------------------------|------------------------------|
| Input primary current       | 21/11                        |
| Upper / lower input current | 69/22 / 49/22                |
| Each output branch current  | 83/17                        |
| Receiver winding current    | 4                            |
| $A_j, T_j$                  | 73/30, 61/10                 |
| $X_{+j}, Q_{+j}$            | 51/10, 11/20                 |
| $X_{-j}, Q_{-j}$            | 41/10, 53/30                 |
| $p_j, v_o$                  | 11/5, 11/5                   |
| Every probe voltage         | 1/5                          |

For each physical capacitor with incidence  $a_e$  and capacitance  $c_e$  in normalized units, define  $r_e = c_e a_e^T C^{-1} [-B, -G_0]$  and  $b_e = c_e a_e^T C^{-1} d(U)$ . In the state box of radius  $\eta = 134/1939$ , its current law and triangle inequality give

$$|i_e| \leq |r_e x_0 + b_e| + \eta \|r_e\|_1 + \sum_{j,k} |c_e a_e^T C^{-1} a_{jk}| (|a_{jk}^T v_0| + \eta \|a_{jk}\|_1). \quad (110)$$

Every selector is bounded over its entire conductance range. Restoring amperes yields these values for each physical component, not a bound on an equivalent capacitor that could conceal branch current.

**Table 15.** All capacitor currents remain below the declared 2 A limit.

| Physical capacitor                         | Current magnitude upper bound, A |
|--------------------------------------------|----------------------------------|
| $A_j$ parasitic                            | 83667/193900                     |
| $T_j$ parasitic                            | 8311/7756                        |
| $X_{+j}$ parasitic                         | 187351/96950                     |
| $Q_{+j}$ parasitic and positive cell, each | 195067/387800                    |
| $X_{-j}$ parasitic                         | 87417/48475                      |
| $Q_{-j}$ parasitic                         | 36296/48475                      |
| Negative cell, $Q_{-j} - A_j$              | 49313/83100                      |
| Primary-node capacitor                     | 3413/9695                        |
| Output-node capacitor                      | 4887/19390                       |
| Each source probe                          | 2073/9695                        |
| Output probe                               | 2207/19390                       |

Copper, source-resistor and receiver currents have magnitude at most  $1 + \eta = 2073/1939$  A. Each secondary bleeder is bounded by  $2073/19390$  A, each probe resistor by  $2073/9695$  A, and each selector by  $3/10 + 2\eta = 8497/19390$  A. Gates, including their capacitors, carry at most 1 A by their independent RC law. Clamps carry zero. The coarser  $d_* = 331/840$  state box would give an insufficient  $10051/2800$  A bound for a capacitor current; that loose bound is not evidence of a physical violation. The sharper majorant proves the original limits without changing the circuit, preparation or interval.

For the observation theorem, group selector terms by  $k$  across all six channels. The permutation exchanging channels 1 and 2 commutes with the base matrix, the forcing and each of the four grouped selector matrices. It exchanges the corresponding primary, secondary, branch and probe coordinates and fixes the output coordinates. Since it sends the prepared difference to its negative, the connected difference trajectory stays in the antisymmetric subspace. This verifies the receiver identity across both constant commands and the entire exponentially varying edge.

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